

University of Tehran

Department of Mechanical Engineering

**Design Report:
Cooling Tower Fan
Drive Transmission System**

Machine Elements Design Project

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February 24, 2026

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Chapter 1

Introduction and Design Objective

This report documents the complete mechanical design process for the power transmission system of an industrial cooling tower blower. The objective is to transfer mechanical power from an electric motor to a large axial-flow fan blade rotating at a prescribed speed of **110 rpm**. The design follows a structured workflow of selection, elimination, preliminary sizing, and comparative analysis, ultimately producing a full mechanical design grounded in AGMA standards.

The cooling tower uses a 4-metre-diameter fan that must move air at a specified flow rate and pressure rise. The transmission must be reliable, compact, and structurally compatible with the tower's elevated mounting frame—a ladder-type steel structure that imposes hard constraints on both the allowable weight and the available installation envelope. A key geometric constraint is the orientation of the fan shaft: **the blower shaft hangs vertically from its support frame**, and the drive system must integrate with this vertical-shaft arrangement.

The design problem is therefore multi-objective: the system must simultaneously satisfy structural safety (bending and wear safety factors), power delivery requirements, and practical installation constraints. A systematic scoring metric was developed to objectively compare competing transmission architectures on the basis of weight, volume, and efficiency.

Chapter 2

Blower Power Estimation

The required output power delivered to the fan blades is the fundamental starting point for all downstream design decisions. The following operating conditions are specified for the cooling tower installation:

Table 2.1: Given Operating Parameters

Parameter	Symbol	Value	Unit
Fan blade diameter	D	4	m
Air exit velocity	V_{out}	12	m/s
Static pressure rise	h_s	0.9	cm H ₂ O
Site altitude	—	300	m
Air density at altitude	ρ_{air}	1.05401	kg/m ³

2.1 Volumetric Flow Rate

The fan blade sweeps a circular cross-section of diameter $D = 4$ m. Since the air exits the fan uniformly at the specified velocity V_{out} , the volumetric flow rate is:

$$Q = V_{\text{out}} \cdot A = V_{\text{out}} \cdot \frac{\pi D^2}{4} = 12 \times \frac{\pi \times 4^2}{4} = \boxed{150.796 \text{ m}^3/\text{s}} \quad (2.1)$$

This large flow rate—approximately 150 m³/s—reflects the scale of the industrial cooling tower in question.

2.2 Total Pressure Head

The fan must overcome two distinct pressure demands simultaneously:

- **Velocity head** h_v : the kinetic energy that must be imparted to the outgoing air stream to accelerate it from rest to $V_{\text{out}} = 12$ m/s.
- **Static head** h_s : the pressure loss through the cooling tower fill, drift eliminators, and structural components—specified as 0.9 cm H₂O.

Both heads are expressed in inches of water column (in H₂O), the standard convention in cooling tower and HVAC engineering. The velocity head is derived from Bernoulli's principle, converting the kinetic energy per unit weight of air into an equivalent height of

water:

$$h_v = \frac{\rho_{\text{air}} \cdot V_{\text{out}}^2}{2 g \rho_{\text{water}}} = \frac{1.054 \times 12^2}{2 \times 9.81 \times 1000} = 7.74 \text{ mm H}_2\text{O} = 0.30513 \text{ in H}_2\text{O} \quad (2.2)$$

The factor $\rho_{\text{air}}/\rho_{\text{water}}$ corrects for the fact that the velocity head formula is typically defined for the fluid itself; here we are computing the equivalent water column that represents the same pressure as the moving air stream. Note that the reduced air density at 300 m altitude (compared to sea level) is already captured in $\rho_{\text{air}} = 1.054 \text{ kg/m}^3$.

The static head in inches of water is:

$$h_s = 0.9 \text{ cm H}_2\text{O} = 0.354 \text{ in H}_2\text{O} \quad (2.3)$$

The total pressure head seen by the fan is therefore:

$$h_t = h_v + h_s = 0.30513 + 0.354 = \boxed{0.6595 \text{ in H}_2\text{O}} \quad (2.4)$$

2.3 Air Power Calculation

The total aerodynamic power required to drive the airflow against the combined pressure head is computed as:

$$P_{\text{air}} = Q \cdot \rho_{\text{water}} \cdot g \cdot h_{t,m} = 150.796 \times 1000 \times 9.81 \times 0.01675 \approx 41 \text{ HP} = \boxed{31 \text{ kW}} \quad (2.5)$$

where $h_{t,m} = 0.6595 \times 0.0254 = 0.01675 \text{ m}$ is the total head converted to metres of water. This figure—**31 kW**—represents the mechanical power that must be delivered to the fan blades at the blower shaft, and becomes the reference output power for all subsequent transmission design.

Chapter 3

Transmission Configuration Selection

With a required output of 31 kW at 110 rpm, the next step is to select the architecture of the mechanical transmission. This is not a trivial choice: the physical arrangement of the cooling tower imposes strong geometric constraints, and the large speed reduction ratio required (approximately 13.4:1) eliminates several otherwise attractive options.

3.1 Geometric Constraint: Fan Shaft Orientation

The blower shaft is suspended vertically from the tower's support structure. A motor mounted horizontally—which is the standard, preferred orientation for virtually all industrial AC motors due to shaft loading and lubrication considerations—therefore requires a **90° change in the direction of rotation**. This constraint immediately narrows the candidate mechanisms to those capable of transmitting motion between non-parallel shafts:

- **Worm gear set** — skew shafts, inherently 90° capable in a single compact stage.
- **Bevel gear stage followed by a helical stage** — the bevel stage handles the 90° directional change and first reduction, while the helical stage provides the second reduction.

Alternatively, the motor could be mounted **vertically** above the fan and drive the blower shaft through an inline helical gearbox, entirely eliminating the need for angular redirection.

3.2 Belt and Chain Drives — Eliminated

Before narrowing down to gear-based configurations, belt and chain drives were considered as candidates. These are attractive for several reasons: low unit cost, ability to absorb shock loads through elastic deformation, ease of maintenance, and no requirement for precise shaft alignment. However, a review of manufacturer catalogues for commercially available synchronous belts and roller chains revealed a fundamental incompatibility with this application.

At the motor output speed (1500–3000 rpm depending on pole count) and the required reduction ratio of approximately 13.4, a single-stage belt or chain drive would require an extremely large driven sprocket or pulley—impractical for installation on a cooling tower frame. Furthermore, at the 55 kW power level, the chain tension and belt pretension forces become very large, requiring either an impractically wide drive element or a custom solution outside the standard catalogue range. No single-stage standard-catalog belt or chain drive could accommodate the combination of high power, high speed, and large ratio within the geometric constraints of the cooling tower frame.

Eliminated: Belt and Chain Drives

Catalogue review confirmed no standard single-stage belt or chain drive is feasible for this power level (55 kW), speed (1475 rpm), and reduction ratio (13.4:1) combination. Both configurations are eliminated.

3.3 Efficiency Model and Motor Power

For all gear-based configurations, the following efficiency values are used consistently throughout:

- Gear pair efficiency (helical or bevel, not worm): $\eta_{\text{gear}} = 0.975$
- Rolling-element bearing efficiency: $\eta_{\text{bearing}} = 0.99$

These values are applied multiplicatively across all components in the power path. For a configuration with n_b bearings and n_g gear pairs, the total gearbox efficiency is:

$$\eta_{\text{gearbox}} = \eta_{\text{bearing}}^{n_b} \times \eta_{\text{gear}}^{n_g} \quad (3.1)$$

The required motor power is then found by accounting for the gearbox losses, a **service factor** $K_s = 1.25$ for uniform/light-shock loading (fan application), and a **motor power safety margin** $n_d = 1.1$:

$$P_{\text{motor,req}} = \frac{P_{\text{out}} \times K_s \times n_d}{\eta_{\text{gearbox}}} \quad (3.2)$$

3.4 Vertical Helical Configuration — Eliminated

Mounting the motor vertically and driving the fan shaft through an inline two-stage helical gearbox was evaluated as a third candidate. This arrangement is mechanically straightforward: helical gears are among the most efficient and well-characterised gear types, and a two-stage configuration can readily achieve the required 13.4:1 ratio.

A two-stage helical gearbox of this type has 3 shafts, 2 gear pairs, and 7 bearings (6 radial + 1 thrust):

$$\eta_{\text{gearbox}}^{\text{helical}} = 0.99^7 \times 0.975^2 = 0.886 \quad (3.3)$$

Applying Equation 3.2:

$$P_{\text{motor,req}} = \frac{31 \times 1.25 \times 1.1}{0.886} = 48.1 \text{ kW} \rightarrow \text{Select 55 kW motor} \quad (3.4)$$

The 55 kW motor is available in both **2-pole** (≈ 3000 rpm) and **4-pole** (≈ 1500 rpm) variants. For the vertical mount, *either* pole configuration requires the motor's full weight of **345 kg** to be supported on a vertical cantilevered structural extension above the gearbox. This creates:

- Substantial overhung bending moment on the gearbox output shaft.
- Demand for a custom, heavy support frame that is costly and difficult to access for maintenance.

- Significant vibration transmission risk from the vertically mounted motor to the gearbox and fan blade assembly.
- Complicated sealing and lubrication arrangements for a non-standard motor orientation.

After consultation with the course instructor (Dr. Masoud Shariat Panahi), the fully vertical helical configuration was eliminated on structural and practical grounds.

Eliminated: Vertical Helical Configuration

Motor weight (345 kg) imposes impractical structural demands on the tower frame when mounted vertically. Configuration rejected after instructor consultation.

3.5 Remaining Candidates

Following the systematic elimination process, two viable configurations remain:

Bevel-Helical Configuration: A bevel gear stage performs the 90° axis change and the first speed reduction; a downstream helical stage provides the second reduction. Motor is horizontal; output shaft is vertical. This requires 3 shafts, 2 gear pairs, and 7 bearings total.

Worm Gear Configuration: A single worm gear set simultaneously achieves the 90° directional change and the full reduction in one compact stage. Motor is horizontal; output shaft is vertical. This requires 2 shafts and 5 bearings total.

Both configurations are carried forward for full preliminary design and comparative scoring.

Chapter 4

Scoring Methodology for Configuration Comparison

To provide an objective basis for the final configuration selection, a composite scoring metric was defined. The three most important design parameters for this installation—as specified in the project requirements—are **weight**, **volume**, and **efficiency**.

4.1 Scoring Formula

The scoring formula normalises weight and volume against the heavier and larger of the two candidate designs, then adds the gearbox efficiency directly:

$$\text{Score} = \left(1 - \frac{W}{W_{\max}}\right) + \left(1 - \frac{V}{V_{\max}}\right) + \eta_{\text{gearbox}} \quad (4.1)$$

where $W_{\max}$ and $V_{\max}$ are the weight and volume of the heaviest and largest candidate respectively. A design that simultaneously has zero weight, zero volume, and perfect efficiency (all physically impossible, but serving as the upper bound) would score exactly 3.0. The normalized structure ensures each of the three criteria contributes comparably to the total.

Volume definition: The volume V refers specifically to the smallest rectangular bounding box (cuboid) that fully encloses the gearbox. The motor volume is *not* included, since the motor is identical for all candidate configurations and would not affect the relative comparison.

Rationale for this metric: The gearbox sits on an elevated ladder frame where dead weight is a primary structural concern, physical envelope is limited by the tower geometry, and operating efficiency directly affects long-term energy operating cost over a 20-year design life. All three criteria are therefore genuinely important and appropriate to include.

Chapter 5

Worm Gear Preliminary Design

5.1 The Worm Efficiency Problem: An Iterative Challenge

The worm gear configuration presents a fundamental design challenge that distinguishes it from all other gear types: **the efficiency of a worm gear set cannot be determined before the geometry is established**, but the required motor power—and hence the feasibility of a given design—depends directly on efficiency. This circular dependency means the design must begin with an assumed efficiency, complete the geometry, verify the assumption, and iterate if necessary.

Worm gear efficiency depends primarily on three geometric and tribological parameters:

- The **lead angle** λ (helix angle of the worm thread at the pitch cylinder).
- The **pressure angle** ϕ .
- The **coefficient of friction** μ at the sliding tooth contact.

The governing efficiency equation is:

$$e_w = \frac{\cos \phi - \mu \tan \lambda}{\cos \phi + \mu / \tan \lambda} \quad (5.1)$$

The lead angle in turn depends on the worm geometry through:

$$\tan \lambda = \frac{N_w \cdot p}{\pi \cdot d_w} \quad (5.2)$$

where N_w is the number of worm starts, p is the axial pitch, and d_w is the worm pitch diameter. A larger lead angle increases efficiency (the worm thread becomes more inclined and the sliding action more closely resembles rolling). However, λ also determines the pressure angle requirement and affects all force calculations downstream.

The friction coefficient is assumed as $\mu = 0.05$, representative of a well-lubricated phosphor-bronze worm wheel meshing with a hardened steel worm—the standard material pairing for industrial worm drives that balances cost, machinability of the wheel, and wear resistance of the worm.

For the expected geometry of this application (moderate-to-large pitch, moderate worm diameter), the worm mesh efficiency can be estimated to lie in the range:

$$e_w \in [0.775, 0.94] \quad (5.3)$$

This range, combined with the output power requirement and the design factors, gives:

$$P_{\text{motor,req}} = \frac{31 \times 1.25 \times 1.1}{e_w} = \frac{42.625}{e_w} \in [45.3, 55.0] \text{ kW} \quad (5.4)$$

Crucially, the entire range falls below 55 kW—the next standard motor size above the minimum. Therefore, **a 55 kW motor is required regardless of where the actual efficiency falls within this range**, which means the motor selection is insensitive to the exact efficiency value at this stage. The geometry design can proceed without resolving the circularity for motor selection purposes; the efficiency must still be verified at the end to confirm the chosen motor is genuinely sufficient.

5.2 Motor Selection and Reduction Ratio

A **4-pole, 55 kW motor (225MX4)** is selected, with rated speed $n_w = 1475$ rpm. The 4-pole configuration is preferred over 2-pole for two reasons: first, the lower input speed requires a smaller reduction ratio from the worm (making the gearbox more compact); second, higher-speed worms generate more frictional heat and require more careful thermal management.

The required total reduction ratio is:

$$i = \frac{n_{\text{motor}}}{n_{\text{fan}}} = \frac{1475}{110} = 13.41 \quad (5.5)$$

Per the lecture notes, for reduction ratios in the range 10–20, the recommended number of worm starts is $N_w = 3$. This provides a good balance between efficiency (more starts increase the lead angle and improve efficiency) and load capacity (fewer starts per tooth pass). The number of worm wheel teeth follows directly:

$$N_G = N_w \times i = 3 \times 13.41 \approx \mathbf{41} \text{ teeth} \quad (5.6)$$

The value is rounded to 41 (rather than 40) to provide a slightly better contact ratio and to avoid coincident tooth mesh frequencies.

5.3 Worm Diameter Selection Philosophy

For each candidate axial pitch p , the center distance c is determined by solving the geometric constraint that the sum of the pitch radii equals the center distance. With $d_G = N_G \cdot p / \pi$ and the AGMA recommended minimum worm diameter $d_w = c^{0.875} / 2$:

$$2c = d_w + d_G = \frac{c^{0.875}}{2} + \frac{N_G \cdot p}{\pi} \quad (5.7)$$

This transcendental equation in c is solved numerically for each pitch value. Once c is known, the worm diameter is set to its **minimum AGMA allowable value**:

$$d_w = \frac{c^{0.875}}{2} \quad (5.8)$$

This minimum-diameter choice serves two objectives simultaneously:

1. **Compactness and weight:** A smaller worm diameter means smaller center distance and a smaller, lighter gearbox.
2. **Efficiency:** From Equations 5.1 and 5.2, decreasing d_w increases $\tan \lambda$ and therefore increases λ , which directly increases e_w . Minimising d_w therefore simultaneously minimises size *and* maximises efficiency.

There is a lower practical limit to d_w beyond the AGMA formula: the gear face width is $b_G = 2/3 \cdot d_w$, which directly controls the allowable tooth force. However, this structural check is applied *after* the geometry is established—at the comparison stage it suffices to use the minimum AGMA value as a consistent baseline.

5.4 Phase 1: Small-to-Medium Pitch Investigation ($p = 6, 20, 40$ mm)

Three axial pitches spanning a wide range were selected for the first pass to understand the general relationship between pitch, tooth forces, and safety factor. All three cases use $P_{\text{in}} = 55$ kW (the rated motor power, representing the worst-case loading) and $n_w = 1475$ rpm.

5.4.1 Key Design Equations

The allowable worm wheel tangential force (wear criterion, AGMA):

$$W_{tG,\text{allow}} = 1.32 \times 10^{-2} \cdot C_s \cdot d_G^{0.8} \cdot b_G \cdot C_m \cdot C_v \times 10^{-3} \quad [\text{kN}] \quad (5.9)$$

The actual worm wheel tangential force from the worm:

$$w_{tG} = \frac{\cos \phi \cos \lambda - \mu \sin \lambda}{\cos \phi \sin \lambda + \mu \cos \lambda} \cdot W_{t,\text{worm}} \quad (5.10)$$

where the worm tangential force is:

$$W_{t,\text{worm}} = \frac{6 \times 10^7 \cdot P_{\text{in}} \cdot K_s}{\pi \cdot d_w \cdot n_w} \times 10^{-3} \quad [\text{kN}] \quad (5.11)$$

The bending stress on the worm wheel tooth:

$$\sigma_{\text{bending}} = \frac{10^3 \cdot w_{tG}}{p \cdot b_G \cdot Y} \quad [\text{MPa}] \quad (5.12)$$

where Y is the Lewis form factor (from standard tables, based on ϕ), C_s is the materials/size factor (interpolated from AGMA tables based on b_G), C_v is the velocity factor (interpolated from AGMA tables based on sliding velocity V_s), and $C_m = 0.794$ is the ratio correction factor.

The sliding velocity is:

$$V_s = \frac{n_w \cdot d_w}{19100 \cdot \cos \lambda} \times 200 \quad [\text{m/min}] \quad (5.13)$$

5.4.2 Phase 1 Results

Table 5.1: Phase 1 Worm Design Results ($p = 6, 20, 40$ mm)

Parameter	$p = 6$ mm	$p = 20$ mm	$p = 40$ mm
Center distance c [mm]	46.32	150.63	297.51
Worm diameter d_w [mm]	14.34	40.24	72.99
Gear diameter d_G [mm]	78.30	261.01	522.03
Lead angle λ [°]	21.78	25.39	27.62
Pressure angle ϕ [°]	20	20	20
Gear face width b_G [mm]	9.56	26.83	48.66
Mesh efficiency e_w	0.8637	0.8765	0.8824
Sliding velocity V_s [m/min]	238.5	687.9	1272.4
Velocity factor C_v	0.5064	0.3136	0.2281
Size factor C_s	1065.4	1048.2	1026.3
Actual gear force w_{tG} [kN]	134.19	40.85	20.57
Allowable gear force [kN]	1.77	7.93	17.83
Bending SF	0.0037	0.113	0.815
Wear SF	0.0132	0.194	0.867
Overall SF (min)	0.004 ×	0.113 ×	0.815 ×

None of the three cases meets the required safety factor of 1.5. The physical reason is clear from examining the trend: as pitch decreases, the gear teeth become smaller. Smaller teeth—and in particular a smaller face width $b_G = 2/3 \cdot d_w$ —can carry less load. Meanwhile, the smaller worm diameter results in a higher sliding velocity V_s , which penalises the velocity factor C_v and further reduces the allowable load capacity. The actual tooth force w_{tG} at the same time increases dramatically as the tooth size shrinks (same power, much smaller moment arm).

The velocity factor C_v drops from 0.506 at $p = 6$ mm to 0.228 at $p = 40$ mm—a factor of 2.2 reduction in allowable load capacity from velocity penalty alone. This is a direct consequence of the high worm speed (1475 rpm) and the increasing sliding velocity as geometry scales up.

The motor power check confirms all three designs are within the 55 kW envelope ($P_{\text{req}} \in [49.35, 48.30]$ kW for the three cases), confirming the motor selection is adequate.

Conclusion from Phase 1: The viable pitch range lies above 40 mm. A second phase with larger pitches is required.

5.5 Phase 2: Larger Pitch Range ($p = 45, 55, 65$ mm)

Table 5.2: Phase 2 Worm Design Results ($p = 45, 55, 65$ mm)

Parameter	$p = 45$ mm	$p = 55$ mm	$p = 65$ mm
Center distance c [mm]	334.03	406.89	479.57
Worm diameter d_w [mm]	80.78	96.00	110.84
Gear diameter d_G [mm]	587.28	717.79	848.30
Lead angle λ [°]	28.01	28.68	29.25
Gear face width b_G [mm]	53.85	64.00	73.90
Mesh efficiency e_w	0.8833	0.8848	0.8860
Sliding velocity V_s [m/min]	1413.1	1690.1	1962.1
Velocity factor C_v	0.2149	0.1941	0.1773
Size factor C_s	1021.1	1011.0	1001.1
Bending SF	1.141	2.021	3.256
Wear SF	1.111	1.691	2.382
Overall SF (min)	1.111 $\times$	1.691 $\checkmark$	2.382 $\checkmark$

Both $p = 55$ mm and $p = 65$ mm satisfy the minimum safety factor requirement of 1.5. The $p = 45$ mm case falls just short (SF = 1.111, governed by wear), confirming that the lower bound of viable pitches lies between 45 mm and 55 mm.

The efficiency values across this range are very close to each other (0.883–0.886), reflecting that the lead angle λ changes only slightly in this regime (28.0°–29.3°). The dominant effect of increasing pitch is the growth of the gear wheel diameter d_G (from 587 to 848 mm), which increases allowable load capacity but also increases gearbox size and mass.

5.6 Representative Design Selection for Comparison

For the purpose of comparing the worm configuration against the bevel-helical alternative at this stage, the $p = 55$ mm case is selected as the **representative worm design**. This choice is deliberate and intentionally *non-optimal*:

- It sits at the lower end of the viable pitch range, providing the most compact and lightweight estimate of the worm configuration.
- As a design that just satisfies the safety factor, it represents the realistic “best case” for the worm in terms of size and weight—not the most over-designed version.
- Symmetrically, the bevel-helical design is also a preliminary (not fully optimised) design. Using a non-optimised representative for both sides ensures the comparison is on equal footing.
- If the margin between the two configurations turns out to be small, a more refined optimisation would be warranted for both.

The representative worm design characteristics:

Worm Representative Design Summary (p

- Worm mesh efficiency: $e_w = 0.8848$
- Overall SF: 1.691 (governed by wear)
- Gearbox mass (worm + wheel + shafts + bearings): **203 kg**
- Gearbox bounding volume: **0.046 m³**
- Total gearbox efficiency (including 5 bearings): $\eta = 0.871 \times 0.99^5 \approx \mathbf{0.82}$

The slightly lower mesh efficiency used for the final gearbox calculation (0.871 vs. the 0.8848 from minimum-diameter MATLAB analysis) reflects a small upward adjustment to the worm diameter made to satisfy the shaft minimum clearance requirement of 25 mm from the casing wall—the only practical deviation from the strict minimum AGMA diameter.

Chapter 6

Bevel-Helical Configuration Design (AGMA Standard)

6.1 Power Transmission and Component Selection

6.1.1 System Efficiency Analysis

To determine the required input power from the motor, the overall mechanical efficiency of the bevel-helical gearbox is calculated based on all internal components. The drive configuration consists of:

- **Bearings:** 6 standard rolling-element radial bearings and 1 thrust bearing (7 total)
- **Gear stages:** 1 helical gear pair and 1 bevel gear pair

Using the specified efficiency values ($\eta_{\text{gear}} = 0.975$, $\eta_{\text{bearing}} = 0.99$):

$$\eta = (0.99)^7 \times (0.975)^2 = 0.886 \quad (6.1)$$

6.1.2 Motor Power Requirement

The required motor power is calculated considering the gearbox output power, the service factor, and the design safety margin:

$$P_m = \frac{P_{\text{out}} \times k_s \times n_d}{\eta} = \frac{35.4 \times 1.25 \times 1.1}{0.886} = 48.1 \text{ kW} \quad (6.2)$$

6.1.3 Motor Selection and Technical Specifications

Based on the calculated requirement of 48.1 kW, the **225MX4 3-phase AC motor** is selected. This motor provides a rated power of 55 kW with adequate margin.

Table 6.1: 225MX4 Motor Technical Specifications

Specification	Symbol	Value
Rated power	P_n	55 kW
Rated speed	H_n	1475 rpm
Efficiency (full load)	—	93.5%
Power factor	$\cos \phi$	0.84
Rated current	I_n	101.08 A
Weight	—	345 kg
Max. torque ratio	M_m/M_n	2.8
Frame size	—	IEC 225M
Shaft diameter	D	60 mm
Shaft length	E	140 mm
Centre height	H	225 mm
Foot hole distances	$A \times B$	356×311 mm

For component design checks—where the objective is to size each gear, shaft, and bearing to handle the worst credible combination of service loads—the **design input power** applies both the service factor and the safety factor to the rated motor power:

$$P_{\text{in,design}} = P_{\text{rated}} \times n_d \times K_s = 55 \times 1.5 \times 1.25 = 103.1 \text{ kW} \quad (6.3)$$

6.1.4 Transmission Ratios

The total reduction ratio is:

$$i_{\text{total}} = \frac{n_{\text{in}}}{n_{\text{out}}} = \frac{1475}{110} = 13.41 \quad (6.4)$$

This ratio is distributed across the two stages as follows:

- **Bevel stage:** $i_{\text{bevel}} = 3.5$
- **Helical stage:** $i_{\text{helical}} = 13.41/3.5 = 3.83$

A lower ratio is intentionally assigned to the bevel stage. Bevel gears are more complex to manufacture, more sensitive to alignment errors, and generate significant thrust loads due to the conical geometry. By keeping the bevel ratio low, the bevel gear diameters remain small, the cantilevered loads on the bevel pinion shaft are minimised, and noise and vibration from the direction-change stage are reduced. The helical stage—which is mechanically more forgiving, cheaper to manufacture, and more easily aligned—carries the larger share of the speed reduction where torque is already partially multiplied.

6.2 Bevel Gear Stage Design

6.2.1 Operating Parameters and Power Flow

All force and stress calculations use the design input power (Eq. 6.3). The power at each point in the bevel stage is:

- **Bevel pinion power:** $P_{\text{in}} \times \eta_{\text{bearing}}^2 = 103.1 \times 0.99^2 = 100.95 \text{ kW}$
- **Bevel gear power:** $100.95 \times \eta_{\text{gear}} = 100.95 \times 0.975 = 98.43 \text{ kW}$
- **Pinion speed:** 1475 rpm
- **Gear speed:** $1475/3.5 = 421.43 \text{ rpm}$
- **Design life:** 38,400 hours
- **Load cycles (pinion):** $n_L = 1475 \times 60 \times 38400 = 3.398 \times 10^9 \text{ cycles}$

6.2.2 Minimum Tooth Count and Cone Angles

The minimum number of pinion teeth to avoid undercut at $\phi = 20^\circ$ is obtained from the AGMA criterion:

$$N_{\min} = \frac{2}{(1 + 2m_G) \sin^2 \phi} \left(m_G + \sqrt{m_G^2 + (1 + 2m_G) \sin^2 \phi} \right) \quad (6.5)$$

Rounding up for manufacturing allowance:

- Pinion teeth: $N_p = 17$
- Gear teeth: $N_g = 17 \times 3.5 = 59.5 \rightarrow \mathbf{60}$

The pitch cone angles are:

$$\gamma_p = \arctan\left(\frac{1}{3.5}\right) = 16^\circ \quad (6.6)$$

$$\gamma_g = \arctan(3.5) = 74^\circ \quad (6.7)$$

6.2.3 AGMA Factors and Constants

Established for AGMA Quality Grade $Q_v = 6$ and reliability $R = 0.95$:

Table 6.2: Bevel Stage AGMA Factors

Factor	Symbol	Value
Overload factor	K_A	1.25
Load distribution factor	$K_{H\beta}$	$1.1 + 5.6 \times 10^{-6} b^2$
Geometry factor (pinion)	Y_J	0.24
Geometry factor (gear)	Y_J	0.19
Elastic coefficient	Z_E	$191\sqrt{\text{MPa}}$
Reliability factor	Y_Z	0.895

6.2.4 Bending Stress Analysis

The AGMA bending stress equation for bevel gears balances the actual stress against the allowable value:

$$\sigma_F = \frac{W_t}{b} \cdot \frac{K_A K_v m_e}{Y_x K_{H\beta} Y_\beta Y_J} \leq \frac{\sigma_{F,\lim} Y_{NT}}{S_F K_\theta Y_Z} \quad (6.8)$$

Solving for the minimum required mean cone module:

- Pinion bending: $m_e = 6.67$ mm
- Gear bending: $m_e = 7.08$ mm

6.2.5 Surface Contact (Wear) Stress Analysis

The AGMA pitting resistance equation:

$$\sigma_H = Z_E \sqrt{\frac{W_t}{b \cdot m_e \cdot N_p \cdot Z_I} K_A K_v K_{H\beta} Z_x Z_{xc}} \leq \frac{\sigma_{H,\lim} Z_{NT} Z_W}{S_H K_\theta Z_Z} \quad (6.9)$$

Minimum modules for pitting resistance:

- Gear contact: $m_e = 8.90$ mm
- Pinion contact: $m_e = 9.28$ mm

Contact (wear) governs, as is typical for long-life gear applications.

6.2.6 Final Selected Bevel Geometry

The maximum required module of 9.28 mm is rounded to the next standard value: $m_e = 10$ mm.

Table 6.3: Final Bevel Gear Stage Specifications

Parameter	Symbol	Pinion	Gear	Unit
Number of teeth	N	17	60	—
Module	m_e	10	10	mm
Pitch diameter	d	170	600	mm
Face width	b	100	100	mm
Pitch cone angle	γ	16	74	°
Pressure angle	ϕ	20	20	°

6.3 Helical Gear Stage Design

6.3.1 Operating Parameters and Power Flow

The power entering the helical stage has passed through four bearings and one gear mesh from the design input:

$$P_{\text{helical pinion}} = P_{\text{in,design}} \times \eta_{\text{bearing}}^4 \times \eta_{\text{gear}} = 103.1 \times 0.99^4 \times 0.975 = 96.47 \text{ kW} \quad (6.10)$$

- Pinion speed: 421.43 rpm (output of bevel stage)
- Gear (fan) speed: 110 rpm
- Pressure angle: $\phi_n = 20^\circ$
- Helix angle: $\psi = 20^\circ$

6.3.2 Gear Tooth Numbers

Using the same minimum-teeth criterion as for the bevel stage:

- Pinion: $N_p = 17$
- Gear: $N_g = 17 \times (421.43/110) \approx \mathbf{65}$

6.3.3 AGMA Factors

Table 6.4: Helical Stage AGMA Factors

Factor	Symbol	Value
Overload factor	K_o	1.25
Safety factors	S_H, S_F	1.5
Geometry factor (pinion)	Y_J	0.47125
Geometry factor (gear)	Y_J	0.545

6.3.4 Stress Analysis and Final Module

Bending stress results:

- Pinion bending module: $m_e = 8.05 \text{ mm}$
- Gear bending module: $m_e = 7.53 \text{ mm}$

Surface contact results (again governing):

- Gear contact module: $m_e = 10.36 \text{ mm}$
- Pinion contact module: $m_e = 11.01 \text{ mm}$

The maximum transverse module required is 11.01 mm. The normal module follows from:

$$m_n = m_t \cdot \cos \psi = 11.01 \times \cos 20^\circ = 10.35 \text{ mm} \quad (6.11)$$

Rounding to the next standard value: $m_n = 11$ mm.

Table 6.5: Final Helical Gear Stage Specifications

Parameter	Symbol	Pinion	Gear	Unit
Number of teeth	N	17	65	—
Normal module	m_n	11	11	mm
Pitch diameter	d	199	760	mm
Face width	b	99.5	99.5	mm
Normal pressure angle	ϕ_n	20	20	°
Helix angle	ψ	20	20	°

The large helical gear diameter of 760 mm is a direct consequence of the high torque at the fan shaft (high power, low speed) and the large module required for pitting resistance.

6.4 Shaft Design (ASME Code)

6.4.1 Design Methodology

All three shafts are sized using the ASME shaft design formula for solid steel shafts subjected to combined fluctuating bending and torsion:

$$d = \left[\frac{5.1}{\tau_{\text{all}}} \sqrt{(K_m M)^2 + (K_t T)^2} \right]^{1/3} \quad (6.12)$$

Standard design inputs:

- Allowable shear stress: $\tau_{\text{all}} = 41$ MPa (commercial steel with keyways, per ASME standard)
- Bending factor: $K_m = 2.0$ (minor shocks + fatigue)
- Torsion factor: $K_t = 1.5$ (minor shocks)
- Size/derating factor: $B = 1.0$ (baseline)

A minimum clearance of 25 mm between all gear faces and the internal casing wall is maintained to accommodate standard manufacturing tooling during assembly.

6.4.2 Shaft 1 — Input Shaft (Motor → Bevel Pinion)

This shaft operates at the highest speed (1475 rpm) but carries the lowest torque in the gearbox. The bending moment arises from the combined radial and axial forces acting on the bevel pinion.

- Power: $P = 100.95$ kW
- Maximum bending moment: $M_{\text{max}} = 301,242.8$ N mm

- Maximum torque: $T_{\max} = 653,560 \text{ N mm}$

Applying Equation 6.12:

$$d_1 = \left[\frac{5.1}{41} \sqrt{(2 \times 301,242.8)^2 + (1.5 \times 653,560)^2} \right]^{1/3} = \mathbf{52.3 \text{ mm}} \quad (6.13)$$

6.4.3 Shaft 2 — Intermediate Shaft (Bevel Gear → Helical Pinion)

This shaft carries both the driven bevel gear and the driving helical pinion. It is subjected to the most complex three-dimensional loading: bevel gear forces act in a different plane from the helical gear forces, producing combined bending moments that must be resolved vectorially. This is the most structurally demanding shaft in the gearbox despite being at intermediate speed.

- Power: 98.43 kW (bevel output) / 96.47 kW (helical input)
- Maximum bending moment: $M_{\max} = 460,870 \text{ N mm}$
- Maximum torque: $T_{\max} = 2,230,273.7 \text{ N mm}$

$$d_2 = \left[\frac{5.1}{41} \sqrt{(2 \times 460,870)^2 + (1.5 \times 2,230,273.7)^2} \right]^{1/3} = \mathbf{75.6 \text{ mm}} \quad (6.14)$$

6.4.4 Shaft 3 — Output Shaft (Helical Gear → Fan)

This is the lowest speed (110 rpm), highest torque shaft in the system. It connects directly to the cooling tower fan blade hub and must also support the weight of the blower assembly. The shaft extends 40–50 mm above the top surface of the gearbox casing to provide the fan mounting interface.

- Maximum bending moment: $M_{\max} = 469,824.04 \text{ N mm}$
- Maximum torque: $T_{\max} = 8,165,183.16 \text{ N mm}$

$$d_3 = \left[\frac{5.1}{41} \sqrt{(2 \times 469,824.04)^2 + (1.5 \times 8,165,183.16)^2} \right]^{1/3} = \mathbf{115.2 \text{ mm}} \quad (6.15)$$

The progression from $d_1 = 52.3 \text{ mm}$ to $d_3 = 115.2 \text{ mm}$ vividly illustrates how rapidly torque builds through the transmission stages.

6.5 System Efficiency, Volume, and Mass Estimation

6.5.1 Overall Wire-to-Air Efficiency

The complete system efficiency from electrical input at the motor terminals to mechanical output at the fan blades includes the motor's own conversion efficiency:

$$\eta_{\text{system}} = \eta_{\text{gear, helical}} \times \eta_{\text{gear, bevel}} \times \eta_{\text{bearing}}^7 \times \eta_{\text{motor}} \quad (6.16)$$

$$\eta_{\text{system}} = 0.975 \times 0.975 \times 0.99^7 \times 0.935 = \mathbf{82.84\%} \quad (6.17)$$

6.5.2 Gearbox Volume Estimation

The gearbox housing dimensions are determined by stacking the gear pitch diameters and face widths, with 25 mm clearances applied on all sides, and accounting for spatial overlap between the bevel and helical stages along the gearbox length direction:

Table 6.6: Bevel-Helical Gearbox Dimensional Estimation

Dimension	Derivation	Value
Length	$25 + 600 + 760 + 25 = 280.5$	1129.5 mm
Width	$760 + 25 + 25$	810 mm
Height	$25 + 99.5 + 25 + 27.6 + 25 + 2(35)$	272.1 mm
Gearbox volume	$1.1295 \times 0.810 \times 0.2721$	0.2489 m³

For reference, the motor occupies a separate bounding box of: $835 \times 435 \times 555$ mm = 0.2016 m³.

6.5.3 Mass Estimation

All gear and shaft masses are estimated using the standard density of structural steel: $\rho = 7.85 \times 10^{-6}$ kg/mm³.

Table 6.7: Bevel-Helical Gearbox Mass Breakdown

Component	Mass [kg]
Shaft 1 (Input)	3.64
Shaft 2 (Intermediate)	9.58
Shaft 3 (Output)	9.58
Bevel pinion ($d_{\text{avg}} = 142.44$ mm)	12.02
Bevel gear ($d_{\text{avg}} = 503.87$ mm)	43.15
Helical pinion	24.30
Helical gear	354.33
Total gearbox internals (excl. casing)	456.6 kg
Motor (225MX4)	345 kg

The helical gear alone accounts for 354 kg—**77% of the total gearbox internal mass**—a direct consequence of its large 760 mm diameter, 99.5 mm face width, and steel density. This is a fundamental disadvantage of the bevel-helical configuration in the weight criterion, and it stems directly from the large reduction ratio and power level of the second gear stage.

Chapter 7

Configuration Comparison and Selection

7.1 Scoring Calculation

With both preliminary designs complete, the scoring formula (Eq. 4.1) is applied. The bevel-helical configuration provides the reference maximum values since it is heavier and larger:

Table 7.1: Configuration Comparison Scoring

Criterion	Worm Gear	Bevel-Helical
Gearbox mass [kg]	203	456.6
Gearbox volume [m ³]	0.046	0.2489
Gearbox efficiency η	0.82	0.886
Weight score ($1 - W/W_{\max}$)	$1 - 203/456.6 = \mathbf{0.556}$	0
Volume score ($1 - V/V_{\max}$)	$1 - 0.046/0.2489 = \mathbf{0.815}$	0
Efficiency score η	0.820	0.886
Total Score	2.191	0.886

7.2 Analysis and Decision

The worm configuration achieves a total score of **2.191** compared to **0.886** for the bevel-helical configuration—a margin of approximately 1.3 points, or roughly 150% superior on the composite metric. The result is decisive and not sensitive to small errors in the preliminary estimates.

The physical explanation is straightforward: the worm gearbox achieves the entire speed reduction *and* the 90° direction change in a *single gear mesh*, whereas the bevel-helical configuration requires *two separate gear stages*, each contributing its own gear mass, shaft mass, and housing volume. The efficiency disadvantage of the worm (0.82 vs. 0.886 at gearbox level) is real but modest in absolute terms ($\approx 8\%$ relative), and it is overwhelmingly offset by the compactness advantage:

- Worm is $2.25\times$ **lighter** (203 vs. 456.6 kg gearbox internals)
- Worm is $5.4\times$ **more compact** (0.046 vs. 0.2489 m³)

For a cooling tower installation on an elevated structural frame where dead weight and spatial envelope are primary constraints, the worm configuration is the clear and logical choice.

Final Selection: Worm Gear Configuration

The worm gear gearbox is selected for detailed design. The scoring margin is large enough that no further comparison is needed at this stage. The following chapters present the optimisation of the worm design parameters and the full final design.

Chapter 8

Worm Gear Parameter Optimisation

8.1 Motivation: Why Optimisation Is Necessary

The preliminary screening (Chapter 5) identified the worm gear configuration as the superior transmission architecture and established that axial pitches in the range $p \in [55, 65]$ mm meet the safety factor requirement. However, that initial search was deliberately non-exhaustive: it used the strict AGMA minimum worm diameter $d_w = c^{0.875}/2$ as the only value tested at each pitch, and it only evaluated three pitch candidates in that range. Two important questions remain unanswered:

1. **Which specific pitch within the viable range is best?** The preliminary scan covered only three discrete values and the winner ($p = 55$ mm) was chosen for convenience, not for optimality.
2. **Is the minimum AGMA diameter truly the best worm diameter for every pitch?** It was argued qualitatively that smaller d_w gives better efficiency and smaller size, but the structural constraint (safety factor ≥ 1.5) may *force* the diameter above the AGMA minimum at some pitches—and the optimum for each pitch is the smallest d_w that simultaneously satisfies all constraints.

The goal of this chapter is therefore to:

- Sweep the full pitch range $p = 40$ mm to 70 mm in 1 mm steps (31 designs).
- At each pitch, **find the minimum feasible worm diameter**—the smallest d_w that satisfies both the AGMA geometric bound and the safety factor requirement.
- Compare all 31 optimised designs against each other using the same scoring philosophy introduced in Chapter 4, adapted with a *pairwise local normalisation* to eliminate scoring bias.
- Identify the single best pitch–diameter combination.

Core Optimisation Principle

For any given pitch p , the optimal worm diameter is the **minimum** value of d_w that satisfies all constraints simultaneously. This is optimal because: (a) smaller d_w increases the lead angle λ , which increases mesh efficiency e_w ; (b) smaller d_w reduces gear face width $b_G = 2/3 d_w$ and worm length, shrinking the gearbox volume and mass. Finding the minimum-feasible d_w at each pitch therefore produces the **Pareto-optimal design** for that pitch: no alternative at the same pitch is simultaneously lighter, more compact, and more efficient.

8.2 Algorithmic Overview

The optimisation algorithm has three distinct phases, illustrated in the flowchart of Figure ??:

1. **Geometric initialisation** — establish the absolute minimum allowable d_w from the AGMA centre-distance formula.
2. **Iterative diameter adjustment** — starting from the geometric minimum, iteratively scale d_w upward until the safety factor converges to exactly 1.5, or confirm that the minimum diameter itself already satisfies the requirement.
3. **Pairwise tournament scoring** — after all 31 designs have converged, rank them using a head-to-head scoring competition.

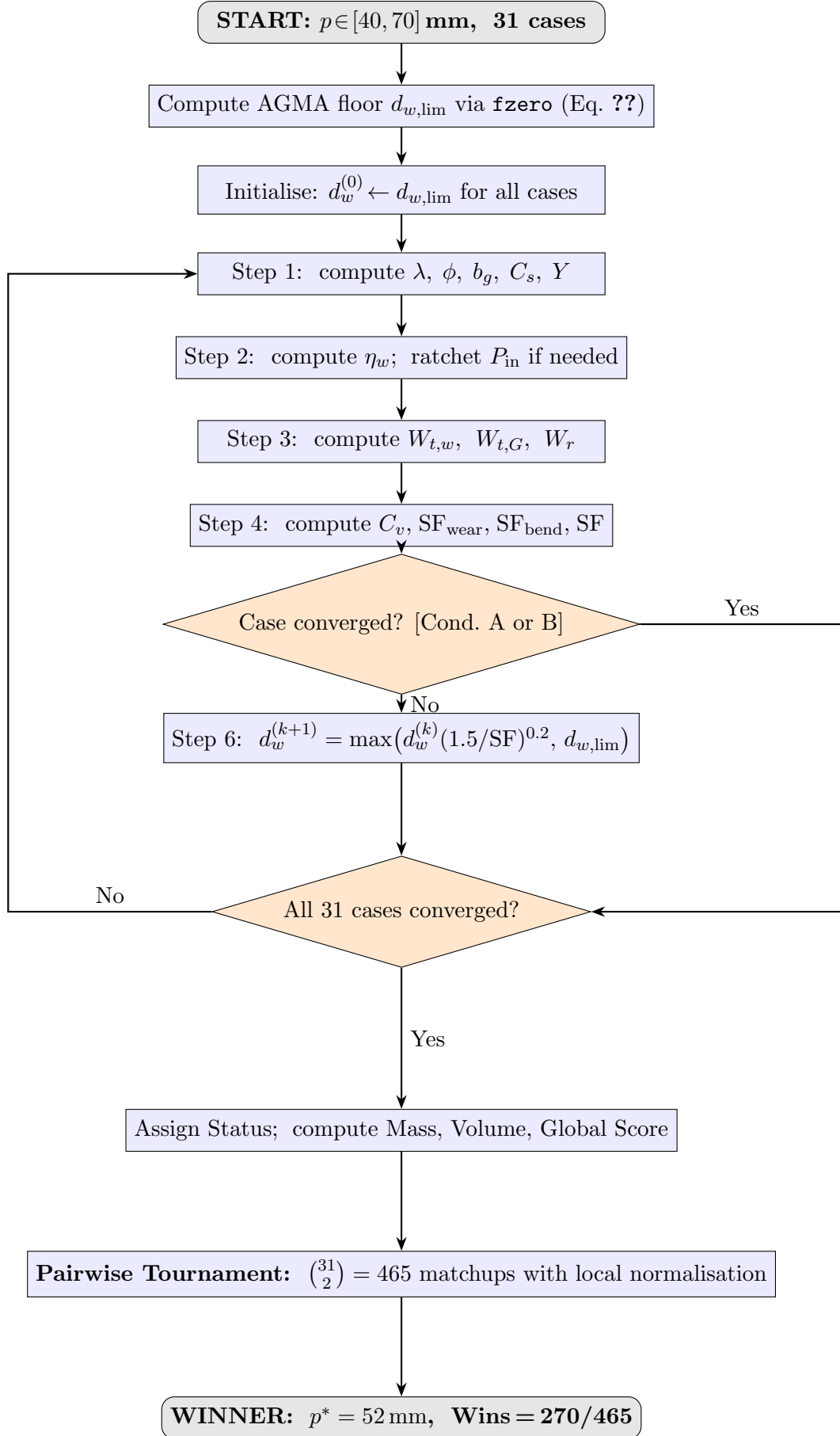


Figure 8.1: Flowchart of the iterative optimisation algorithm. Each of the 31 pitch cases runs its inner loop independently until Condition A ($|SF - 1.5| < 10^{-3}$) or Condition B (clamped at the AGMA geometric floor) is satisfied. The outer loop terminates when all cases have converged, after which the pairwise tournament identifies the global optimum.

8.3 Phase 1 — Geometric Lower Bound on d_w

8.3.1 The AGMA Worm Diameter Constraint

AGMA specifies that the worm pitch diameter must satisfy:

$$d_w \geq \frac{c^{0.875}}{2} \quad (8.1)$$

where c is the centre distance between worm and gear shafts. This empirical bound exists to ensure the worm has sufficient torsional and bending rigidity relative to the span between its support bearings. If d_w were made arbitrarily small at a given centre distance, the worm shaft would deflect excessively under the tooth load, destroying the mesh geometry.

8.3.2 Implicit Equation for the Minimum Diameter

The complication is that c itself depends on d_w :

$$c = \frac{d_w + d_G}{2} \quad (8.2)$$

Substituting Eq. 8.1 at its equality into Eq. 8.2 and noting $d_G = N_G p / \pi$ (which depends only on the *given* pitch):

$$d_{w,\min} = \frac{c^{0.875}}{2} \quad \text{with} \quad c = \frac{d_{w,\min} + d_G}{2} \quad (8.3)$$

This is an implicit nonlinear equation in $d_{w,\min}$ alone (since d_G is known from the pitch). Rearranging into root-finding form:

$$f(d_w) = d_w - \left(\frac{d_w + d_G}{2} \right)^{0.875} / 2 = 0 \quad (8.4)$$

In the MATLAB code this is implemented for each pitch as a separate `fzero` call:

Listing 8.1: Geometric lower bound computation

```
for i = 1:n_cases
    fun_lim = @(x) x - ((x + d_g(i))/2).^0.875 / 2;
    d_w_limit(i) = fzero(fun_lim, 50); % initial guess: 50 mm
    c_initial(i) = (d_w_limit(i) + d_g(i))/2;
end
d_w_initial = d_w_limit; % start iterations at minimum
```

The `fzero` function finds the root of $f(d_w)$ numerically; no closed-form solution exists. The initial guess of 50 mm is suitable for the pitch range of interest ($p = 40\text{--}70$ mm) and converges reliably.

8.4 Phase 2 — Iterative Diameter Optimisation

8.4.1 Initialisation and Starting Point

Once $d_{w,\min}$ is known for every pitch, all 31 designs are initialised at their respective minimum diameters:

$$d_w^{(0)} = d_{w,\min}$$

This starting point has a crucial physical meaning: it is the design with the *highest possible efficiency and smallest possible size* at the given pitch. The question the iteration answers is: *is this design also safe enough?* If $SF \geq 1.5$ at the minimum diameter, no upward adjustment is needed and the minimum is the optimum. If $SF < 1.5$, the diameter must be increased until the structure can carry the load safely.

8.4.2 Within-Iteration Computations

At every iteration, the following quantities are recomputed from the current d_w :

1. **Lead angle:**

$$\lambda = \arctan\left(\frac{N_w \cdot p}{\pi \cdot d_w}\right) \quad (8.5)$$

2. **Pressure angle ϕ** — selected from the standard set $\{14.5^\circ, 20^\circ, 25^\circ\}$ based on λ :

- $\lambda \leq 15^\circ \Rightarrow \phi = 14.5^\circ$
- $15^\circ < \lambda \leq 30^\circ \Rightarrow \phi = 20^\circ$
- $\lambda > 30^\circ \Rightarrow \phi = 25^\circ$

3. **Mesh efficiency:**

$$e_w = \frac{\cos \phi - \mu \tan \lambda}{\cos \phi + \mu / \tan \lambda} \quad (8.6)$$

4. **Size and velocity factors C_s, C_v** from AGMA table interpolations based on face width $b_G = 2/3 d_w$ and sliding velocity V_s .

8.4.3 Motor Power Re-evaluation

This step is a refinement *not* present in the preliminary screening analysis. As d_w increases from its minimum, the lead angle λ decreases (fewer of the worm thread's slope is presented to the driven gear), which *reduces* efficiency. In some cases, particularly at small pitches where d_w must be significantly increased beyond its minimum to satisfy $SF \geq 1.5$, the efficiency can drop enough that the 55 kW motor is no longer adequate to deliver the required 31 kW output with the specified safety margins.

The check is:

$$P_{\text{motor,req}} = \frac{31}{e_w} \times 1.25 \times 1.1 \quad (8.7)$$

If $P_{\text{motor,req}} > P_{\text{in}}$, the available input power is increased to $P_{\text{motor,req}}$ for that design case. This flags the case as **power_increased** in the output table, signalling that such a design

would in practice require either a larger motor or a higher-rated variant of the 55 kW class. The corresponding code:

Listing 8.2: Dynamic motor power re-evaluation

```
motor_power_req = 31 ./ e_w * 1.25 * 1.1;
exceeded = motor_power_req > p_in;
p_in(exceeded) = motor_power_req(exceeded);    % raise if needed
power_increased(exceeded) = true;              % flag for output
```

This self-consistency check is important: without it, a design might show $SF \geq 1.5$ only because it was computed with insufficient input power, giving a falsely favourable result.

8.4.4 Force and Safety Factor Computation

With the (possibly updated) P_{in} , the actual tooth forces and safety factors are computed:

Worm tangential force:

$$W_{t,w} = \frac{6 \times 10^7 \cdot P_{in} \cdot K_s}{\pi \cdot d_w \cdot n_w} \times 10^{-3} \quad [\text{kN}] \quad (8.8)$$

Gear wheel tangential force (transformed through mesh geometry):

$$w_{t,G} = \frac{\cos \phi \cos \lambda - \mu \sin \lambda}{\cos \phi \sin \lambda + \mu \cos \lambda} \cdot W_{t,w} \quad (8.9)$$

Wear (surface durability) safety factor:

$$SF_{\text{wear}} = \frac{W_{t,G,\text{allow}}}{w_{t,G}} \quad \text{where} \quad W_{t,G,\text{allow}} = 1.32 \times 10^{-2} \cdot C_s \cdot d_G^{0.8} \cdot b_G \cdot C_m \cdot C_v \times 10^{-3} \quad (8.10)$$

Bending safety factor:

$$SF_{\text{bending}} = \frac{\sigma_{\text{allow}}}{\sigma_{\text{bending}}} \quad \text{where} \quad \sigma_{\text{bending}} = \frac{10^3 \cdot w_{t,G}}{p \cdot b_G \cdot Y} \quad (8.11)$$

Governing safety factor:

$$SF = \min(SF_{\text{bending}}, SF_{\text{wear}}) \quad (8.12)$$

8.4.5 The Proportional Update Rule

If the design has not converged, d_w is updated using a proportional correction based on the ratio of the required to actual safety factor:

$$d_w^{(k+1)} = d_w^{(k)} \cdot \left(\frac{SF_{\text{req}}}{SF^{(k)}} \right)^{0.2} \quad (8.13)$$

The logic of this rule is straightforward:

- If $SF < SF_{\text{req}}$: the ratio exceeds 1.0, so d_w *increases*, growing the face width and lowering tooth stresses.
- If $SF > SF_{\text{req}}$: the ratio is less than 1.0, so d_w *decreases*, shrinking the design toward the minimum.
- If $SF = SF_{\text{req}}$: the ratio is exactly 1.0, d_w does not change — the design is at optimum.

The exponent 0.2 is a **damping factor**. Because SF is a nonlinear function of d_w (it depends on C_v , C_s , e_w , and force calculations in a coupled way), an unmoderated update (exponent = 1) would cause large oscillations and potential divergence. The exponent 0.2 takes only one-fifth of the “full” correction per iteration, ensuring smooth, stable convergence in all 31 cases within the 2000-iteration budget.

The lower bound is always enforced:

$$d_w^{(k+1)} = \max(d_w^{(k+1)}, d_{w,\text{limit}}) \quad (8.14)$$

This clamping is critical: it prevents the algorithm from ever producing a worm diameter that violates the AGMA structural requirement, regardless of what the safety factor calculation demands.

Listing 8.3: Proportional update with geometric clamping

```
d_w_target = d_w .* (SF_req ./ SF).^0.2;      % proportional step
d_w_next   = max(d_w_target, d_w_limit);      % clamp at AGMA minimum
d_w_next(converged) = d_w(converged);         % freeze converged cases
d_w = d_w_next;
```

Note that converged cases are frozen: once a design reaches its optimum, its d_w is fixed and it does not participate in further updates.

8.4.6 Convergence Criteria

A design is marked as converged under two distinct conditions:

1. Target convergence:

$$|SF - 1.5| < 0.001$$

The safety factor has converged to exactly the required value. This means d_w is the true optimal minimum—any smaller value would give $SF < 1.5$, and any larger value would be unnecessarily conservative.

2. Clamped convergence:

$$|d_w - d_{w,\text{limit}}| < 0.1 \text{ mm} \quad \text{AND} \quad SF > 1.5$$

The design is at the AGMA geometric minimum and *already* has $SF > 1.5$. No further reduction is possible (the AGMA floor prevents it), so this diameter is optimal. The design is “over-built” relative to the safety factor target, but this is forced by geometry, not conservatism.

Listing 8.4: Convergence check logic

```

for i = 1:n_cases
    if ~converged(i)
        at_limit = abs(d_w(i) - d_w_limit(i)) < 0.1;
        if abs(SF(i) - SF_req) < convergence_tol    % Condition A
            converged(i) = true;
        elseif (SF(i) > SF_req) && at_limit          % Condition B
            converged(i) = true;
        end
    end
end
end

```

8.4.7 Design Status Classification

After the iterative loop, each of the 31 designs is classified:

Table 8.1: Design Status Classification

Status	Meaning
Valid (Target)	Converged to SF = 1.5 exactly. True optimal minimum d_w .
Clamped (>1.5)	d_w is at AGMA minimum; SF $\geq$ 1.5 but cannot be reduced further.
Valid (P Incr)	SF satisfied but required a motor power increase. Design is valid but would need a larger motor in practice.
Invalid Ratio	AGMA ratio $d_w/c^{0.875}$ fell outside the bounds $[1/2.0, 1/1.07]$, indicating a geometrically infeasible design.
FAILED	Did not converge within 2000 iterations.

8.5 Phase 3 — Mass, Volume, and Bounding-Box Estimation

Once all designs have converged, the key comparison metrics are computed. Gearbox dimensions are estimated based on the dominant gear sizes plus a uniform 25 mm clearance (for housing wall and tooling access):

$$L_{\text{box}} = \frac{d_G + 50}{1000} \quad [\text{m}] \quad (8.15)$$

$$W_{\text{box}} = \frac{b_G + 50}{1000} \quad [\text{m}] \quad (8.16)$$

$$H_{\text{box}} = \frac{d_w + d_G + 50}{1000} \quad [\text{m}] \quad (8.17)$$

The gearbox bounding volume is:

$$V = L_{\text{box}} \cdot W_{\text{box}} \cdot H_{\text{box}} \quad (8.18)$$

Mass is estimated by modelling the worm as a solid steel cylinder and the wheel as a solid bronze cylinder:

$$m_{\text{worm}} = \frac{\pi}{4} d_w^2 \cdot L_{\text{box}} \cdot \rho_{\text{steel}} \quad m_{\text{gear}} = \frac{\pi}{4} d_G^2 \cdot b_G \cdot \rho_{\text{bronze}} \quad (8.19)$$

where $\rho_{\text{steel}} = 7850 \text{ kg/m}^3$ and $\rho_{\text{bronze}} = 8800 \text{ kg/m}^3$. Using bronze density for the wheel is appropriate: worm wheels in industrial drives are almost universally made of phosphor-bronze or aluminium-bronze, both for wear resistance (soft material conforms to the hardened steel worm) and for the favourable tribological properties that keep μ low.

8.6 Phase 4 — Pairwise Tournament Scoring

8.6.1 Why Pairwise (Local) Normalisation?

The scoring formula from Chapter 4 (Eq. 4.1) requires normalising weight and volume against the *maximum* values in the set. When applied globally across all 31 designs, this creates a systematic bias: the heaviest and largest design (highest pitch) always scores exactly zero on both the weight and volume terms, while intermediate designs score well on both terms simply because they are smaller than the extreme. A design at $p = 70 \text{ mm}$ might score poorly not because it is bad, but simply because it is the reference maximum.

Pairwise (local) normalisation solves this by evaluating each design's score *only relative to its current opponent*. In a head-to-head matchup between designs i and j :

$$\text{Score}_i^{(ij)} = \left(1 - \frac{V_i}{\max(V_i, V_j)}\right) + \left(1 - \frac{m_i}{\max(m_i, m_j)}\right) + e_{w,i} \quad (8.20)$$

Design i wins matchup (i, j) if $\text{Score}_i^{(ij)} > \text{Score}_j^{(ij)}$. The final ranking is determined by the total number of wins across all $N(N-1)/2$ matchups:

$$\text{Wins}(i) = \sum_{j \neq i} \mathbf{1}[\text{Score}_i^{(ij)} > \text{Score}_j^{(ij)}]$$

For 31 designs, this amounts to $31 \times 30/2 = 465$ pairwise matchups. The design with the most wins is selected.

Listing 8.5: Pairwise tournament implementation

```

for i = 1:N_total
    for j = i+1:N_total
        local_max_vol = max(Vol_box(i), Vol_box(j));
        local_max_mass = max(Total_Mass(i), Total_Mass(j));

        score_i = (1 - Vol_box(i)/local_max_vol) + ...
                  (1 - Total_Mass(i)/local_max_mass) + e_w(i);
        score_j = (1 - Vol_box(j)/local_max_vol) + ...
                  (1 - Total_Mass(j)/local_max_mass) + e_w(j);

        if score_i > score_j, Wins(i) = Wins(i) + 1;
        elseif score_j > score_i, Wins(j) = Wins(j) + 1;
        end
    end
end
end

```

8.6.2 Tournament Results

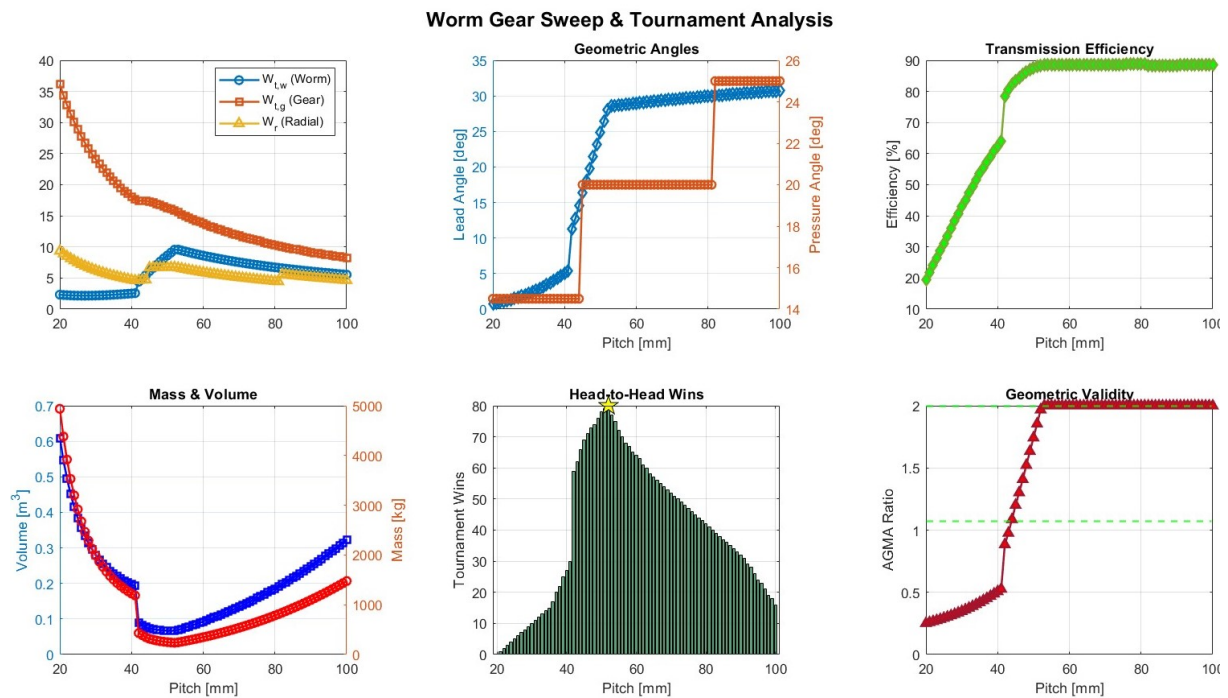


Figure 8.2: Design analysis dashboard: (a) forces vs. pitch, (b) lead and pressure angle vs. pitch, (c) efficiency vs. pitch, (d) mass and volume vs. pitch, (e) tournament wins per pitch with winner highlighted, (f) AGMA geometric validity ratio.

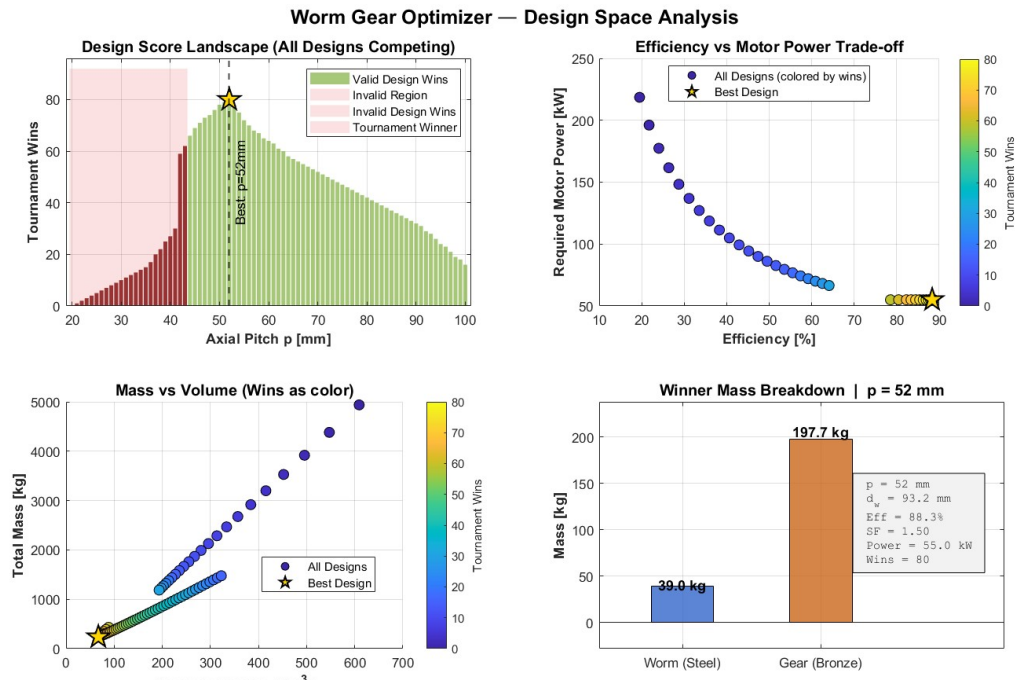


Figure 8.3: Design space analysis: (a) tournament wins across pitch range with invalid region shaded red; (b) efficiency vs. required motor power, coloured by wins; (c) mass vs. volume, coloured by wins; (d) mass breakdown of winning design (worm vs. wheel).

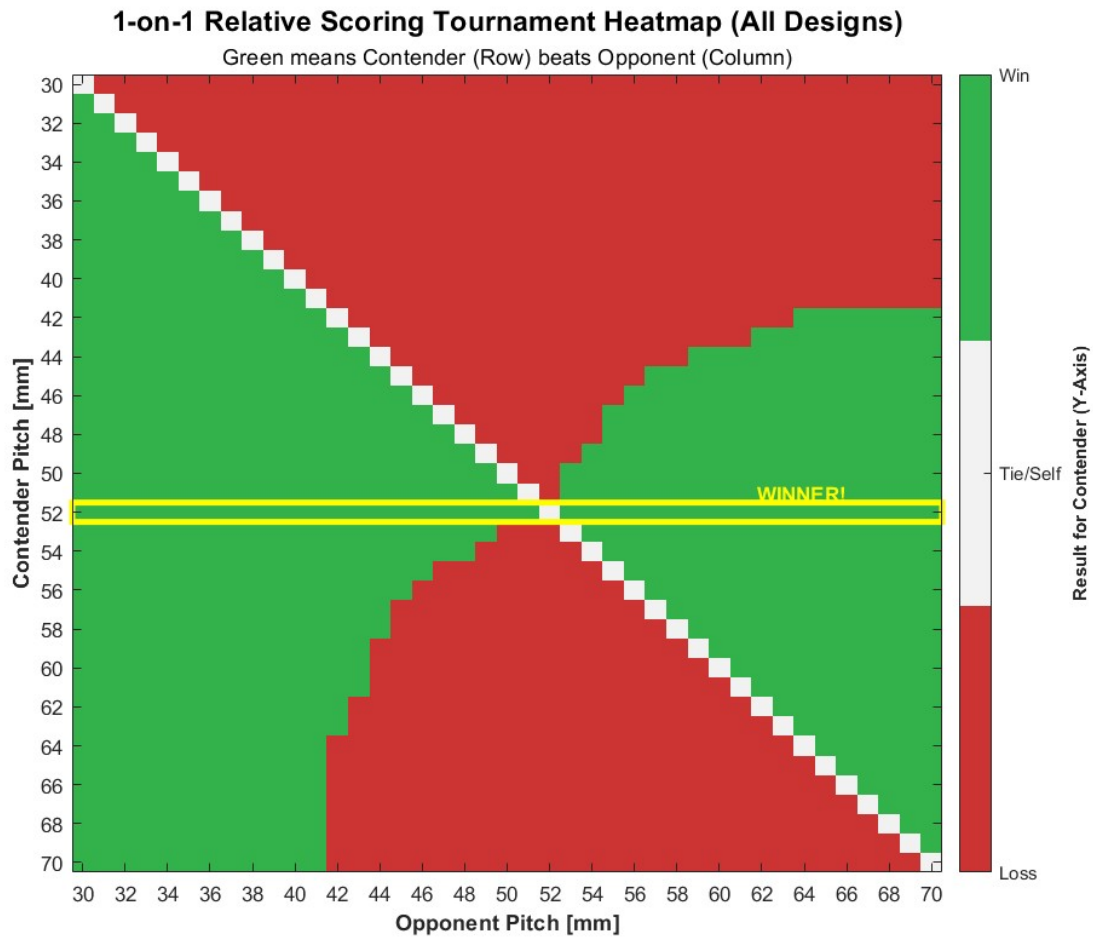


Figure 8.4: Pairwise tournament heatmap: rows are contenders, columns are opponents. Green cells indicate a win for the row design; red indicates a loss. The winner ($p = 52$ mm) is highlighted with a yellow border.

8.7 Optimisation Results

Tournament Winner: p

Axial pitch p :	52 mm
Worm diameter d_w :	93.23 mm
Gear diameter d_G :	678.64 mm
Centre distance c :	385.93 mm
Lead angle λ :	28.04°
Pressure angle ϕ :	20°
Mesh efficiency e_w :	88.34%
Governing safety factor:	1.50 (wear governs)
Sliding velocity V_s :	1627 m/min
Velocity factor C_v :	0.198
Total gearbox mass:	≈ 234 kg
Bounding-box volume V :	≈ 0.067 m ³
AGMA ratio:	1.966
Status:	Valid (Target)

The winning design ($p = 52$ mm) achieves the highest number of pairwise wins by striking the best balance among all three criteria. Designs at lower pitches (40–50 mm) cannot meet the safety factor requirement at the AGMA minimum d_w and must grow their diameter significantly, increasing mass and volume while degrading efficiency. Designs at higher pitches (58–70 mm) easily satisfy the safety factor but produce larger gear diameters d_G (directly proportional to pitch), which inflates both mass and volume faster than the efficiency gain compensates. The 52 mm pitch sits at the *transition* between these two regimes, where the penalty on each side is minimised.

The fact that the winning design converges with the status **Valid (Target)**—meaning SF converged to exactly 1.50—confirms that this is not merely the smallest of the passable designs, but the one whose constrained optimum (minimum d_w at SF = 1.5) produces the best composite score. This $p = 52$ mm, $d_w = 93.23$ mm design is therefore carried forward as the **selected worm configuration for full detailed design** in Chapter 8.

Chapter 9

Final Design: Component Specifications and Structural Validation

9.1 Material Selection

All rotating shafts in the selected worm gear transmission are manufactured from **AISI 4140 Steel (Quenched and Tempered)**. This grade was specifically chosen over commercial steel to enable compact shaft cross-sections: its high strength allows the ASME formula to yield small minimum diameters, which combined with the bearing bore requirements ultimately governs the final stepped shaft geometry.

Table 9.1: AISI 4140 Q&T Mechanical Properties

Property	Symbol	Value
Yield strength	S_y	1640 MPa
Ultimate tensile strength	S_{ut}	1770 MPa
Allowable shear stress	τ_{all}	$\min(0.3 S_y, 0.18 S_{ut}) = \mathbf{318.6}$ MPa

The worm wheel is manufactured from **centrifugal cast phosphor-bronze** (calculated mass: 197.85 kg). This material pairing — hardened steel worm against bronze wheel — is the standard for industrial worm drives: the bronze conforms slightly to the steel worm under load, improving contact and reducing friction, which directly benefits the efficiency e_w that was central to the optimisation in Chapter 7.

9.2 Bearing Selection

9.2.1 Worm Shaft Bearings: Tapered Roller Bearings

The worm shaft experiences a significant axial thrust force in addition to the radial gear mesh loads. This combined loading eliminates deep-groove ball bearings (which can carry limited axial loads) and calls for a bearing type with dedicated axial load capacity. **Tapered roller bearings** were selected because:

- They carry simultaneous radial *and* axial forces efficiently through their conical geometry.
- They can be pre-loaded against each other in an opposed arrangement, eliminating axial play critical for worm mesh accuracy.

- Their robust construction suits the moderate-speed, high-load conditions of the 1475 rpm worm shaft.

Using the SKF Product Select application, two different bearings were chosen for the two sides of the worm shaft. An asymmetric arrangement is justified because the two sides see different resultant load magnitudes: the side with the higher axial reaction component receives the larger bearing.

Table 9.2: Worm Shaft Tapered Roller Bearing Selection

Bearing	Position	Bore d	OD D	Width T
SKF 32207	Coupling side	35 mm	72 mm	24.25 mm
SKF 31309	Free side	45 mm	100 mm	27.25 mm



Figure 9.1: SKF 32207 (left) and SKF 31309 (right) tapered roller bearings selected for the worm shaft. Product data from SKF catalogue.

3.6. Bearing rating life					
Designation		Bearing rating life		SKF life modification factor	Contamination factor
		Basic	SKF		
		$L_{10h} (h)$	$L_{10mh} (h)$	a_{skf}	η_c
Left	■ 31309	10100	72500	7.15	0.59
Right	■ 32207	26200	128000	4.91	0.42

Figure 9.2: SKF Product Select application output confirming a calculated bearing rated life of 72,500 hours for the selected worm shaft bearing arrangement.

Rated Life Verification

The project specification requires a minimum operational life of:

$$L_{\text{req}} = 10 \text{ years} \times 320 \text{ days/year} \times 12 \text{ h/day} \times 1.5 \text{ (service factor)} = 57,600 \text{ hours}$$

The SKF Product Select application returns a rated life of **72,500 hours** for the selected bearing pair under the actual operating loads. This exceeds the project requirement by a margin of approximately 26%, confirming the bearing selection is adequate.

9.2.2 Bearing Housing: Custom Tapered-Bearing Housing

Tapered roller bearings require a dedicated housing to maintain their pre-load setting, provide the correct shoulder contact surface, and allow adjustment during assembly. A **custom two-part housing** was designed and incorporated *inside* the gearbox casing. This internal placement prevents contamination ingress and contributes to the compact overall envelope. The two housing halves are joined by **M14 screws** and the assembled housing unit is bolted to the inner wall of the gearbox shell using **M8 bolts**.

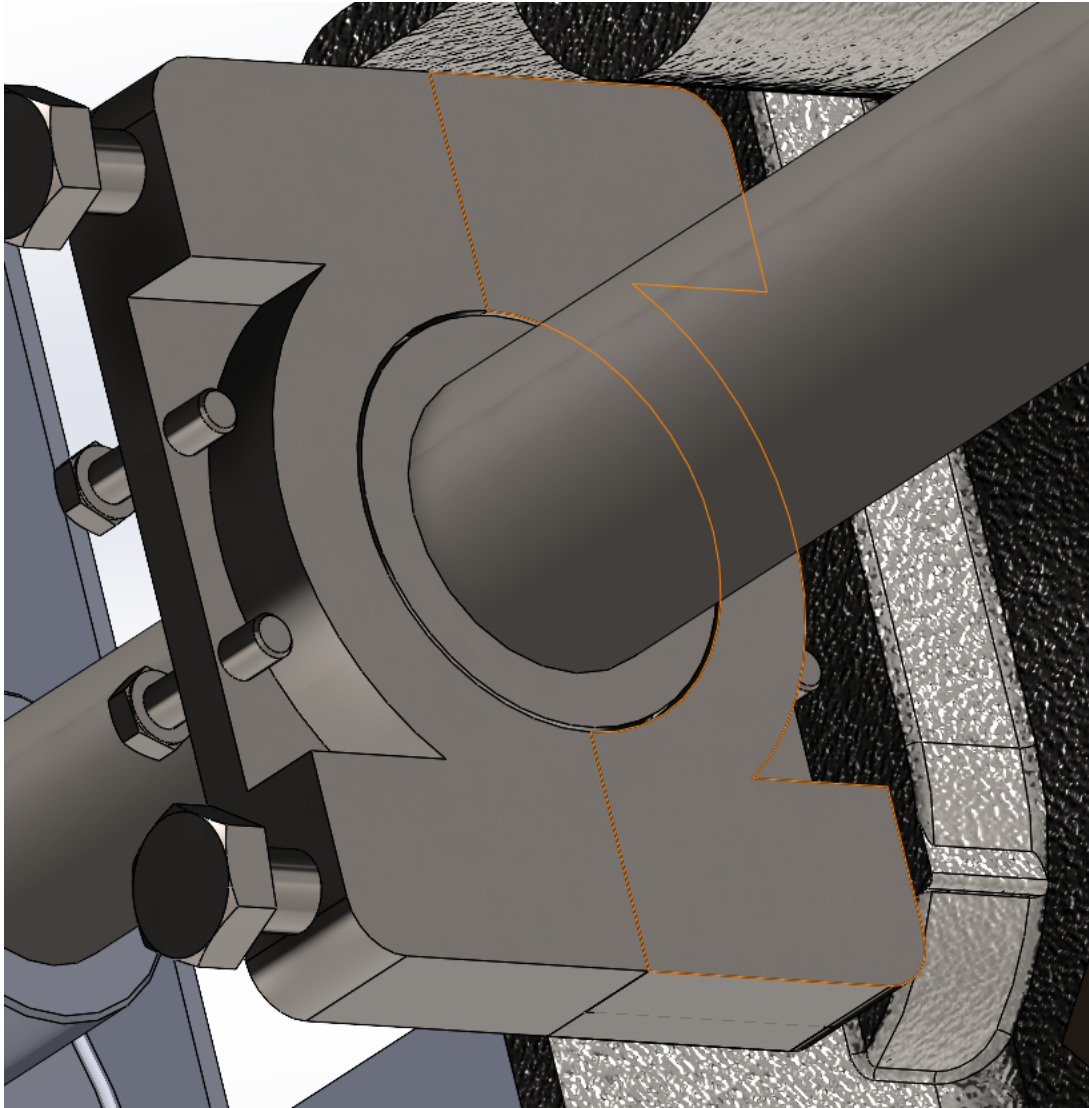


Figure 9.3: Custom two-piece integrated tapered roller bearing housing for the worm shaft. The split design allows the bearing pre-load to be set during assembly. Assembled unit mounts to the gearbox interior walls via M8 bolts.

M8 Bolt Feasibility Check

The M8 mounting bolts must resist the reaction forces transmitted from the worm shaft bearings into the housing and gearbox shell. The dominant loading comes from the axial thrust at the worm mesh, which for the $p = 52$ mm design equals the gear tangential force: $W_{t,G} = 15.84$ kN. With four M8 Grade 8.8 bolts per housing:

$$\sigma_{\text{bolt}} = \frac{W_{t,G}/4}{A_{\text{tens,M8}}} = \frac{15840/4}{36.6} = 108.2 \text{ MPa} \leq 0.9 \times S_y = 576 \text{ MPa} \quad \checkmark \quad (9.1)$$

$$\tau_{\text{bolt}} = \frac{W_{r,G}/4}{A_{\text{tens,M8}}} = \frac{6720/4}{36.6} = 45.9 \text{ MPa} \leq \frac{0.577 S_y}{2} = 185 \text{ MPa} \quad \checkmark \quad (9.2)$$

Both tensile and shear safety factors comfortably exceed 1.0, confirming M8 Grade 8.8 bolts are suitable for this housing connection.

9.2.3 Worm Shaft Sealing: SKF HMSA10RG

To retain the gearbox oil and prevent external contamination from entering along the worm shaft, an **SKF HMSA10RG spring-loaded radial lip seal** is installed at the shaft exit location.

- **Model:** HMSA10RG
- **Dimensions:** 40 × 55 × 7 mm (bore × OD × width)
- **Feature:** Spring-loaded lip ensures consistent contact pressure and compensates for minor shaft runout over the service life.



Figure 9.4: SKF HMSA10RG spring-loaded lip seal (40×55×7 mm) used at the worm shaft exit to retain lubricant.

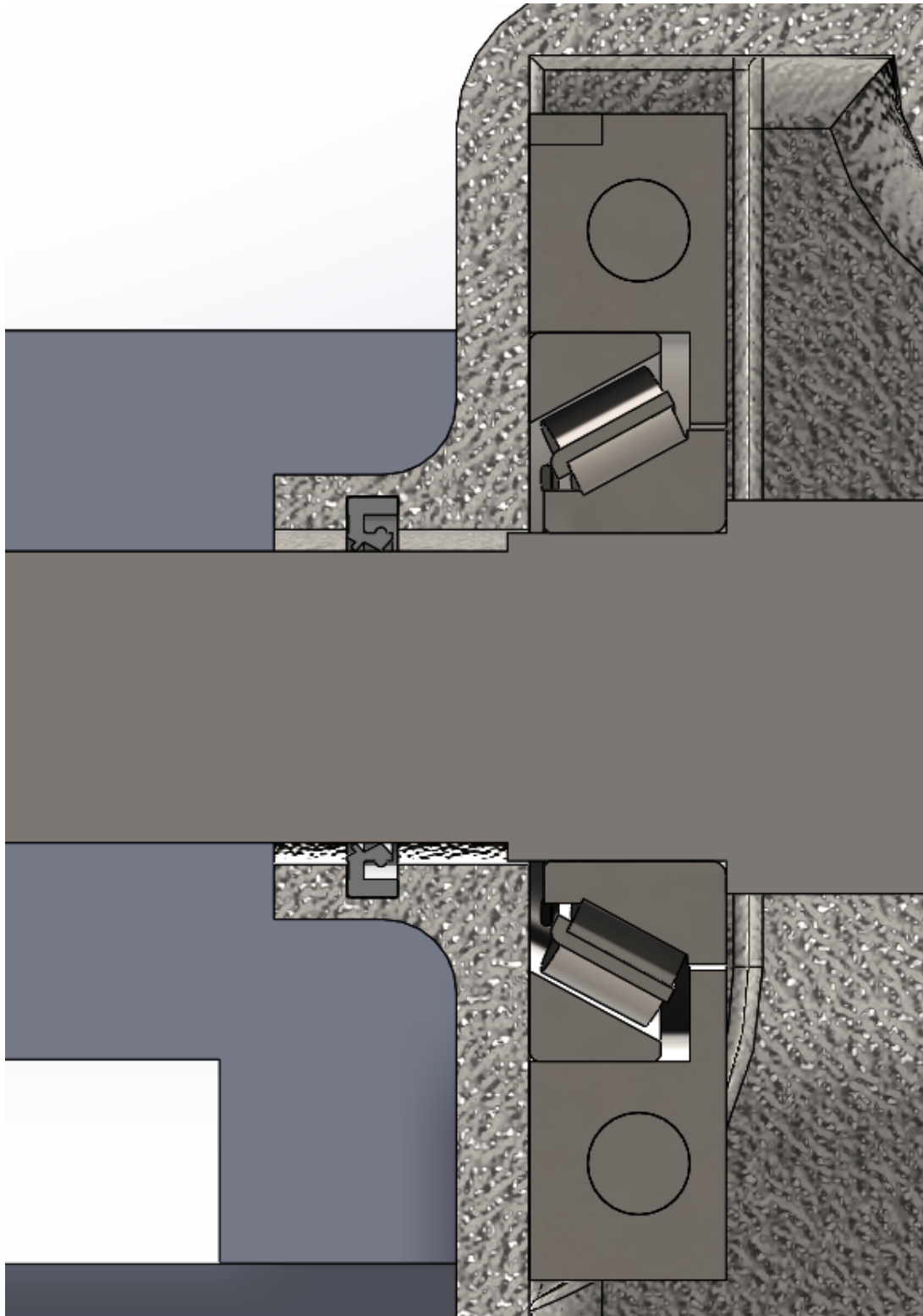


Figure 9.5: Sectioned 3D SolidWorks view of the worm shaft housing at the input end, showing the SKF HMSA10RG spring-loaded lip seal seated against the shaft shoulder. The spring element maintains consistent radial preload on the sealing lip throughout the service life.

9.2.4 Gear Shaft Bearings: Angular Contact Ball Bearings SKF 7311

The gear (output) shaft rotates at the much lower speed of 110 rpm and carries the weight of the bronze worm wheel plus the fan blade assembly. The forces on this shaft include:

- Axial force: $W_a = 9.55 \text{ kN}$ (gear axial = worm tangential)
- Tangential force: $W_t = 15.84 \text{ kN}$
- Radial force: $W_r = 6.72 \text{ kN}$

Angular contact ball bearings (SKF 7311 BECBY) were selected via SKF Product Select. The angular-contact design handles the combined axial and radial loading characteristic of worm gear output shafts:

Table 9.3: SKF 7311 BECBY Bearing Specifications

Parameter	Value
Bore diameter d	55 mm
Outer diameter D	120 mm
Width B	29 mm
Contact angle	40°



Image may differ from product. See technical specification for details.

7311 BECBM

Einreihiges Schrägkugellager

Diese einreihigen Schrägkugellager nehmen Radial-Axial-Kombibelastungen mit einseitig wirkendem Axialbelastungsanteil auf. Sie können mit hohen Drehzahlen betrieben werden (je nach Ausführung sogar mit sehr hohen Drehzahlen). Sie sind besser als Rillenkugellager für starke, einseitig wirkende Axialkräfte geeignet.

- Hohe Nenndrehzahlen
- Aufnahme relativ hoher Radiallasten und hoher, einseitig wirkender Axialbelastungen

Figure 9.6: SKF 7311 BECBY angular contact ball bearing for the gear (output) shaft. Product data from SKF catalogue.

For the gear shaft bearings, **SKF FLN series standard plummer block housings** were selected rather than a custom housing. These market-available units integrate the bearing seat, seal, and lubrication provisions into a compact package, providing a reliable and cost-effective mounting solution for the low-speed output shaft.

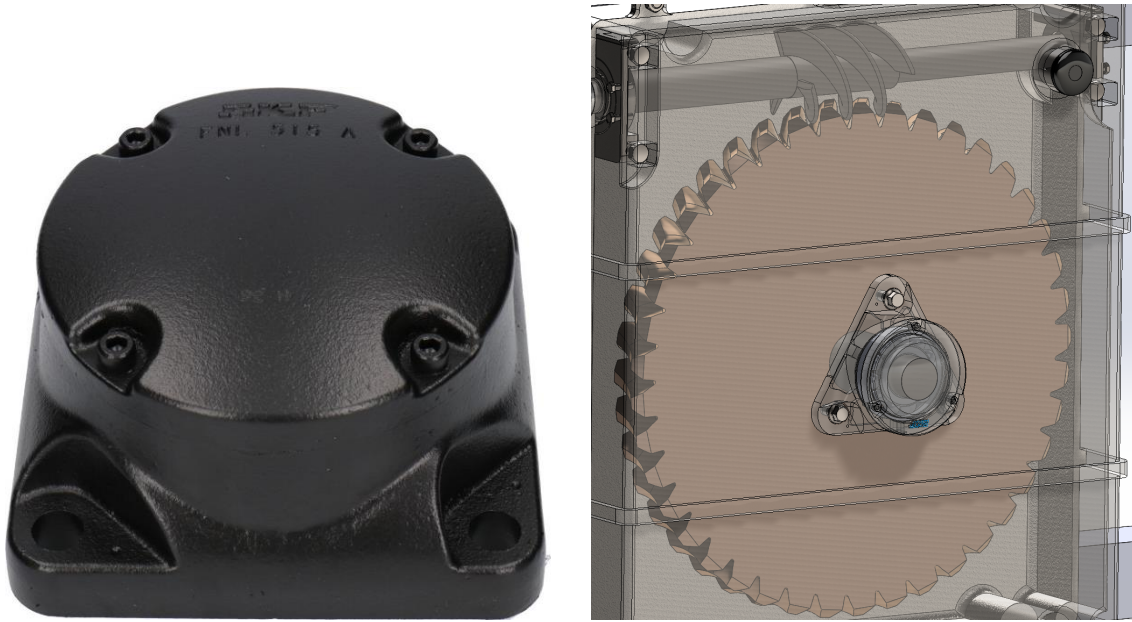


Figure 9.7: SKF FLN series flanged pillow-block housing (left, catalogue specification) and SolidWorks assembly model showing the housing installed on the gear shaft within the gearbox casing (right).

9.3 Shaft Design

9.3.1 Design Methodology Recap

All shafts are sized using the ASME shaft equation:

$$d_{\min} = \left[\frac{5.1}{\tau_{\text{all}}} \sqrt{(K_m M)^2 + (K_t T)^2} \right]^{1/3} \quad (9.3)$$

with $\tau_{\text{all}} = 318.6 \text{ MPa}$ (AISI 4140 Q&T), $K_m = 2.0$, $K_t = 1.5$. The minimum diameter computed is a structural lower bound; the actual machined shaft profile is governed by the *larger* of this minimum and the bearing bore diameter requirements. A **minimum 20% shoulder step** rule ($d_{\text{shoulder}} \geq 1.2 d_{\text{bearing seat}}$) is enforced at every bearing-to-shoulder transition to ensure adequate axial retention force on the bearing inner ring and to prevent stress concentration at the transition from becoming critical.

Fillets are applied at all shoulders to reduce stress concentration factors K_t . Where bearing geometry constrains the fillet radius (the maximum fillet radius that fits within the bearing chamfer), the bearing manufacturer's minimum clearance radius specification is used as the upper bound for the fillet. For all other shoulders, a fillet radius of at least 2 mm is applied.

9.3.2 Worm Shaft

The worm shaft is the high-speed element of the gearbox (1475 rpm, input power 55 kW). Its critical cross-section occurs at the worm thread root, where bending from the transverse gear forces combines with the transmitted torque.

ASME Minimum Diameter at Critical Section

Applying the torque and bending moment from the optimised $p = 52$ mm worm design at the critical section:

$$d_{\min, \text{worm}} = \left[\frac{5.1}{318.6} \sqrt{(K_m M_{\text{worm}})^2 + (K_t T_{\text{worm}})^2} \right]^{1/3} = \mathbf{28.23 \text{ mm}} \quad (9.4)$$

This value is notably small compared to the 41-MPa designs earlier in the report—a direct benefit of specifying AISI 4140 Q&T, whose allowable shear stress is $318.6/41 = 7.8\times$ higher than commercial steel.

Stepped Shaft Geometry Governed by Bearings

The actual shaft profile is entirely governed by bearing bore requirements rather than the structural minimum. The bearing selection established:

- **Coupling side** (SKF 32207, bore = 35 mm): shaft at bearing seat = 35 mm $\gg$ 28.23 mm ✓
- **Free side** (SKF 31309, bore = 45 mm): shaft at bearing seat = 45 mm $\gg$ 28.23 mm ✓

With the 20% shoulder rule, the shaft must step up:

$$d_{\text{shoulder}, 32207} \geq 1.2 \times 35 = 42 \text{ mm} \quad (9.5)$$

$$d_{\text{shoulder}, 31309} \geq 1.2 \times 45 = 54 \text{ mm} \quad (9.6)$$

Since the worm thread section must accommodate the shoulder steps from both sides, the worm body diameter is set to be at least 54 mm (matching the larger shoulder requirement). This produces a shaft profile where *every* cross-section has a diameter strictly greater than the 28.23 mm structural minimum. No additional stress analysis is required for secondary sections — if the critical section at the worm root already satisfies the ASME criterion with large margin, the larger-diameter shoulder and bearing-seat sections are implicitly safe.

The worm thread section length is set to:

$$L_{\text{worm}} = 3 \times p = 3 \times 52 = 156 \text{ mm} \quad (9.7)$$

This three-pitch length ensures sufficient thread engagement and is consistent with the standard worm design practice for $N_w = 3$ starts.

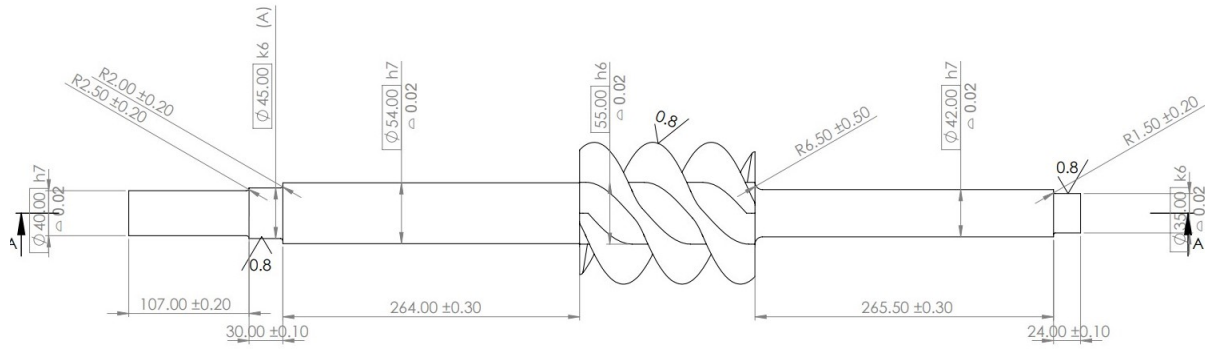


Figure 9.8: Worm shaft stepped profile — 2D detail drawing. Diameter steps shown with all shoulder transitions, fillet radii per bearing chamfer specifications. The Rotex 160 coupling extension is visible at the left end. All dimensions in millimeters.

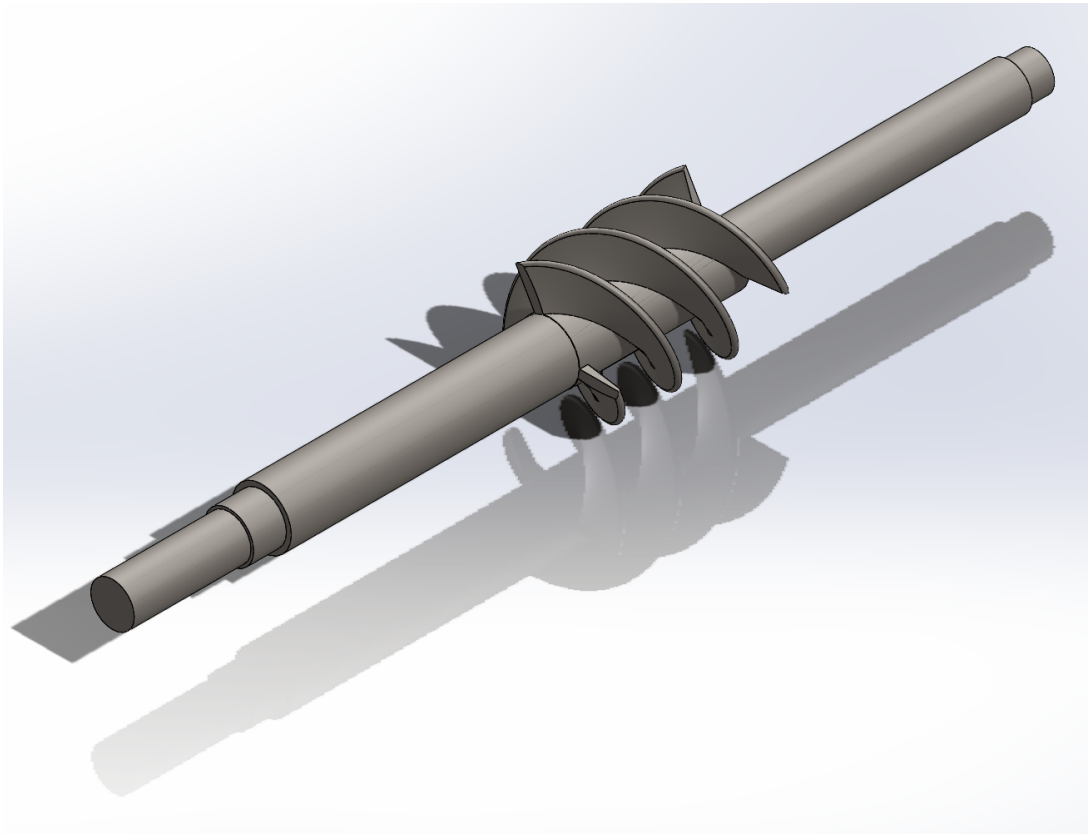


Figure 9.9: 3D SolidWorks model of the complete worm shaft assembly showing all stepped diameters, bearing seats, coupling extension, and worm thread geometry.

Coupling Extension Validation

The shaft must be extended beyond the 32207 bearing to accommodate the Rotex 160 coupling hub (bore 40 mm, hub machined down to 40 mm at the shaft end). The coupling end section has a reduced diameter of 40 mm. Applying Eq. 9.3 at this section (the coupling end carries pure torsion, no bending):

$$d_{\min, \text{coupling}} = \left[\frac{5.1}{318.6} \sqrt{(K_t T_{\text{worm}})^2} \right]^{1/3} = 12.85 \text{ mm} \quad (9.8)$$

The actual shaft diameter at the coupling seat is 40 mm $\gg$ 12.85 mm $\checkmark$. The shoulder creates a positive axial stop for the coupling hub.

9.3.3 Gear (Output) Shaft

The gear shaft is the low-speed, high-torque element (110 rpm, output torque $T_{\max} = 5,374.8$ N·m). Its load analysis, bearing forces, and bending moments are detailed in Section 9.9 of this report. The ASME minimum diameter at the critical section is:

$$d_{\min, \text{gear}} = 50.73 \text{ mm} \quad (9.9)$$

The stepped shaft geometry is again governed by the SKF 7311 bearing bores and the 20% shoulder rule:

Table 9.4: Gear Shaft Stepped Diameter Profile

Section	Location	Diameter	Basis
d_1	Bearing seat (lower, 7311)	55 mm	Bearing bore
d_2	Shoulder below gear	66 mm	1.2×55 mm
d_3	Central gear seat	80 mm	Key / hub fit
d_4	Bearing seat (upper, 7311)	55 mm	Bearing bore

All four sections satisfy $d \geq d_{\min} = 50.73$ mm $\checkmark$. The critical section ($d_1 = d_4 = 55$ mm) exceeds the structural minimum by 8%, and the gear seat ($d_3 = 80$ mm) exceeds it by 58%, confirming that once the bearing geometry is satisfied the shaft is structurally adequate everywhere.

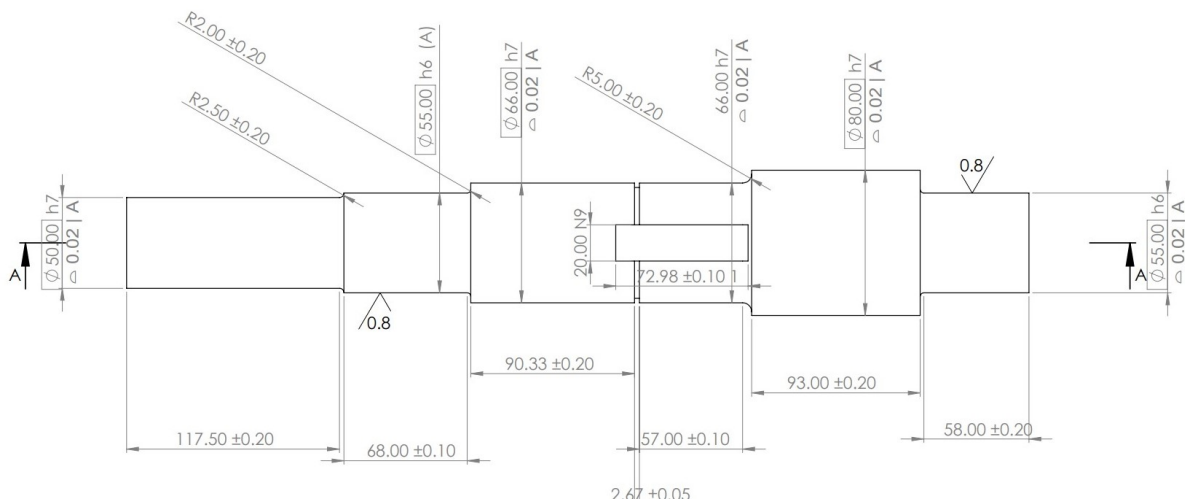


Figure 9.10: Gear (output) shaft stepped profile — 2D detail drawing. Three-key configuration at $d_3 = 80$ mm gear seat visible. Bearing seats at $d_1, d_4 = 55$ mm. Omega coupling extension at the upper end. All dimensions in millimeters.

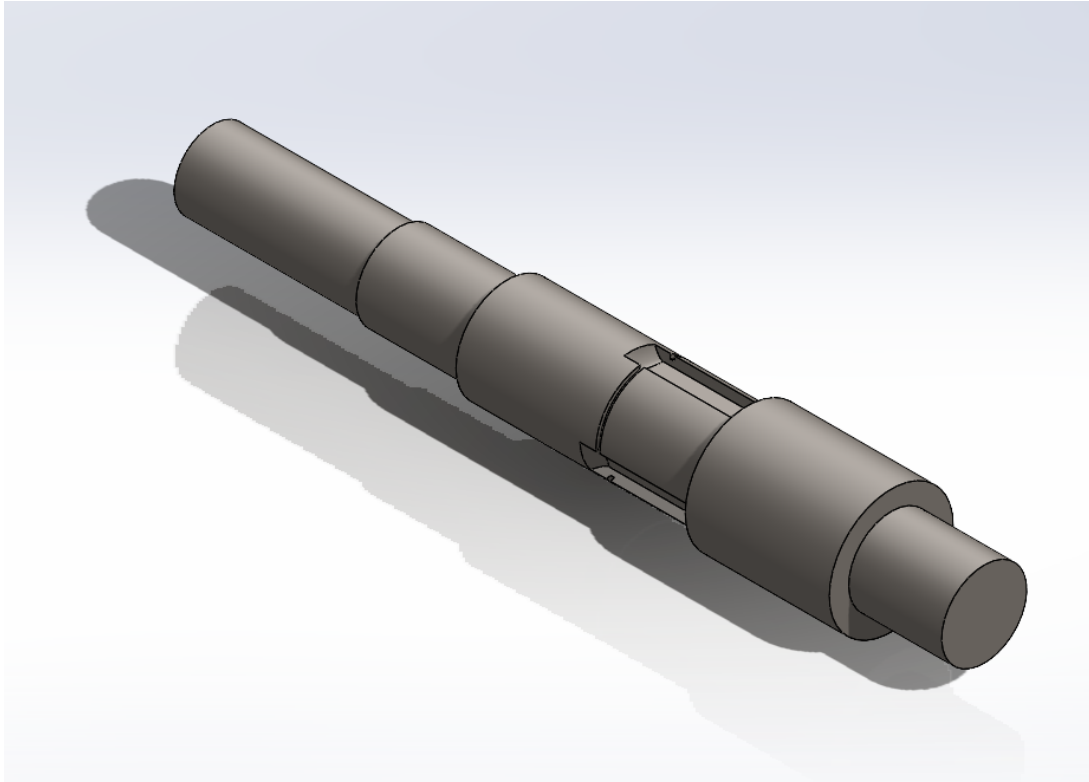


Figure 9.11: 3D SolidWorks model of the gear shaft showing the three parallel keyways at the $d_3 = 80$ mm central gear seat and retaining ring groove.

Output Coupling Extension Validation

The output shaft is extended above the upper bearing to connect to the Rexnord Omega 425 coupling. The shaft is machined down to 50 mm at the coupling seat, creating a positive shoulder. The structural check for this pure-torsion section:

$$d_{\min, \text{output coupling}} = 23.3 \text{ mm} \quad (9.10)$$

The actual coupling seat diameter is 50 mm $\gg$ 23.3 mm $\checkmark$. The 120 mm hub length of the Omega 425 accommodates the required engagement.

9.4 Worm Wheel Detailed Design

The worm wheel geometry is defined by the optimised parameters from Chapter 7:

Table 9.5: Worm Wheel Final Geometry

Parameter	Symbol	Value
Number of teeth	N_G	41
Axial pitch	p	52 mm
Pitch diameter	d_G	678.64 mm
Face width	b_G	62.15 mm ($2/3 d_w$)
Pressure angle	ϕ	20°
Material	—	Centrifugal cast phosphor-bronze
Mass	—	197.85 kg

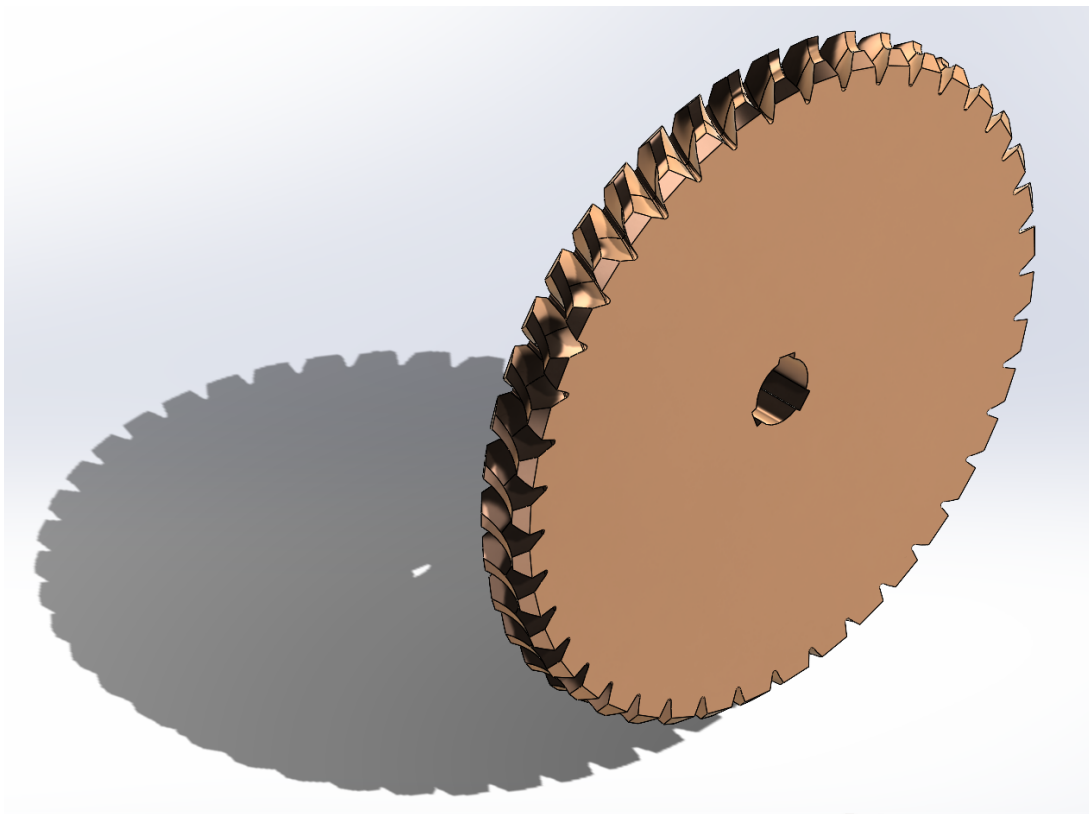


Figure 9.12: 3D SolidWorks model of the final worm wheel showing the 41-tooth hobbled profile, hub geometry, and three-key mounting interface.

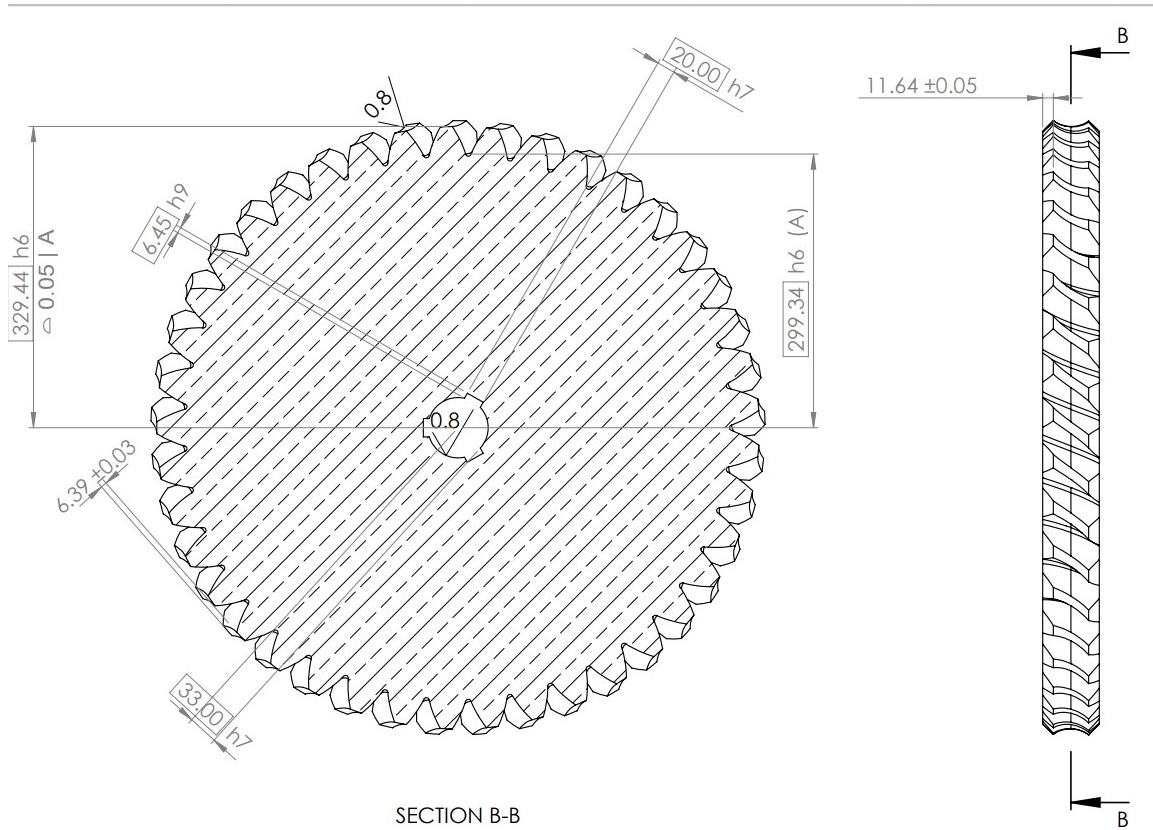


Figure 9.13: 3D SolidWorks detail drawing of the worm wheel with all critical dimensions, tolerances, and manufacturing notes.

9.5 Key and Keyway Design

9.5.1 Gear Hub: Three Parallel Keys

The 197.85 kg bronze worm wheel transmits the full output torque $T_{\max} = 5,374.8 \text{ N}\cdot\text{m}$ through its hub to the shaft. At the 66 mm gear seat, standard metric key dimensions apply: $b = 20 \text{ mm}$, $h = 11 \text{ mm}$. The hub material (cast bronze) governs bearing (crushing) failure at $S_y = 200 \text{ MPa}$:

$$l_{\text{bearing}} = \frac{4T \cdot FS}{d \cdot h \cdot S_{y,\text{hub}}} = \frac{4 \times 5374.8 \times 1}{0.066 \times 0.011 \times 200 \times 10^6} = 148 \text{ mm} \quad (9.11)$$

To distribute the load and maintain shaft integrity, **three parallel keys** are used, each of length $148/3 \approx 49.3 \text{ mm}$, manufactured by **end milling** to create closed-end slots preventing axial migration.

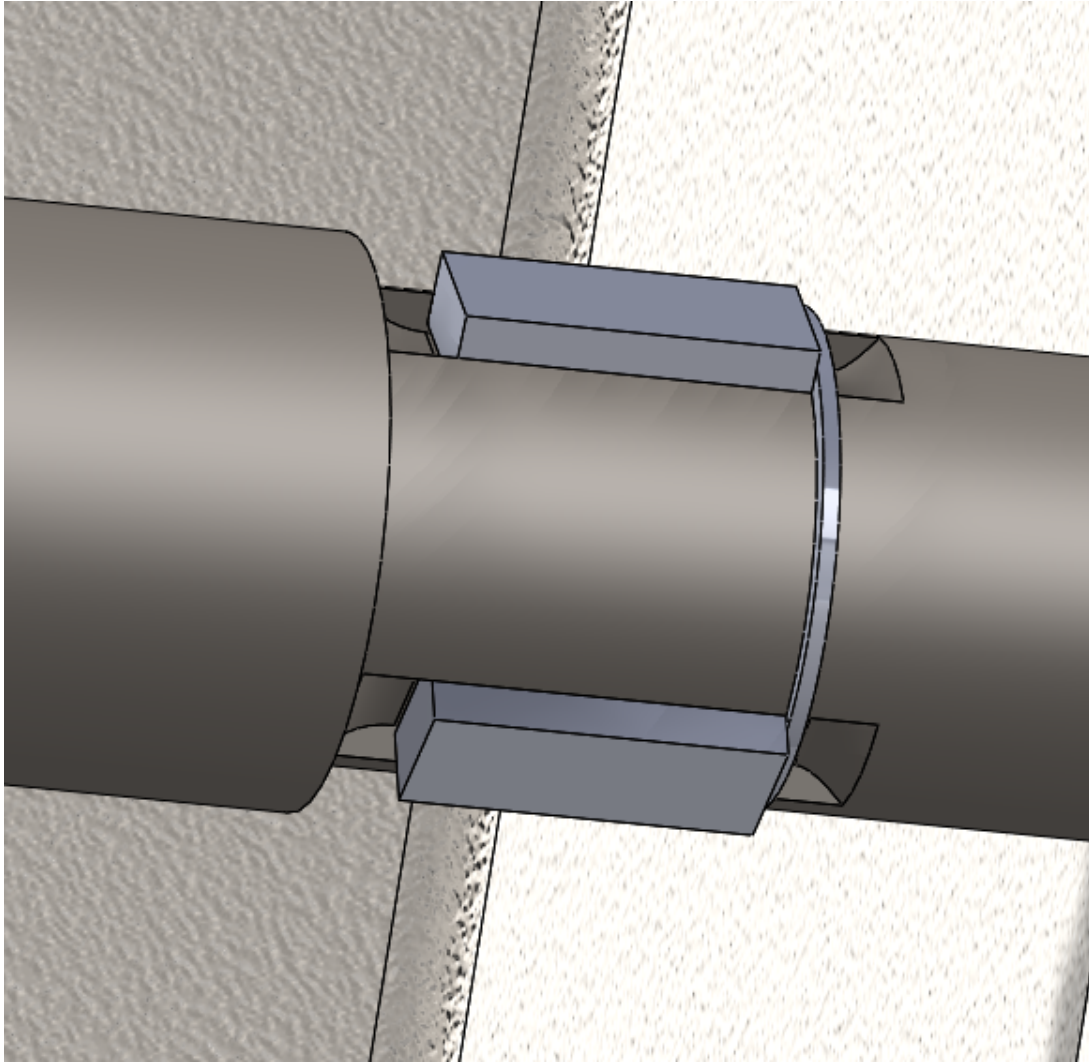


Figure 9.14: SolidWorks detail of the three-key arrangement on the gear shaft hub seat at $d_3 = 80$ mm. Keys are spaced at 120° intervals; each keyway slot is 49.3 mm long and produced by end milling.

9.5.2 Input Coupling (Rotex 160) Key

At the 40 mm input shaft end, the Rotex 160 hub (nodular cast iron, $S_y = 450$ MPa) requires a 12×8 mm key. Bearing failure governs:

$$l_{\text{bearing}} = \frac{4 \times 801 \times 1.5}{0.040 \times 0.008 \times 450 \times 10^6} = 33.3 \text{ mm} \quad (9.12)$$

Per the rule $l \geq d$: **selected length = 40 mm**. End milled.

9.5.3 Output Coupling (Omega 425) Key

At the 50 mm output shaft coupling seat, the Omega 425 hub ($S_y = 450$ MPa) requires a 16×10 mm key. Bearing failure (hub) governs over shear failure (shaft):

$$l_{\text{bearing}} = \frac{4 \times 5374.8}{0.050 \times 0.010 \times 450 \times 10^6} = 95.5 \text{ mm} \quad (9.13)$$

The Omega 425 hub length is 114 mm, fully accommodating the 95.5 mm key. **Selected length = 95.5 mm.** End milled.

9.6 Coupling Selection

9.6.1 Motor Input: Rotex Size 160

Table 9.6: Rotex 160 Coupling Specifications

Parameter	Value
Nominal torque capacity T_{KN}	877 N·m
Maximum torque $T_{K \max}$	1755 N·m
Design torque T_d	801 N·m < 877 ✓
Maximum speed	4750 rpm $\gg$ 1475 rpm ✓
Outer diameter	160 mm
Bore range	30–75 mm
Motor hub bore (machined)	60 mm (motor shaft 60 mm)
Gearbox input hub bore (machined)	40 mm (creates shaft shoulder)

The design torque was computed with service factor 1.25, safety factor 1.5, and temperature factor 1.2 (for 50°C ambient):

$$T_d = T_{\text{nominal}} \times 1.25 \times 1.5 \times 1.2 = 356 \times 1.25 \times 1.5 \times 1.2 = 801 \text{ N·m}$$

9.6.2 Fan Output: Rexnord Omega Size 425

Table 9.7: Rexnord Omega 425 Coupling Specifications

Parameter	Value
Torque capacity	5580 N·m
Maximum design torque	5374.8 N·m < 5580 ✓
Maximum speed	2000 rpm $\gg$ 110 rpm ✓
Hub length L	114 mm
Bore range	35–155 mm
Fan shaft bore	40 mm
Gearbox output hub bore	50 mm (creates shaft shoulder)

9.7 Gearbox Structure and Lubrication

9.7.1 Gearbox Casing Design

The gearbox is enclosed in a **two-part split casing**: an upper half and a lower half joined along a horizontal parting plane to allow assembly and disassembly without disturbing the shaft positions. This is standard practice for industrial gearboxes. The casing material is cast iron or fabricated steel plate (to be detailed in assembly drawings). The casing incorporates integral mounting bosses for the tapered bearing housings and FLN plummer blocks.

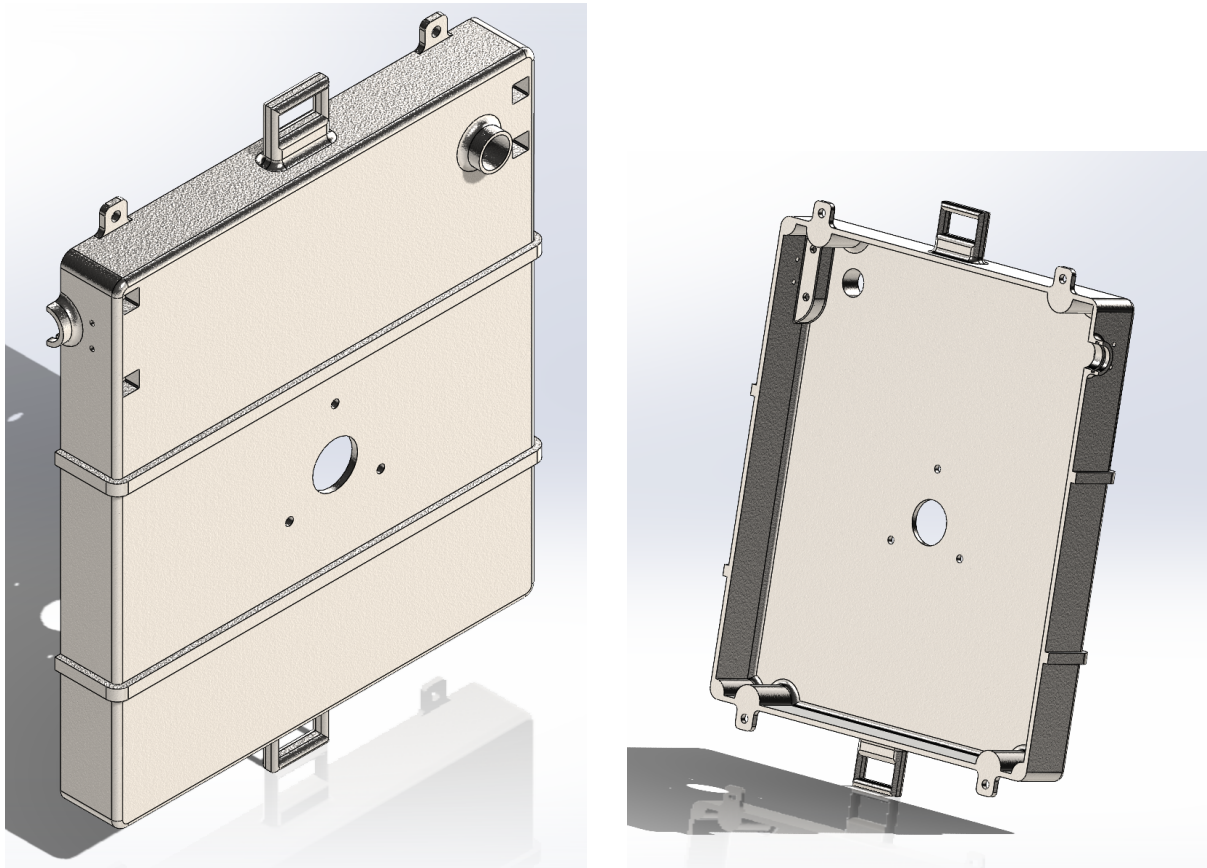


Figure 9.15: 3D SolidWorks models of the gearbox casing: front view (left) and back view (right) showing the split parting line and internal mounting bosses for bearing housings.

9.7.2 Lubrication System

The gearbox employs **splash lubrication** with the oil level maintained at **2/3 the height of the lower casing half**. This level ensures both the worm-gear mesh and the gear shaft bearings are adequately submerged for forced lubrication while preventing excessive churning losses.

The casing joint is sealed using a combination of:

- **Sealing epoxy** applied along the full parting line perimeter
- Mechanical clamping via the casing cover bolts

This dual-sealing approach was inspired by automotive oil pan (car crankcase) designs, which successfully retain lubricant under dynamic conditions.



Figure 9.16: Automotive crankcase oil pan sealing detail (inspiration for gearbox casing joint design). The gasket + sealant + bolt clamping approach is directly adapted to the split casing parting line.

Specified lubricant: ISO VG 320 EP gear oil (worm gear standard). **Estimated oil volume:** ≈ 25 litres (determined from lower casing half displacement to $2/3$ height).

9.7.3 Mounting on Cooling Tower Frame: Welded L-Brackets

The gearbox is mounted on the cooling tower's ladder frame using **welded L-brackets** fabricated from mild steel (S235, $S_y = 235$ MPa). The bracket dimensions are:

- Plate thickness: $t = 10$ mm
- Plate width: $b = 50$ mm
- Horizontal leg length: $L = 80$ mm (cantilever arm)

Each bracket is designed to carry a vertical force of $F = 1$ kN, derived from the estimated gearbox mass distributed across the number of mounting points.

Bracket Bending and Shear Check

The horizontal leg acts as a cantilever fixed at the weld. The critical cross-section is at the weld root:

Direct shear stress:

$$\tau_{\text{direct}} = \frac{F}{A} = \frac{1000}{10 \times 50} = 2.0 \text{ MPa} \quad (9.14)$$

Bending stress at weld root:

$$\sigma_{\text{bend}} = \frac{M}{S} = \frac{F \cdot L}{b t^2 / 6} = \frac{1000 \times 80}{50 \times 10^2 / 6} = 96.0 \text{ MPa} \quad (9.15)$$

Von Mises equivalent stress:

$$\sigma_{\text{VM}} = \sqrt{\sigma_{\text{bend}}^2 + 3 \tau_{\text{direct}}^2} = \sqrt{96.0^2 + 3 \times 2.0^2} = 96.1 \text{ MPa} \quad (9.16)$$

Allowable stress with $SF = 2.0$: $\sigma_{\text{allow}} = S_y / SF = 235 / 2 = 117.5 \text{ MPa}$. Since $96.1 < 117.5 \text{ MPa}$ ✓, the bracket section is adequate.

Fillet Weld Design

The bracket is attached to the frame by a fillet weld with leg size $a_w = 4 \text{ mm}$ (throat $= 0.707 \times 4 = 2.83 \text{ mm}$). Using E43xx electrode ($F_u = 430 \text{ MPa}$), the allowable weld shear stress is:

$$\tau_{w,\text{allow}} = 0.3 \times 430 = 129 \text{ MPa}$$

Minimum weld length from strength:

$$L_{w,\text{min}} = \frac{F}{\tau_{w,\text{allow}} \times a} = \frac{1000}{129 \times 2.83} = 2.7 \text{ mm} \quad (9.17)$$

Applying the workmanship minimum (weld length $\geq 4 \times a_w = 16 \text{ mm}$) raises the practical requirement to approximately **15–16 mm**. The weld stress at this length:

$$\tau_w = \frac{1000}{2.83 \times 15} = 23.6 \text{ MPa} \ll 129 \text{ MPa} \checkmark \quad (SF_{\text{weld}} = 5.5) \quad (9.18)$$

In practice, the full bracket width of 50 mm is welded, reducing the weld stress to $\tau_w = 7.1 \text{ MPa}$ ($SF = 18$). This generous margin ensures the weld is not a limiting factor under any realistic loading scenario, including dynamic and vibration loads from the fan operation.

9.7.4 Motor Mounting: Plate-and-Rib Configuration

The 345 kg motor is mounted horizontally on a **plate-and-rib welded structure** connected to the gearbox base frame. This configuration consists of a flat plate spanning the frame cross-members, reinforced with longitudinal and transverse ribs to resist bending under the motor weight and the transmitted coupling reaction forces. The assembly is fastened to the base using **M13 bolts** verified to withstand the combined static weight and dynamic coupling loads.

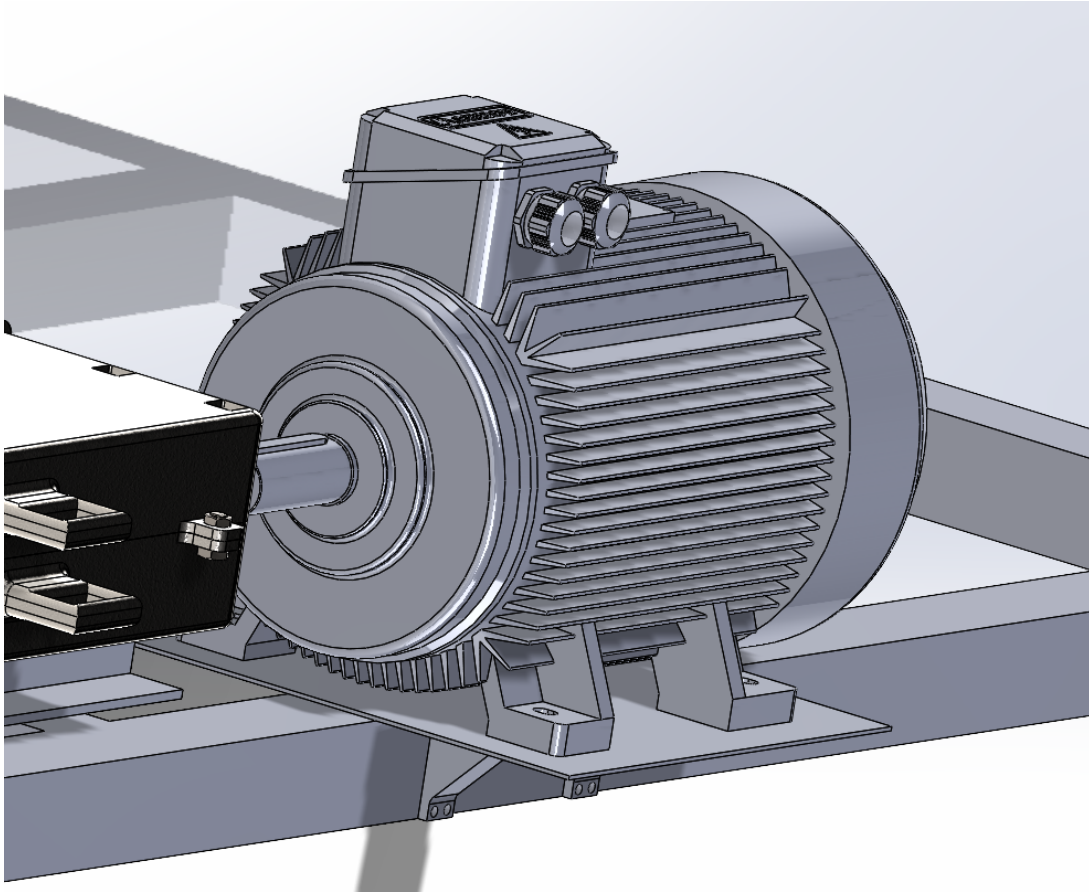


Figure 9.17: Motor mounting plate-and-rib structure. Ribs prevent plate bending under motor weight (345 kg). M13 bolts connect the assembly to the cooling tower frame base.

9.8 Complete Gearbox Assembly

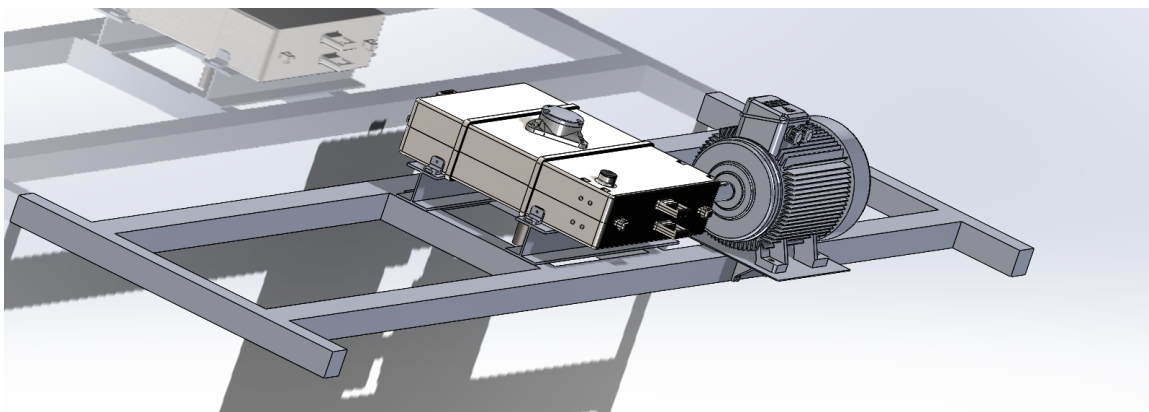


Figure 9.18: Complete 3D SolidWorks assembly model of the worm gear transmission showing all components: worm shaft with tapered bearings and housing, gear shaft with FLN housings, worm wheel, couplings, motor mounting, and split casing.

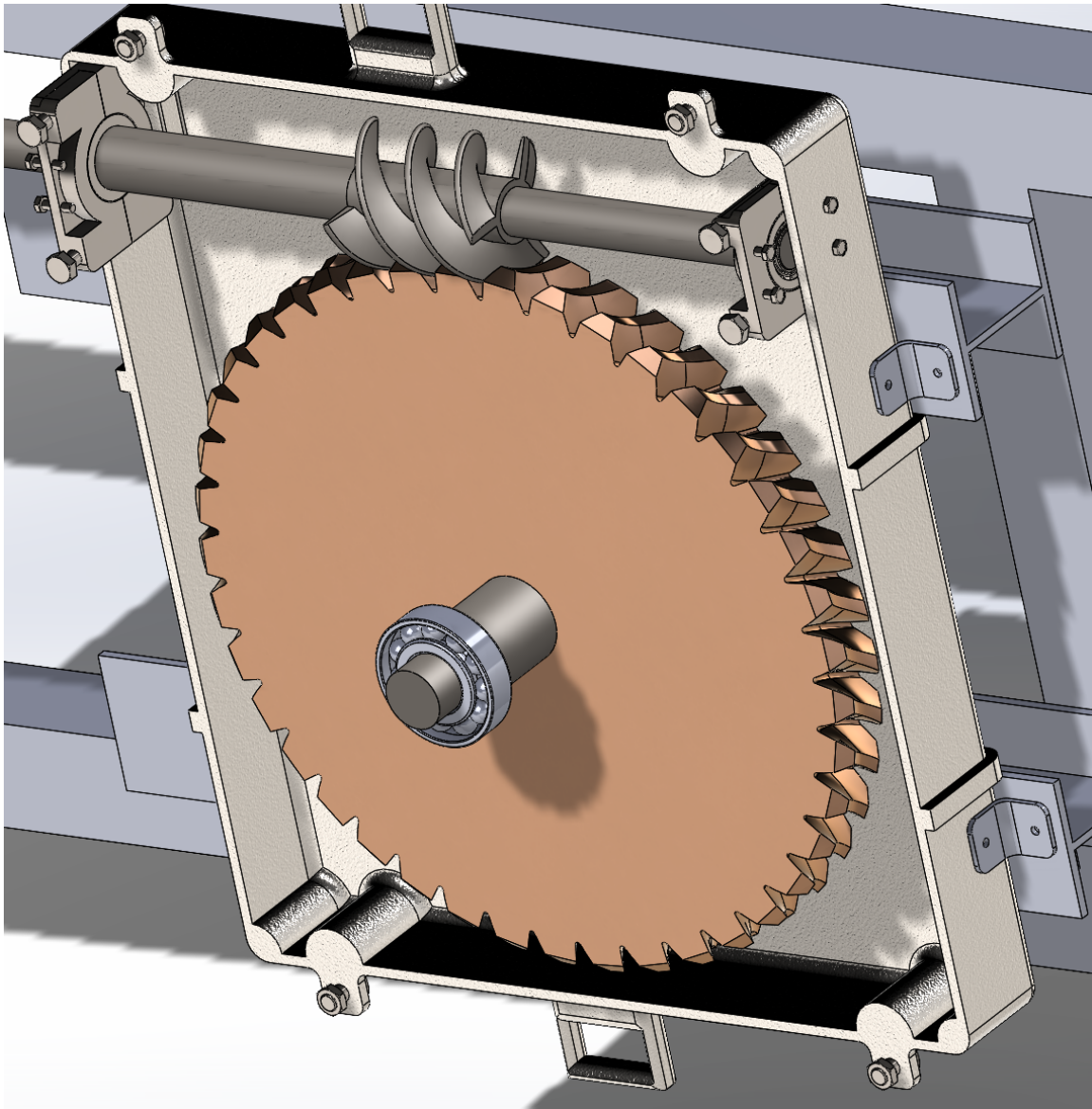


Figure 9.19: 3D SolidWorks model with upper casing removed, showing the complete internal assembly, bearing arrangements, lubrication level (2/3 lower casing), and all shaft geometries.

9.9 Output Shaft Load Analysis and Bearing Design

The following section presents the complete load analysis for the gear (output) shaft, as the highest-torque component in the drive chain.

9.9.1 Material and Life Requirement

- Worm wheel: Centrifugal cast bronze, mass 197.85 kg
- Shaft: AISI 4140, $S_y = 1640$ MPa, $\tau_{all} = 318.6$ MPa
- Design life: $10 \times 320 \times 12 \times 1.5 = 57,600$ hours

9.9.2 Gear Forces and Bending Moments

Based on transmitted power and gear pitch diameter $d_G = 678.64$ mm:

Table 9.8: Worm Wheel Force Components

Force	Symbol	Value
Tangential (circumferential)	W_t	15.84 kN
Axial (thrust)	W_a	9.55 kN
Radial (separating)	W_r	6.72 kN

Assuming symmetrical bearing placement, the bearing reactions are:

$$F_{1r} = F_{2r} = \frac{W_r}{2} = 3.36 \text{ kN}, \quad F_{1t} = F_{2t} = \frac{W_t}{2} = 7.92 \text{ kN}$$

The moment arm from bearing centreline to gear centre is $57.575 + b/2 = 57.575 + 14.5 = 72.075$ mm (bearing width $b = 29$ mm):

$$M_{xy} = 3.36 \times 72.075 = 242,172 \text{ N}\cdot\text{mm} \quad (9.19)$$

$$M_{xz} = 7.92 \times 72.075 = 570,834 \text{ N}\cdot\text{mm} \quad (9.20)$$

$$M_{\max} = \sqrt{M_{xy}^2 + M_{xz}^2} = \sqrt{242172^2 + 570834^2} = 620,086 \text{ N}\cdot\text{mm} \quad (9.21)$$

9.9.3 ASME Shaft Sizing

Maximum torque: $T = W_t \times d_G/2 = 15840 \times 339.32 = 5,374,800 \text{ N}\cdot\text{mm}$.

$$d_{\min} = \left[\frac{5.1}{318.6} \sqrt{(2 \times 620,086)^2 + (1.5 \times 5,374,800)^2} \right]^{1/3} = \mathbf{50.73 \text{ mm}} \quad (9.22)$$

Bearing bore 55 mm > 50.73 mm; shoulder 66 mm, gear seat 80 mm — all sections pass ✓.

9.10 Verification of Optimised Design Parameters

The final pitch and worm diameter combination was determined through the iterative optimisation process described in Chapter 7. The convergence results for the winning $p = 52$ mm design are summarised below.

9.11 Summary of Final Design Parameters

Table 9.9: Final Worm Gearbox Design Summary

Component	Description	Key Dimension
Worm gear pair	$p = 52 \text{ mm}$, $N_w = 3$, $N_G = 41$	$d_w = 93.2 \text{ mm}$, $d_G = 678.6 \text{ mm}$
Worm shaft	AISI 4140 Q&T, stepped	28–54 mm range
Gear shaft	AISI 4140 Q&T, stepped	55–80 mm range
Worm bearings	SKF 32207 + SKF 31309 (tapered)	Rated life 72,500 h
Gear bearings	SKF 7311 BECBY ($\times 2$)	55 mm bore
Lip seal	SKF HMSA10RG	40 $\times$ 55 $\times$ 7 mm
Input coupling	Rotex 160	877 N·m rated
Output coupling	Rexnord Omega 425	5580 N·m rated
Mounting	Welded L-bracket (10 $\times$ 50 mm) + 4 mm fillet weld	1 kN/bracket
Material (shafts)	AISI 4140 Q&T	$\tau_{\text{all}} = 318.6 \text{ MPa}$

Appendix: Shaft Coupling Extension Calculations

Detailed ASME shaft calculations for coupling extension sections of the worm shaft (required 12.85 mm, provided 40 mm) and the output shaft (required 23.3 mm, provided 50 mm) are presented here for completeness. Both sections pass with large margins due to the absence of bending moment in the pure-torsion coupling seat.

%% WORM GEAR DESIGN OPTIMIZATION & VISUALIZATION

```
clear; clc; close all;
```

%% ===== USER INPUTS =====

```
p = 20:1:100; % Axial pitches [mm]
p_in = ones(size(p))*55; % Input power [kW]
n_w = ones(size(p))*1475; % Worm speed [rpm]
```

% Common Parameters

```
N_G = 41; N_w = 3; K_s = 1.25; SF_req = 1.5; sigma_allowed = 68.9; mu = 0.05;
```

% Material Densities (kg/m³)

```
rho_steel = 7850; % Worm
rho_bronze = 8800; % Gear
```

%% ===== SETUP =====

```
n_cases = length(p);
pi_val = pi; deg2rad = pi/180;
fprintf('<strong>RUNNING %d DESIGN CASES...\</strong>\n', n_cases);
```

RUNNING 81 DESIGN CASES...

% Lookup Tables

```
V_s_table = [1 1.5 10 20 30 40 60 80 100 150 200 250 300 350 400 450 500 550 600
700 800 900 1000 1200 1400 1600 1800 2000 2200 2400 2600 2800 3000 4000 5000 6000];
C_v_table = [0.649 0.647 0.644 0.638 0.631 0.625 0.613 0.600 0.588 0.558 0.528
0.500 0.472 0.446 0.421 0.398 0.378 0.358 0.340 0.310 0.289 0.269 0.258 0.235 0.216
0.200 0.187 0.175 0.165 0.156 0.148 0.140 0.134 0.106 0.089 0.079];
b_g_table = [75 100 125 150 175 200 225];
C_s_table = [1000 975 940 900 850 800 750];
phi_table = [14.5 20 25]; Y_table = [0.100 0.125 0.150];
```

%% ===== 1. INITIAL DESIGN CALCULATION (GEOMETRIC LIMIT) =====

```
d_g = N_G*p/pi;
d_w_limit = zeros(1, n_cases);
c_initial = zeros(1, n_cases);

for i = 1:n_cases
    fun_lim = @(x) x - ((x + d_g(i))/2).^0.875 / 2;
    d_w_limit(i) = fzero(fun_lim, 50);
    c_initial(i) = (d_w_limit(i) + d_g(i))/2;
end
d_w_initial = d_w_limit;

lambada_init = atan(N_w./(pi*d_w_initial).*p)*360/(2*pi);
phi_init = zeros(1, n_cases);
for i=1:n_cases
    if lambada_init(i)>30, phi_init(i)=25; elseif lambada_init(i)<=15,
phi_init(i)=14.5; else, phi_init(i)=20; end
```

end

```
b_g_init = 2/3*d_w_initial;
b_g_clamped = max(min(b_g_init, max(b_g_table)), min(b_g_table));
C_s_init = interp1(b_g_table, C_s_table, b_g_clamped, 'linear');
Y_init = interp1(phi_table, Y_table, phi_init, 'nearest', 'extrap');
e_w_init = (cos(phi_init*deg2rad)-mu*tan(lambada_init*deg2rad))./(
cos(phi_init*deg2rad)+mu./tan(lambada_init*deg2rad));

W_t_w_init = 6*10^7*p_in*K_s./(pi*d_w_initial.*n_w)*10^-3;
w_t_G_init = (cos(phi_init*deg2rad).*cos(lambada_init*deg2rad)-
mu*sin(lambada_init*deg2rad))./(cos(phi_init*deg2rad).*sin(lambada_init*deg2rad)
+mu*cos(lambada_init*deg2rad)).*W_t_w_init;
W_r_t_init = W_t_w_init .* sin(phi_init*deg2rad) ./
(cos(phi_init*deg2rad).*sin(lambada_init*deg2rad) + mu*cos(lambada_init*deg2rad));

V_s_init = n_w.*d_w_initial./(19100*cos(lambada_init*deg2rad))*200;
V_s_clamped = max(min(V_s_init, max(V_s_table)), min(V_s_table));
C_v_init = interp1(V_s_table, C_v_table, V_s_clamped, 'linear');

w_t_G_Allowed_init =
1.32*10^-2.*C_s_init.*(d_g.^0.8).*d_w_initial.*2/3.*0.794.*C_v_init*10^-3;
SF_wear_init = w_t_G_Allowed_init./w_t_G_init;
Sigma_Bending_init = (10^3)*w_t_G_init./(p*2/3.*d_w_initial.*Y_init);
SF_bending_init = sigma_allowed./Sigma_Bending_init;
SF_initial = min([SF_bending_init; SF_wear_init]);
ratio_initial = 1./((d_w_initial./c_initial.^0.875);

%% ===== 2. ITERATIVE OPTIMIZATION =====
d_w = d_w_initial;
max_iterations = 2000;
power_increased = false(1, n_cases);
converged = false(1, n_cases);
convergence_tol = 0.001;

for iter = 1:max_iterations
    lambada = atan(N_w./(pi*d_w).*p)*360/(2*pi);
    phi = zeros(1, n_cases);
    for i=1:n_cases
        if lambada(i)>30, phi(i)=25; elseif lambada(i)<=15, phi(i)=14.5; else,
phi(i)=20; end
    end

    b_g=2/3*d_w;
    b_g_clamped = max(min(b_g, max(b_g_table)), min(b_g_table));
    C_s = interp1(b_g_table, C_s_table, b_g_clamped, 'linear');
    Y = interp1(phi_table, Y_table, phi, 'nearest', 'extrap');
    e_w = (cos(phi*deg2rad)-mu*tan(lambada*deg2rad))./(cos(phi*deg2rad)+mu./
tan(lambada*deg2rad));
```

```

motor_power_req = 31./e_w*1.25*1.1;
exceeded = motor_power_req > p_in;
p_in(exceeded) = motor_power_req(exceeded);
power_increased(exceeded) = true;

W_t_w=6*10^7*p_in*K_s./(pi*d_w.*n_w)*10^-3;
w_t_G=(cos(phi*deg2rad).*cos(lambada*deg2rad)-mu*sin(lambada*deg2rad))./(
cos(phi*deg2rad).*sin(lambada*deg2rad)+mu*cos(lambada*deg2rad)).*W_t_w;

V_s = n_w.*d_w./(19100*cos(lambada*deg2rad))*200;
V_s_clamped = max(min(V_s, max(V_s_table)), min(V_s_table));
C_v = interp1(V_s_table, C_v_table, V_s_clamped, 'linear');

w_t_G_Allowed = 1.32*10^-2.*C_s.*(d_g.^0.8).*d_w.*2/3.*0.794.*C_v*10^-3;
SF_wear = w_t_G_Allowed./w_t_G;
Sigma_Bending=(10^3)*w_t_G./(p*2/3.*d_w.*Y);
SF_bending=sigma_allowed./Sigma_Bending;
SF = min([SF_bending; SF_wear]);

for i = 1:n_cases
    if ~converged(i)
        at_limit = abs(d_w(i) - d_w_limit(i)) < 0.1;
        if abs(SF(i) - SF_req) < convergence_tol
            converged(i) = true;
        elseif (SF(i) > SF_req) && at_limit
            converged(i) = true;
        end
    end
end
if all(converged), break; end

d_w_target = d_w .* (SF_req ./ SF).^0.2;
d_w_next = max(d_w_target, d_w_limit);
d_w_next(converged) = d_w(converged);
d_w = d_w_next;
end

%% ===== 3. FINAL CALCS & STATUS =====
c_final = (d_w + d_g)/2;
ratio_final = 1./(d_w./c_final.^0.875);
W_r_t = W_t_w .* sin(phi*deg2rad) ./ (cos(phi*deg2rad).*sin(lambada*deg2rad) +
mu*cos(lambada*deg2rad));

Status = strings(n_cases, 1);
for i = 1:n_cases
    at_limit = abs(d_w(i) - d_w_limit(i)) < 0.1;
    valid_ratio = (ratio_final(i) >= 1.04) && (ratio_final(i) <= 2.03);

    if ~valid_ratio, Status(i) = "Invalid Ratio";
    elseif ~converged(i), Status(i) = "FAILED";

```

```

elseif at_limit && SF(i) > (SF_req + 0.1), Status(i) = "Clamped (>1.5)";
elseif power_increased(i), Status(i) = "Valid (P Incr)";
else, Status(i) = "Valid (Target)"; end
end

%% ===== 4. MASS & VOLUME CALCULATION =====
L_worm = (d_g + 50) / 1000;
b_g_final = 2/3 * d_w;
W_box = (b_g_final + 50) / 1000;
H_box = (d_w + d_g + 50) / 1000;

Vol_box = L_worm .* W_box .* H_box;
Mass_worm = (pi/4 * (d_w/1000).^2) .* L_worm * rho_steel;
Mass_gear = (pi/4 * (d_g/1000).^2) .* (b_g_final/1000) * rho_bronze;
Total_Mass = Mass_worm + Mass_gear;

valid_idx = ~strcmp(Status, "Invalid Ratio") & ~strcmp(Status, "FAILED");

% Compute Global Score (without disqualifying invalids, so we can see their raw
score)
Vol_Norm_Global = Vol_box ./ max(Vol_box);
Mass_Norm_Global = Total_Mass ./ max(Total_Mass);
Global_Score = (1 - Vol_Norm_Global) + (1 - Mass_Norm_Global) + e_w;

%% ===== 5. RELATIVE PAIRWISE TOURNAMENT (ALL DESIGNS) =====
fprintf('\n<strong>===== PAIRWISE PITCH TOURNAMENT =====</strong>\n');

```

```

===== PAIRWISE PITCH TOURNAMENT =====

```

```

fprintf('Running 1-on-1 matchups for ALL designs (Valid & Invalid competing)...\n');

```

```

Running 1-on-1 matchups for ALL designs (Valid & Invalid competing)...

```

```

N_total = n_cases;
Wins = zeros(1, N_total);
Tournament_Matrix = zeros(N_total, N_total);

total_matchups = 0;
for i = 1:N_total
    for j = i+1:N_total
        total_matchups = total_matchups + 1;

        % LOCAL NORMALIZATION (Only between these 2 specific designs)
        local_max_vol = max(Vol_box(i), Vol_box(j));
        local_max_mass = max(Total_Mass(i), Total_Mass(j));

        score_1 = (1 - Vol_box(i)/local_max_vol) + (1 - Total_Mass(i)/
local_max_mass) + e_w(i);
        score_2 = (1 - Vol_box(j)/local_max_vol) + (1 - Total_Mass(j)/
local_max_mass) + e_w(j);

```

```

        if score_1 > score_2
            Wins(i) = Wins(i) + 1;
            Tournament_Matrix(i, j) = 1;
            Tournament_Matrix(j, i) = -1;
        elseif score_2 > score_1
            Wins(j) = Wins(j) + 1;
            Tournament_Matrix(i, j) = -1;
            Tournament_Matrix(j, i) = 1;
        end
    end
end

[max_wins, best_idx] = max(Wins);

fprintf('Tournament Complete! %d total matchups played.\n', total_matchups);

```

Tournament Complete! 3240 total matchups played.

```

fprintf('<strong>💎 TOURNAMENT WINNER: Pitch = %d mm (Won %d matchups)</strong>\n',
p(best_idx), max_wins);

```

💎 TOURNAMENT WINNER: Pitch = 52 mm (Won 80 matchups)

```

if ~valid_idx(best_idx)
    fprintf('<strong>⚠ WARNING: The winner is actually an INVALID design! Status:
%s</strong>\n', Status(best_idx));
end
fprintf('    Vol: %.3f m^3 | Mass: %.1f kg | Eff: %.1f%%\n', ...
    Vol_box(best_idx), Total_Mass(best_idx), e_w(best_idx)*100);

```

Vol: 0.067 m³ | Mass: 236.7 kg | Eff: 88.3%

%% ===== 6. OUTPUT TABLES =====

```

OptimizedTableDetailed = table(p', p_in', d_w', d_g', c_final', ...
    lambda', phi', e_w'*100, V_s', C_v', C_s', Y', W_t_w', w_t_G', W_r_t', SF',
    ratio_final', Status, ...
    'VariableNames', {'Pitch_mm', 'Power_kW', 'WormDia_mm', 'GearDia_mm',
    'CenterDist_mm', ...
    'LeadAngle_deg', 'PressAngle_deg', 'Efficiency_Pct', 'SlidingVel_fpm', ...
    'Cv_Factor', 'Cs_Factor', 'Y_Factor', 'W_t_w_WormTangential_kN',
    'W_t_g_GearTangential_kN', 'W_r_RadialForce_kN', ...
    'SafetyFactor', 'AGMA_Ratio', 'Status'});
writetable(OptimizedTableDetailed, 'Detailed_Optimized_Design.csv');

ScoreTable = table(p', d_w', Vol_box', Total_Mass', e_w'*100, Global_Score', Wins',
    Status, ...
    'VariableNames', {'Pitch_mm', 'WormDia_mm', 'Est_Volume_m3', 'Est_Mass_kg',
    'Efficiency_Pct', 'Global_Score_Ref', 'Tournament_Wins', 'Status'});
fprintf('\n<strong>===== MASS, TOURNAMENT WINS & SCORING =====</
strong>\n');

```

===== MASS, TOURNAMENT WINS & SCORING =====

disp(ScoreTable);

Pitch_mm	WormDia_mm	Est_Volume_m3	Est_Mass_kg	Efficiency_Pct	Global_Score_Ref	Tournament_Wins
20	1525	0.6091	4938.2	19.504	0.19504	0
21	1396.8	0.54718	4381.5	21.736	0.43176	1
22	1284.6	0.49551	3917.6	24.031	0.63346	2
23	1185.9	0.45217	3528.7	26.37	0.80676	3
24	1098.5	0.41533	3198.6	28.743	0.95783	4
25	1020.6	0.38391	2917.1	31.135	1.0903	5
26	951.03	0.35694	2675.5	33.532	1.2075	6
27	888.54	0.33366	2466.9	35.92	1.3119	7
28	832.29	0.31351	2286.2	38.286	1.4052	8
29	781.3	0.29591	2128.2	40.625	1.4895	9
30	735.1	0.28057	1990.2	42.92	1.5656	10
31	692.92	0.26704	1868.2	45.172	1.635	11
32	654.44	0.25516	1760.7	47.368	1.6982	12
33	619.18	0.24466	1665.4	49.503	1.7561	13
34	586.77	0.23535	1580.4	51.576	1.8093	14
35	556.97	0.22712	1504.7	53.578	1.8582	15
36	529.43	0.21978	1436.8	55.512	1.9033	17
37	503.96	0.21325	1375.9	57.374	1.945	20
38	480.33	0.20742	1320.9	59.164	1.9836	22
39	458.35	0.2022	1271.1	60.883	2.0195	25
40	437.98	0.19758	1226.5	62.524	2.0525	27
41	418.92	0.19343	1185.7	64.098	2.0833	30
42	202.2	0.088466	430.7	78.529	2.5528	59
43	181.61	0.08289	387.8	80.455	2.5899	62
44	162.04	0.077561	347.24	82.274	2.6251	66
45	145.92	0.07351	315.55	83.37	2.6491	69
46	134.98	0.071495	297.21	84.459	2.667	71
47	125.27	0.06985	281.34	85.41	2.6824	73
48	116.51	0.06848	267.27	86.243	2.6959	74
49	109.26	0.067652	256.63	86.922	2.7062	76
50	103.29	0.067287	248.85	87.472	2.7139	78
51	97.902	0.0671	242.13	87.946	2.7203	79
52	93.153	0.067125	236.66	88.345	2.7253	80
53	92.987	0.06933	244.52	88.456	2.7212	77
54	94.495	0.072425	257.79	88.47	2.7136	75
55	95.999	0.075605	271.52	88.483	2.7057	72
56	97.499	0.078871	285.72	88.496	2.6976	70
57	98.996	0.082225	300.38	88.509	2.6893	68
58	100.49	0.085667	315.53	88.522	2.6807	67
59	101.98	0.089198	331.16	88.534	2.6718	65
60	103.46	0.092819	347.29	88.546	2.6627	64
61	104.95	0.096532	363.92	88.557	2.6534	63
62	106.43	0.10034	381.05	88.569	2.6438	61
63	107.9	0.10423	398.7	88.58	2.6339	60
64	109.38	0.10823	416.88	88.591	2.6238	58
65	110.84	0.11231	435.58	88.602	2.6134	57
66	112.31	0.1165	454.81	88.612	2.6028	56
67	113.78	0.12078	474.59	88.622	2.5918	55
68	115.24	0.12515	494.92	88.632	2.5806	54
69	116.69	0.12963	515.81	88.642	2.5691	53
70	118.15	0.13421	537.26	88.652	2.5574	52
71	119.6	0.13889	559.28	88.662	2.5453	51
72	121.05	0.14367	581.87	88.671	2.533	50
73	122.5	0.14855	605.05	88.68	2.5204	49
74	123.94	0.15353	628.82	88.689	2.5075	48
75	125.38	0.15862	653.19	88.698	2.4943	47

76	126.82	0.16382	678.17	88.707	2.4808	46
77	128.25	0.16912	703.75	88.716	2.467	45
78	129.69	0.17453	729.95	88.724	2.4529	44
79	131.12	0.18005	756.78	88.733	2.4385	43
80	132.55	0.18568	784.24	88.741	2.4238	42
81	133.97	0.19142	812.33	88.749	2.4087	41
82	135.4	0.19727	841.08	88.379	2.3896	40
83	136.82	0.20323	870.47	88.387	2.3739	39
84	138.24	0.20931	900.52	88.395	2.358	38
85	139.65	0.2155	931.24	88.402	2.3416	37
86	141.07	0.2218	962.63	88.41	2.325	36
87	142.48	0.22822	994.69	88.418	2.3081	35
88	143.89	0.23476	1027.4	88.425	2.2908	34
89	145.3	0.24141	1060.9	88.432	2.2731	33
90	146.7	0.24819	1095	88.44	2.2552	32
91	148.1	0.25508	1129.9	88.447	2.2369	31
92	149.51	0.2621	1165.4	88.454	2.2182	29
93	150.91	0.26923	1201.7	88.461	2.1992	28
94	152.3	0.27649	1238.7	88.467	2.1799	26
95	153.7	0.28387	1276.4	88.474	2.1602	24
96	155.09	0.29138	1314.9	88.481	2.1402	23
97	156.48	0.29901	1354.1	88.487	2.1198	21
98	157.87	0.30676	1394.1	88.494	2.099	19
99	159.26	0.31465	1434.8	88.5	2.0779	18
100	160.64	0.32266	1476.3	88.506	2.0564	16

```
writetable(ScoreTable, 'Design_Scoring_Report.csv');
```

```
%% ===== 7. STANDARD 2D DASHBOARD =====
```

```
figure('Name', 'Worm Gear Analysis - 2D Dashboard', 'Color', 'w', 'Position', [50  
50 1400 900]);
```

```
subplot(2,3,1);  
plot(p, W_t_w, '-o', p, w_t_G, '-s', p, W_r_t, '-^', 'LineWidth', 1.5);  
legend('W_{t,w} (Worm)', 'W_{t,g} (Gear)', 'W_r (Radial)', 'Location', 'best'); grid on;
```

```
subplot(2,3,2);  
yyaxis left; plot(p, lambada, '-d', 'LineWidth', 1.5); ylabel('Lead Angle [deg]');  
yyaxis right; plot(p, phi, '-o', 'LineWidth', 1.5); ylabel('Pressure Angle [deg]');  
xlabel('Pitch [mm]'); title('Geometric Angles'); grid on;
```

```
subplot(2,3,3);  
plot(p, e_w*100, '-d', 'LineWidth', 1.5, 'Color', [0.47 0.67 0.19],  
'MarkerFaceColor', 'g');  
xlabel('Pitch [mm]'); ylabel('Efficiency [%]'); title('Transmission Efficiency');  
grid on;
```

```
subplot(2,3,4);  
yyaxis left; plot(p, Vol_box, '-s', 'LineWidth', 1.5, 'Color', 'b'); ylabel('Volume  
[m^3]');  
yyaxis right; plot(p, Total_Mass, '-o', 'LineWidth', 1.5, 'Color', 'r');  
ylabel('Mass [kg]');  
xlabel('Pitch [mm]'); title('Mass & Volume'); grid on;
```

```
subplot(2,3,5);
```

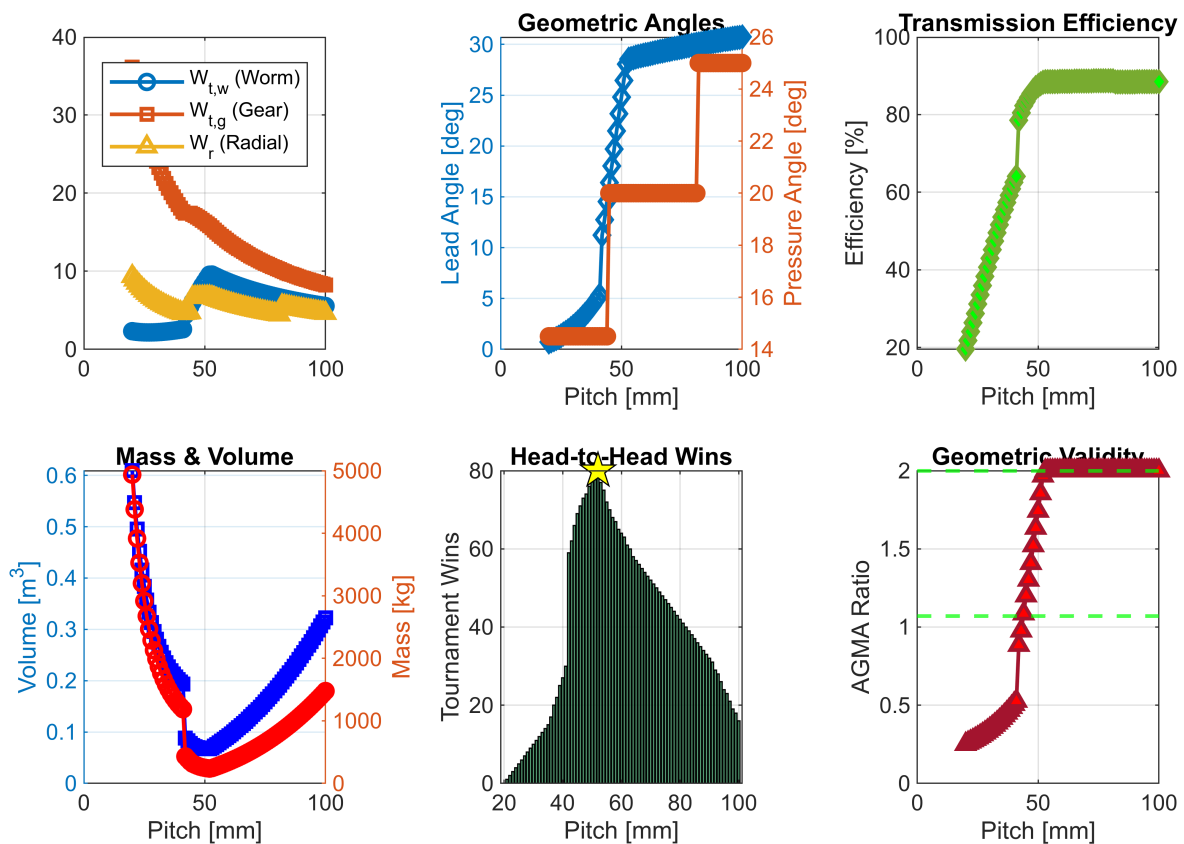
```

bar(p, Wins, 'FaceColor', [0.18 0.55 0.34], 'FaceAlpha', 0.8);
hold on;
plot(p(best_idx), max_wins, 'p', 'MarkerSize', 15, 'MarkerFaceColor', 'y',
'MarkerEdgeColor', 'k');
xlabel('Pitch [mm]'); ylabel('Tournament Wins'); title('Head-to-Head Wins'); grid
on;

subplot(2,3,6);
plot(p, ratio_final, '-^', 'LineWidth', 1.5, 'Color', [0.64 0.08 0.18],
'MarkerFaceColor', 'r');
hold on; yline(1.07, '--g', 'LineWidth', 1.5); yline(2.0, '--g', 'LineWidth', 1.5);
xlabel('Pitch [mm]'); ylabel('AGMA Ratio'); title('Geometric Validity'); grid on;
sgtitle('Worm Gear Sweep & Tournament Analysis', 'FontSize', 15, 'FontWeight',
'bold');

```

Worm Gear Sweep & Tournament Analysis



```

%% ===== 8. 2D DEMONSTRATIVE PLOTS =====

```

```

figure('Name', 'Worm Gear Optimizer - Design Landscape', 'Color', 'w', 'Position',
[80 80 1300 900]);

```

```

%--- Plot 1: Wins across all pitches ---

```

```

subplot(2,2,1);
bar(p, Wins, 'FaceColor', [0.47 0.67 0.19], 'FaceAlpha', 0.65, 'EdgeColor', 'none');
hold on;

```

```

invalid_p = p(~valid_idx);
for i = 1:length(invalid_p)
    xrect = [invalid_p(i)-0.5, invalid_p(i)+0.5, invalid_p(i)+0.5,
invalid_p(i)-0.5];
    yrect = [0 0 max(Wins)*1.15 max(Wins)*1.15];
    fill(xrect, yrect, [0.9 0.2 0.2], 'FaceAlpha', 0.15, 'EdgeColor', 'none');
end
% Re-plot the bars for invalid ones over the shading so they are clearly visible
bar(invalid_p, Wins(~valid_idx), 'FaceColor', [0.6 0.2 0.2], 'EdgeColor', 'none');

plot(p(best_idx), max_wins, 'p', 'MarkerSize', 18, 'MarkerFaceColor', [1 0.84 0],
'MarkerEdgeColor', 'k', 'LineWidth', 1.2);
xline(p(best_idx), '--k', sprintf('Best: p=%dmm', p(best_idx)), 'LineWidth', 1.2,
'LabelVerticalAlignment', 'middle');
xlabel('Axial Pitch p [mm]', 'FontWeight', 'bold');
ylabel('Tournament Wins', 'FontWeight', 'bold');
title('Design Score Landscape (All Designs Competing)', 'FontSize', 12,
'FontWeight', 'bold');
legend('Valid Design Wins', 'Invalid Region', 'Invalid Design Wins', 'Tournament
Winner', 'Location', 'northeast');
grid on; ylim([0 max(Wins)*1.2]);

%--- Plot 2: Efficiency vs Motor Power (Trade-off) ---
subplot(2,2,2);
scatter(e_w*100, p_in, 60, Wins, 'filled', 'MarkerEdgeColor', 'k', 'LineWidth',
0.5);
colormap(gca, parula); cb = colorbar; cb.Label.String = 'Tournament Wins';
hold on;
plot(e_w(best_idx)*100, p_in(best_idx), 'p', 'MarkerSize', 18, 'MarkerFaceColor',
[1 0.84 0], 'MarkerEdgeColor', 'k', 'LineWidth', 1.2);
xlabel('Efficiency [%]', 'FontWeight', 'bold'); ylabel('Required Motor Power [kW]',
'FontWeight', 'bold');
title('Efficiency vs Motor Power Trade-off', 'FontSize', 12, 'FontWeight', 'bold');
legend('All Designs (colored by wins)', 'Best Design', 'Location', 'best'); grid on;

%--- Plot 3: Mass vs Volume ---
subplot(2,2,3);
scatter(Vol_box*1e3, Total_Mass, 60, Wins, 'filled', 'MarkerEdgeColor', 'k',
'LineWidth', 0.5);
colormap(gca, parula); cb2 = colorbar; cb2.Label.String = 'Tournament Wins';
hold on;
plot(Vol_box(best_idx)*1e3, Total_Mass(best_idx), 'p', 'MarkerSize', 18,
'MarkerFaceColor', [1 0.84 0], 'MarkerEdgeColor', 'k', 'LineWidth', 1.2);
xlabel('Gearbox Volume [dm^3]', 'FontWeight', 'bold'); ylabel('Total Mass [kg]',
'FontWeight', 'bold');
title('Mass vs Volume (Wins as color)', 'FontSize', 12, 'FontWeight', 'bold');
legend('All Designs', 'Best Design', 'Location', 'best'); grid on;

%--- Plot 4: Best Design Breakdown ---
subplot(2,2,4);

```

```

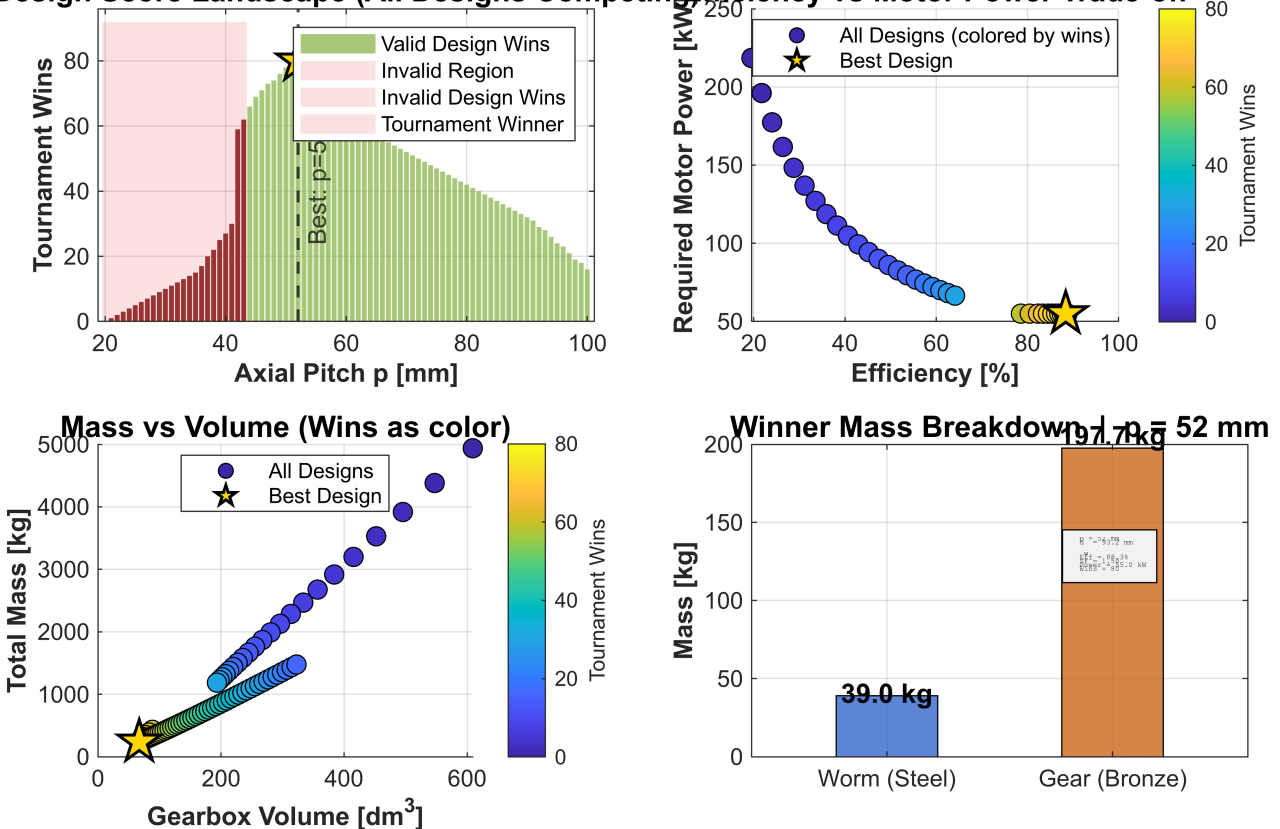
mass_vals = [Mass_worm(best_idx), Mass_gear(best_idx)];
b2 = bar(1:2, mass_vals, 0.45, 'FaceAlpha', 0.8);
b2.FaceColor = 'flat'; b2.CData = [0.2 0.4 0.8; 0.8 0.4 0.1];
hold on;
text(1, Mass_worm(best_idx)*1.04, sprintf('%.1f kg', Mass_worm(best_idx)),
'HorizontalAlignment', 'center', 'FontWeight', 'bold', 'FontSize', 11);
text(2, Mass_gear(best_idx)*1.04, sprintf('%.1f kg', Mass_gear(best_idx)),
'HorizontalAlignment', 'center', 'FontWeight', 'bold', 'FontSize', 11);
set(gca, 'XTick', [1 2], 'XTickLabel', {'Worm (Steel)', 'Gear (Bronze)'});
ylabel('Mass [kg]', 'FontWeight', 'bold');
title(sprintf('Winner Mass Breakdown | p = %d mm', p(best_idx)), 'FontSize', 12,
'FontWeight', 'bold');

annot = sprintf(' p = %d mm\n d_w = %.1f mm\n Eff = %.1f%%\n SF = %.2f\n Power =
%.1f kW\n Wins = %d', ...
p(best_idx), d_w(best_idx), e_w(best_idx)*100, SF(best_idx), p_in(best_idx),
max_wins);
annotation('textbox', [0.78 0.12 0.19 0.22], 'String', annot, ...
'FitBoxToText', 'on', 'BackgroundColor', [0.95 0.95 0.95], ...
'EdgeColor', 'k', 'FontSize', 9, 'FontName', 'Courier');
grid on; xlim([0.4 2.6]);
sgtitle('Worm Gear Optimizer – Design Space Analysis', 'FontSize', 15,
'FontWeight', 'bold');

```

Worm Gear Optimizer — Design Space Analysis

Design Score Landscape (All Designs Competing) **Efficiency vs Motor Power Trade-off**



```

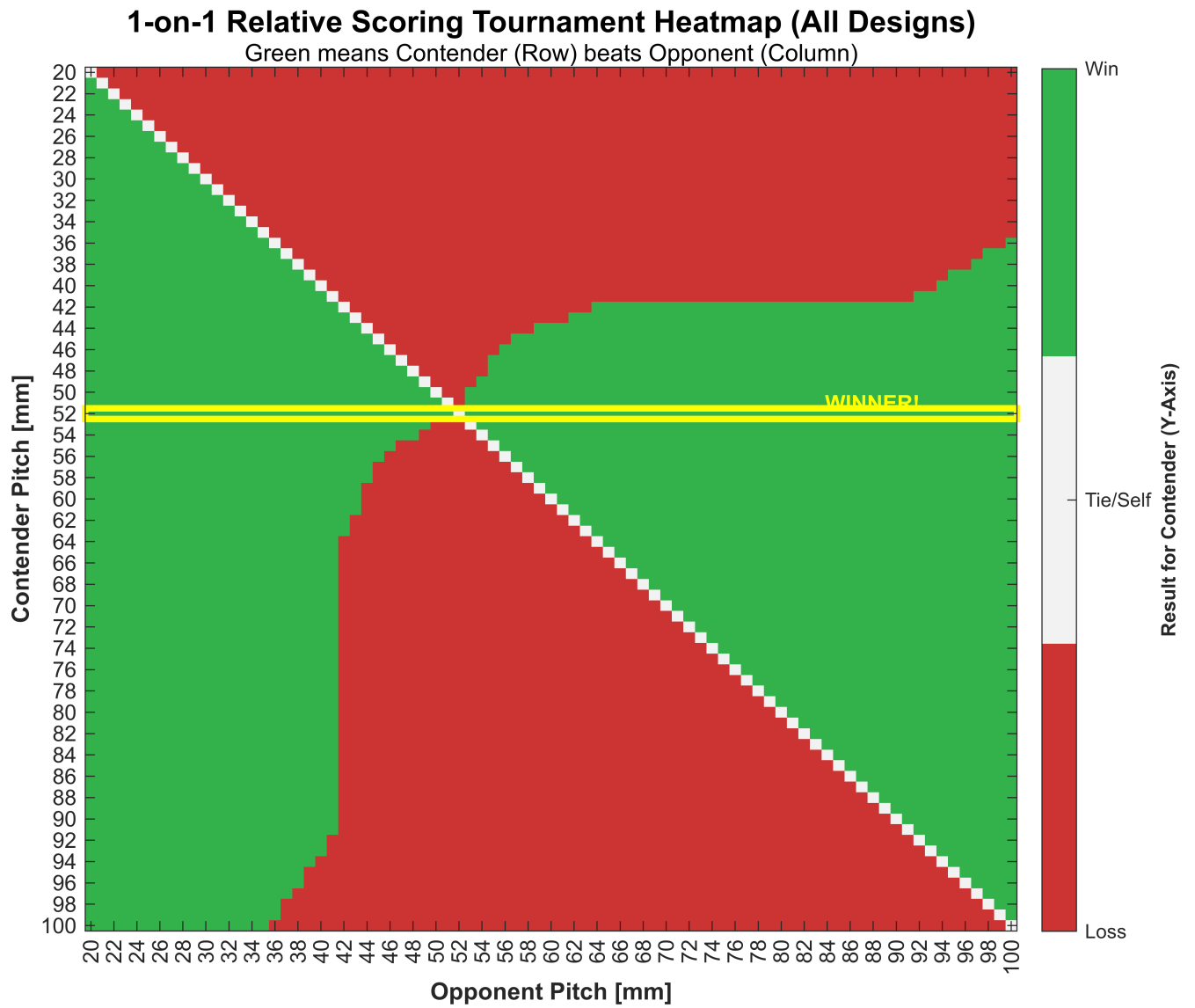
%% ===== 9. PAIRWISE HEATMAP PLOT =====
figure('Name', 'Pairwise Matchup Matrix', 'Color', 'w', 'Position', [200 200 800
650]);
imagesc(Tournament_Matrix);
cmap = [0.8 0.2 0.2; 0.95 0.95 0.95; 0.2 0.7 0.3];
colormap(cmap);
c = colorbar('Ticks', [-1, 0, 1], 'TickLabels', {'Loss', 'Tie/Self', 'Win'});
c.Label.String = 'Result for Contender (Y-Axis)';
c.Label.FontWeight = 'bold';

ticks_to_show = 1:2:N_total;
xticks(ticks_to_show); xticklabels(p(ticks_to_show));
yticks(ticks_to_show); yticklabels(p(ticks_to_show));

title('1-on-1 Relative Scoring Tournament Heatmap (All Designs)', 'FontSize', 14,
'FontWeight', 'bold');
xlabel('Opponent Pitch [mm]', 'FontWeight', 'bold');
ylabel('Contender Pitch [mm]', 'FontWeight', 'bold');
subtitle('Green means Contender (Row) beats Opponent (Column)');

hold on;
rectangle('Position', [0.5, best_idx-0.5, N_total, 1], 'EdgeColor', 'y',
'LineWidth', 3);
text(N_total*0.8, best_idx-1, 'WINNER!', 'Color', 'y', 'FontWeight', 'bold');

```



1- Determine Specifications

Altitude = 300m $n = 110 \text{ rpm}$ $D_{fan} = 4m$ $V_{out} = 12 \frac{m}{sec}$ $P_{stat} = 0.9 \text{ cm H}_2\text{O}$
 $= 0.35433 \text{ in H}_2\text{O}$

Calculations:

$$A = \pi D^2 \times \frac{1}{4} \rightarrow A = 12.566 \text{ m}^2$$

$$Q = V \times A \rightarrow Q = 150.796 \frac{\text{m}^3}{sec} \rightarrow Q = 317519.6 \text{ CFM}$$

At 300m and $50^\circ\text{C} \rightarrow \rho_{air} = 1.05401 \frac{\text{kg}}{\text{m}^3} \rightarrow \rho_{air} = 0.0658 \frac{\text{lb}}{\text{ft}^3}$

Also $\rho_{water} = 62.34 \frac{\text{lb}}{\text{ft}^3} \rightarrow h_v = \frac{V^2 \times 12 \times \rho_{air}}{2 \times g \times \rho_{water}} \rightarrow h_v = \frac{32.37^2 \times 12 \times 0.0658}{2 \times 32.17 \times 62.34}$
 $\rightarrow h_v = 0.30513 \text{ in H}_2\text{O}$

We have: $h_t = h_v + h_s \rightarrow h_t = 0.6595 \text{ in H}_2\text{O}$

$$\text{aph}_{102} = \frac{Q \times h_t \times \rho_{water}}{33000 \times 12} \rightarrow \text{aph}_{102} = 33.17 \text{ hp}$$

Assuming 0.8 fan efficiency:

$$\rightarrow P_{out} = \frac{\text{aph}}{0.8} \rightarrow P_{out} = 41.46 \text{ hp} \xrightarrow{\text{convert}} P_{out} \approx 31 \text{ kw}$$

2- Choosing the right mechanism

Options:

~~Belt-chain drive~~Chosen!
Belted-Helical Gear drive

Worm Gear drive

Major heat loss
9.3 kw loss!
So bad!

Helical-Belted Gearbox:

The expected efficiency: $0.99^{6+1} \times 0.975 \times 0.975 \approx 0.904$ | Assuming two stage would be enough
Two bearings for each shaft

$$\rightarrow P_{in} = 34.63 \text{ kw} \quad \text{Minimum required Power}$$

Chosen Motor:

Baldor ECTM431AT 60 hp 1780 rpm $\rightarrow \text{ratio} = 16.182$ | Two stages would be enough

Specifically engineered for cooling towers - epoxy coated for corrosion
 Moisture Protection and high efficiency!

Pre Design Analysis - Worm Gear

$P_{out} = 31 \text{ kW}$ at $n = 110 \text{ rpm}$

Design Procedure:

Assumed efficiency: $e_w = 0.7$ $k_a = 1.25$ $SF = 1.5$

$\rightarrow P_{in} = P_{out} \times 1.25 \times \frac{1.5}{0.7} = 55 \text{ kW}$

Choosing a motor:

Searching using AI $\rightarrow$ There are some available 55 kW motors

based on dimensions and rpm $\rightarrow$ $250 \text{ M2 } n_{in} = 2970 \text{ rpm}$ All around 3000 rpm
 $225 \text{ M12 } n = 1475 \text{ rpm}$

$\rightarrow m_G = \frac{2970}{110} = 27 \rightarrow m_{G2} = 13.4$

Choosing Material: centrifugal cast bronze (Best Available)

Number of Threads: $N_w = 2 \rightarrow N_G = 54$ $N_w = 3 \rightarrow N_G = 41$

Choosing worm $P_n \rightarrow$ Calculating for 3 entries:

$P_n = 6 \text{ mm}, d_2 = 18 \text{ mm}$

$d_G = 103.13 \text{ mm}$

$d_{G2} = 79.30 \text{ mm}$

$P_n = 20 \text{ mm}, d_2 = 60 \text{ mm}$

$d_G = 343.77 \text{ mm}$

$d_{G2} = 261.01 \text{ mm}$

$P_n = 40 \text{ mm}, d_2 = 120 \text{ mm}$

$d_G = 687.55 \text{ mm}$

$d_{G2} = 522.03 \text{ mm}$

Choosing d_w and C :

Idea: $d_w = \frac{C^{0.875}}{1.5}$ and $d_w + d_G = 2C \rightarrow$ Solving for C :

$C = 64.30 \text{ mm}$

$C_2 = 16.84 \text{ mm}$

$d_w = 25.47 \text{ mm}$

$d_{w2} = 7.15 \text{ mm}$

$C = 207.37 \text{ mm}$

$C_2 = 158.56 \text{ mm}$

$d_w = 70.967 \text{ mm}$

$d_{w2} = 56.11 \text{ mm}$

$C = 407.91 \text{ mm}$

$C_2 = 311.70 \text{ mm}$

$d_w = 128.28 \text{ mm}$

$d_{w2} = 101.375 \text{ mm}$

Let $\lambda = \frac{d_w}{\pi d_G}$

$\lambda = 8.52^\circ \rightarrow \phi = 14^\circ \text{ on}$

$\lambda_2 = 15.70^\circ \rightarrow \phi = 20^\circ$

$e_w = 0.737$

$e_{w2} = 0.828$

$\lambda = 10.17^\circ \rightarrow \phi = 14^\circ \text{ on}$

$\lambda_2 = 18.77^\circ \rightarrow \phi = 20^\circ$

$e_w = 0.769$

$e_{w2} = 0.847$

$\lambda = 11.23^\circ \rightarrow \phi = 14^\circ \text{ on}$

$\lambda_2 = 20.65^\circ \rightarrow \phi = 20^\circ$

$e_w = 0.785$

$e_{w2} = 0.859$

undergo no change!

Best!

oil $\mu = 0.05$

So based on these result we require lower rpm motors? elsewhere

Result Summary:

Two Pole Motor: (smallest motors from the catalogue were chosen)

$$225M \times 2 \quad (2965 \text{ rpm}) \quad 55 \text{ kW} \quad M = 320 \text{ kg} \quad V = 0.158 \text{ m}^3$$

$$450 \times 435 \times 805 \text{ mm}$$

Design 1 ($P = 6 \text{ mm}$)

$$d_w = 25.47 \text{ mm}$$

$$d_G = 1030.13 \text{ mm}$$

$$N_w = 2$$

Design 2 ($P = 20 \text{ mm}$)

$$d_w = 70.97 \text{ mm}$$

$$d_G = 343.77 \text{ mm}$$

$$N_w = 2$$

Design 3 ($P = 40 \text{ mm}$)

$$d_w = 128.18 \text{ mm}$$

$$d_G = 687.55 \text{ mm}$$

$$N_w = 2$$

4-pole Motor:

$$225M \times 4 \quad (1475 \text{ rpm}) \quad 55 \text{ kW} \quad M = 345 \text{ kg} \quad V = 0.164 \text{ m}^3$$

Design 1 ($P = 6 \text{ mm}$)

$$e = 0.828$$

$$d_w = 20.38 \text{ mm}$$

$$d_G = 77.30 \text{ mm}$$

$$N_w = 3$$

Design 2 ($P = 20 \text{ mm}$)

$$e = 0.849$$

$$d_w = 56.11 \text{ mm}$$

$$d_G = 261.01 \text{ mm}$$

$$N_w = 3$$

Design 3 ($P = 40 \text{ mm}$)

$$e = 0.859$$

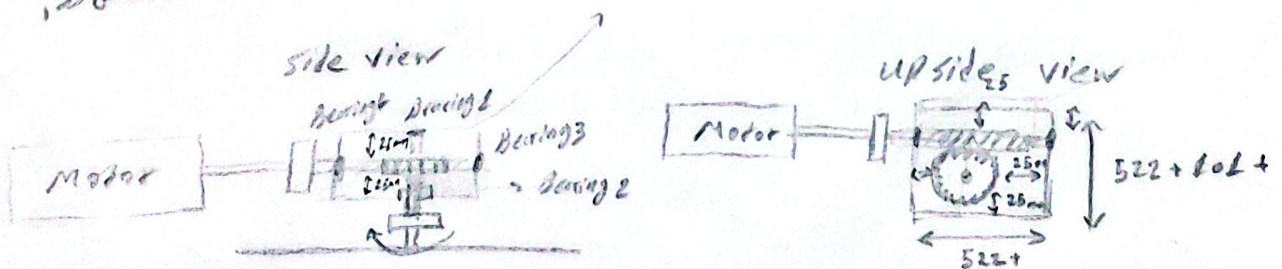
$$d_w = 101.375 \text{ mm}$$

$$d_G = 522.03 \text{ mm}$$

$$N_w = 3$$

Since All $P > 4 \text{ mm} \rightarrow b = 2 \times \frac{d_w}{3}$

bearing is provided as per size of shaft



$$\rightarrow V_{rot} = V_A = (d_w + d_G + 2 \times 25 \text{ mm}) \times (d_G + 2 \times 25 \text{ mm}) \times (d_w + 2 \times 25 \text{ mm})$$

$$\rightarrow \text{At full power} \rightarrow P = 55 \text{ kW at } 1475 \text{ rpm} \rightarrow T_{req} = 5.252 \text{ Nm}$$

shaft and Bearing Design:

$$W_t^W = \frac{6 \times 10^7 \times P \times n}{\pi \times d \times n} = W_r^G \rightarrow W_a^W = W_t^W \times \frac{\cos \phi_n \cos \lambda - \mu \sin \lambda}{\cos \phi_n \sin \lambda + \mu \cos \lambda}$$

$$W_t^W = \frac{6 \times 10^7 \times P \times \lambda}{\pi d_w \times \pi W} , \quad W_a^W = W_t^G = W_t^W \times \frac{\cos \phi \cos \lambda - \mu \sin \lambda}{\cos \phi \sin \lambda + \mu \cos \lambda}$$

Also $W_t^G (\text{Allowable}) = 1.32 \times 10^{-2} \times C_s \times d_G^{0.8} \times b_e^G \times C_m \times C_v \rightarrow v_s = \frac{\pi W \times d_w}{12600 \pi \cos \lambda}$

Strongest Available material was chosen: Centrifugal cast Bronze!

Design 1 ($P=40mm$)

$W_t^W = 12.12 \text{ kN}$

$W_t^G = 20.56 \text{ kN}$

$C_s = 1000, C_m = 0.794$

$v_s = 1272 \frac{\text{ft}}{\text{min}} \rightarrow C_v = 0.228$

$W_t (\text{Allowed}) = 17.36 \text{ kN}$

$\rightarrow n_d = 0.92 \text{ | Fails!}$

Design 2 ($P=45mm$)

$W_t^W = 11.02 \text{ kN}$

$W_t^G = 18.295 \text{ kN}$

$C_s = 1000, C_m = 0.794$

$v_s = 1412 \frac{\text{ft}}{\text{min}} \rightarrow C_v = 0.215$

$W_t (\text{Allowed}) = 19.91 \text{ kN}$

$\rightarrow n_d = 1.09 \text{ | Not enough}$

Design 3 ($P=50mm$)

$W_t^W = 10.065 \text{ kN}$

$W_t^G = 16.48 \text{ kN}$

$C_s = 1000, C_m = 0.794$

$v_s = 1552 \frac{\text{ft}}{\text{min}} \rightarrow C_v = 0.203$

$W_t^G (\text{Allowed}) = 22.39 \text{ kN}$

$\rightarrow n_d = 1.36 \text{ | Not enough though!}$

Well it may seem that I should further increase the module but i won't do that. see before all this i chose to have the smallest allowable worm diameter. by increasing the worm diameter which can be done for all of them by a factor of $\frac{2}{1.07}$ all the safety factors can be increased. though increasing the diameter effects the efficiency but also lets me do use a lower pitch value which leads to smaller wormgear. which is quite important since the worm gear is made out of bronze which has a higher density. Also smaller pitch value leads us to less required length for worm shaft aswell giving us lower volume and lower mass. Also I would like to note that change in pitch effects the high density worm gear volume and thus its mass by a power of 2 while the shafts diameter effects that by power of one. In a nutshell a balance must be made between all these factors. To do this I want to find a case with the lowest pitch value and highest respective allowed worm shaft diameter value and I already have the utmost working design with highest pitch value and lowest allowable diameter. I presume the optimized combination is somewhere in between! So it is obvious that at any pitch the best practice is to minimize d_w . we have three cases already lets see the trend

Four-Pole (1475 rpm): Two Pole were worse in every way so i only go with two

Design 1 (6mm)

Design 2 (20mm)

Design 3 (60mm)

$$W_t^W = \frac{6 \times 10^7 \times P_{in}}{\pi \times d_w \times n_w} \rightarrow$$

$$W_t^W = 34.74 \text{ kN}$$

$$W_t^W = 12.69 \text{ kN}$$

$$W_t^W = 7.02 \text{ kN}$$

$$W_t^W = W_a^W \rightarrow W_a^W = W_t^W \times \frac{\cos \gamma \cos \lambda - \mu \sin \lambda}{\cos \rho_n \sin \lambda + \mu \cos \lambda} \quad \text{Also } W_a^W = W_t^G \rightarrow$$

$$W_t^G = 102.95 \text{ kN}$$

$$W_t^G = 31.67 \text{ kN}$$

$$W_t^G = 16.00 \text{ kN}$$

$$W_t^G (\text{Allowable}) = 1.32 \times 10^{-2} \times C_s \times d_G^{0.8} \times b_G^G \times C_m \times C_v$$

I choose the strongest material for now and all $b_G = \frac{2}{3} d_w$ (centrifugal cast)

$$C_s = 1000$$

$$C_s = 1000$$

$$C_s = 1000$$

$$C_{mG} = 0.8$$

$$C_{mG} = 0.8$$

$$C_{mG} = 0.8$$

$$v_s = 320.99 \frac{\text{ft}}{\text{min}}$$

$$v_s = 1157.79 \frac{\text{ft}}{\text{min}}$$

$$v_s = 1673.24 \frac{\text{ft}}{\text{min}}$$

$$C_v = 0.450$$

$$C_v = 0.235$$

$$C_v = 0.2$$

$$d_G = 72.30 \text{ mm}$$

$$d_G = 261.06 \text{ mm}$$

$$d_G = 522.003 \text{ mm}$$

$$d_w = 20.38 \text{ mm}$$

$$d_w = 56.66$$

$$d_w = 60.375 \quad \eta_d = 1.53$$

$$W_t^G (\text{Allowed}) = 2.13 \text{ kN} \times \quad W_t (\text{Allowed}) = 7.96 \text{ kN} \times \quad W_t (\text{Allowed}) = 21.3 \text{ kN} \checkmark$$

So we expect modules $20 \text{ mm} < m < 4 \text{ mm}$ to work well !!!

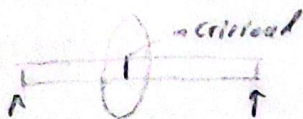
Final Design for the winner:

Shaft:

$$d = \sqrt{\frac{5.1}{P_t} \sqrt{(K_m M)^2 + (K_t T)^2}}$$

Material chosen: AISI 1140

$$\rightarrow P_t = 300 \text{ MPa}$$



$$\rightarrow T_{out} = 4.77 \text{ kN.m} \quad , \quad W_t^W = \frac{W_t^W \sin \gamma}{\cos \rho_n \sin \lambda + \mu \cos \lambda}$$

$$M_{max} = (25 \text{ mm} + \frac{b_G}{2}) \times W_t^W \times \frac{1}{2} \rightarrow W_t^W = 6.348 \text{ kN} \rightarrow$$

$$M_{max} = 239.509 \text{ N.m}$$

$$\rightarrow V = 58.31 \text{ liter}$$

Bearings

$$M_{ff} = 345 \text{ kg} + \left(\frac{\pi d_G^2}{4} \times \left(\frac{2}{3} d_w \right) + \frac{\pi d_w^2}{4} \times (2 \times 25 + 25) \right) \times \rho + 2 m_g = 163 \text{ kg}$$

estimate

Verifying Final Design:

$$P = 52 \text{ mm} \quad d_w = 93.23 \text{ mm} \quad d_g = 678.63 \quad N_w = 3 \quad \phi = 20^\circ$$

$$\rightarrow C = \frac{1}{2} (d_w + d_g) = \underline{385.93 \text{ mm}} \quad \text{Range} = 91.66 \sim 171.92 \text{ mm} \quad \text{verified}$$

$$\tan \lambda = \frac{P \times N_w}{d_w \times \pi} \rightarrow \lambda = \underline{28.04^\circ} \rightarrow e_w = \underline{0.883} \quad \text{Power verified} \\ 55 \text{ hp} \quad 1475 \text{ rpm}$$

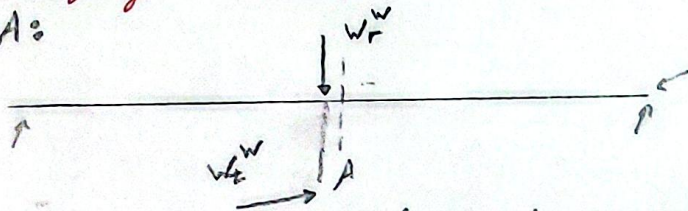
$$\rightarrow W_t^W = \frac{6 \times 10^7 \times P \times k_s}{\pi d_w N_w} \rightarrow \underline{W_t^W = 9.548 \text{ kN}}$$

$$W_t^G = W_a^W = W_t^W \times \frac{\cos \phi \cos \lambda - \lambda \sin \lambda}{\cos \phi \sin \lambda + \lambda \sin \lambda} \rightarrow \underline{W_t^G = W_a^W = 15.85 \text{ kN}}$$

$$W_r^G = W_r^W = W_t^W \times \frac{\sin \phi}{\cos \phi \sin \lambda + \lambda \sin \lambda} \rightarrow \underline{W_r^G = W_r^W = 6.726 \text{ kN}}$$

Designing worm shaft

A:



bending moment:

$$F_{tot} = \sqrt{W_t^{W2} + W_r^{W2}} = \underline{11.08 \text{ kN}}$$

$$T = W_t^W \times \frac{d_w}{2} = \underline{445.08 \text{ N.m}}$$

$$L_{shaft} = d_g + 50 = 738.63 \text{ mm}$$

$$\rightarrow M = F_{tot} \times (d_g + 2 \times 25) \times \frac{1}{2} \times \frac{1}{2} \rightarrow \underline{M = 2127.6 \text{ N.m}}$$

Worm Geometry:

$$a = \frac{P_z}{b} = \underline{16.55 \text{ mm}}$$

$$b = \underline{19.15 \text{ mm}}$$

$$d_p = 93.23 \text{ mm}$$

$$d_o = 126.33 \text{ mm}$$

$$d_o = d_p + 2 \times a$$

$$d_r = d_p - 2 \times b \rightarrow$$

$$\underline{d_r = 54.93 \text{ mm}}$$

Design Continued:

$$\text{AISI 4140} \rightarrow P_s = 381 \text{ MPa}$$

$$k_m = k_t = 1.2 \quad \text{Approx}$$

$$d = B \sqrt[3]{\frac{5.1}{P_t} / (k_m M)^2 + (k_t T)^2} \rightarrow B = 1 \rightarrow \underline{d_A = 32.61} < d_r = 54.93 \text{ mm}$$

$$B_{min} = \frac{1}{0.6} = \sqrt[3]{\frac{1}{1 - (\frac{d_i}{d_o})^3}} \rightarrow \frac{d_i}{d_o} = 0.93 \rightarrow \begin{matrix} d_o = d_r = 54.93 \text{ mm} \\ d_i = 51.217 \text{ mm} \end{matrix} \quad \begin{matrix} \text{do be discussed} \\ \text{later} \end{matrix}$$

Choosing a bearing: $L_{10} = 1175 \text{ rpm} \times 12 \text{ h} \times 320 \text{ days} \times 10 \text{ years} \times 3.32 \times 10^2$ $F_r = 3.363 \text{ kN}$
 $F_a = 15.55 \text{ kN}$ $F_r = 6.726 \text{ kN}$ on two bearings: $F_a = 7.725 \text{ kN}$
 total $A = 0.609$
 B for an estimate on minimum required diameter $\rightarrow d \approx 20 \text{ mm}$



I choose fully sealed bearings for input and output and open bearings for inner bearings:

To provide sealing I use a double row bearing

Choosing: 3304A-2RS1 $d = 20 \text{ mm}$ $B = 22.2 \text{ mm}$ $C = 23.6 \text{ kN}$ $C_0 = 14.6 \text{ kN}$

$$V = 1$$

$$\frac{F_a}{C_0} = 0.512 \rightarrow e = 0.437 \text{ and } \frac{F_a}{V F_r} = 2.36 \rightarrow X = 0.56 \quad Y = 1.005$$

$$\rightarrow F_e = X V F_r + Y F_a \rightarrow F_e = 19.847 \text{ kN}$$

$$19.847 \times 2^{\frac{1}{3}} = C_{10} \times (10^6 \times A)^{\frac{1}{3}} \rightarrow L = 8.3 \times 10^6 \text{ not good at all!}$$

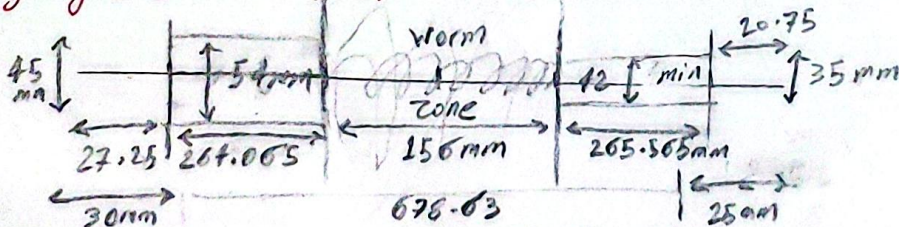
Choosing bigger bearing: we require $\sim 200 = C_{10 \text{ min}}$

So the chosen bearings are

31309 and 32207 $\rightarrow$ No we must for the worm shaft design begin:

$$d = 45 \text{ mm} \quad d = 35 \text{ mm} \quad B = 27.25 \quad B = 20.75 \text{ mm}$$

Designing the worm shaft $d_G + 2 \times 25 + 5 \rightarrow d_{w} = 733.63 \text{ mm}$



Also $\rightarrow$ 31309 $\rightarrow$ Shaft fillet radius $\rightarrow r_{\text{max}} = 2 \text{ mm}$

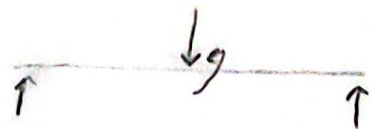
Housing fillet radius $\rightarrow r_{\text{max}} = 1.5 \text{ mm}$

32207 $\rightarrow$ Shaft fillet radius $\rightarrow r_{\text{max}} = 1.5 \text{ mm}$

Housing fillet radius $\rightarrow r_{\text{min}} = 1.5 \text{ mm}$

$$d = B^{3/4} \sqrt[3]{\frac{5.1}{P_t} \sqrt{(k_m M)^2 + (k_t T)^2}}$$

$$F_m = 11.68 \text{ kW and } T = 445.08 \text{ N.m} \quad P_t = 381 \text{ MPa}$$



Worm Section:

$$\rightarrow d = 32.01 > d_r = 54.93 \text{ mm} \quad \text{OK}$$

Left Step:

$$\rightarrow \frac{F_m}{2} \times \left(\frac{27.25}{2} + 264.005 \right) = M \rightarrow M = 1621.740 \text{ N.m}$$

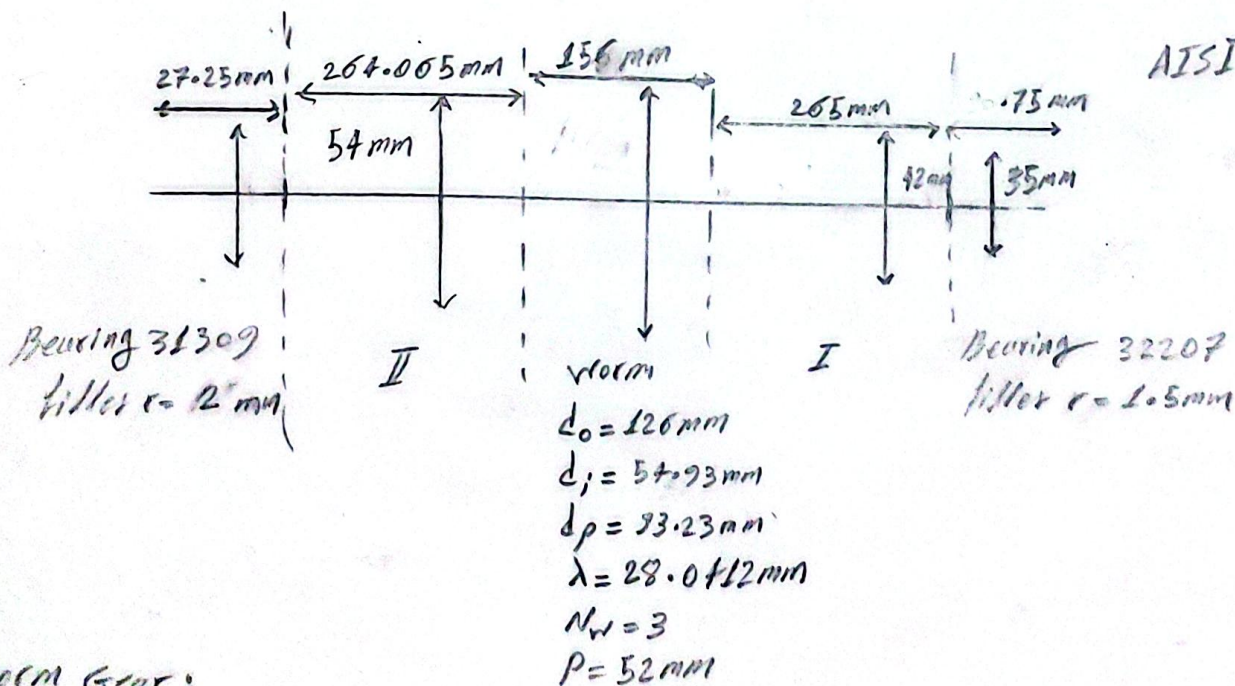
$$\rightarrow B = 1 \rightarrow d_{req} = 28.23 \text{ mm} < d_{now} = 54 \text{ mm} \quad \text{OK and } 45$$

Right Step:

$$M = \frac{F_m}{2} \times \left(\frac{20.75}{2} + 265.565 \right) \rightarrow M = 1611.5 \text{ N.m}$$

$$\rightarrow B = 1 \rightarrow d_{req} = 28.48 \text{ mm} > 42 \text{ mm OK} \quad \text{and } 35 \text{ mm OK}$$

Final Worm-Gear Dimensions:



Worm Gear:

$$d_g = 678.63 \text{ mm}$$

$$N_g = 41$$

$$d_g = \frac{2}{3} \times d_w = 62.15$$

Centrifugal cast bronze

$P_{\text{int, gearbox}} = 31 \text{ kW}$, $\eta = (0.99)^7 \times (0.975)^2 = 0.886$
 تعداد پینیون : 6 $\rightarrow$ bevel-helical
 تعداد گستر : 1
 نسبت چرخه : 1 $k_s = 1.25$ (uniform-light shock)
 نسبت تمرکز : 1 $n_d = 1.5$

$P_m = \frac{P_{\text{int}} \times k_s \times n_d}{\eta} = \frac{31.4 \times 1.25 \times 1.5}{0.886} = 48.1 \text{ kW}$
 موتور مدل : 225 Mx4
 55 kW , 1475 rpm

$P_{\text{in}} = P_m \times n_d \times k_s = 55 \times 1.5 \times 1.25 = 103 \text{ kW}$

$r_{\text{pm, in}} = 1475 \text{ rpm}$, $r_{\text{pm, out}} = 110 \text{ rpm} \rightarrow \text{ratio} = 13.41$
 bevel ratio = 3.5
 helical ratio = 3.83

bevel : (ratio: 3.5) , $P_{\text{pinion}} = P_{\text{in}} \times (0.99)^2$, $P_{\text{gear}} = P_{\text{in}} \times (0.99)^2 \times 0.975$

$n_p = 1475 \text{ rpm}$, $n_g = 421.43 \text{ rpm}$, $P_{\text{in}} = 103 \text{ kW}$, $\phi = 20^\circ$

$N_{\text{min}} = \frac{2}{(1+2m) \sin^2 \phi} (m + \sqrt{m^2 + (1+2m) \sin^2 \phi}) \rightarrow N_{\text{min}} = 15.24 \rightarrow N_{\text{min}} = 17$

$N_p = 17$ ratio 3.5 $\rightarrow N_g = 60$

$b = \min(10m_c, A_0/3)$, $m_G = 3.5 = \tan \delta_g = \cot \delta_p$

$W_t = \frac{6 \times 10^7 P_{\text{inbp}}}{\pi d_p n_p} = \frac{6 \times 10^7 \times P_{\text{inbp}}}{\pi m_e N_p n_p}$, $k_A = 1.25$ (uniform-light shock) , γ_x

$k_{H\beta} = 1.01 + 5.6 (10^{-6}) b^2$, $\gamma_\beta = 1$ (اندازه های مستقیم) , $k_\theta = 1$ ($15^\circ < 12^\circ$)

$F_{\text{lim}} = 207 \text{ MPa}$, $\gamma_{NT} = 1.683 N^{-0.0323}$

$Q_v = 6$, $\gamma_{\delta p} = 0.24$, $\gamma_{\delta g} = 0.19$
 bending-pinion-bevel : $m_c = 6.67 \text{ mm}$
 bending-gear-bevel : $m_c = 7.07 \text{ mm}$

المسألة الثانية :

steel - steel $\rightarrow 191 \sqrt{\frac{N}{mm^2}}$, $Z_I = 0.08$, Z_H (خيالات براس بازه b) ,

حين كورسيت ميت $\rightarrow Z_{xc} = 1.5$ (properly crowned teeth)
نظام

حين كورسيت ميت $\rightarrow \sigma_{Hlim} = 1380 \text{ mpa}$, $Z_E = \sqrt{\frac{1}{2}}$, $Z_N = 1$

حل بآسيب $\rightarrow$ $\left\{ \begin{array}{l} \text{contact - bevel - pinion : } m_c = 9.28 \text{ mm} \\ \text{contact - bevel - gear : } m_c = 8.9 \text{ mm} \end{array} \right.$

$\rightarrow m_c = \max(6.67, 7.08, 9.28, 8.9 \text{ mm}) \rightarrow m_c = 9.28 \text{ mm} \xrightarrow[m_c]{\text{تقريباً}} \underline{\underline{m_c = 10 \text{ mm}}}$

bevel gear : $d_g = 600 \text{ mm}$, $\gamma_g = 74^\circ$, $N_g = 60$, $\phi = 20^\circ$, $b = 100 \text{ mm}$

bevel pinion : $d_p = 170 \text{ mm}$, $\delta_p = 16^\circ$, $N_p = 17$, $\phi = 20^\circ$, $b = 100 \text{ mm}$

وہاں $\rightarrow z_N = 2.466N^{-0.056}$

$$\therefore I_w = 17 \times m_t$$

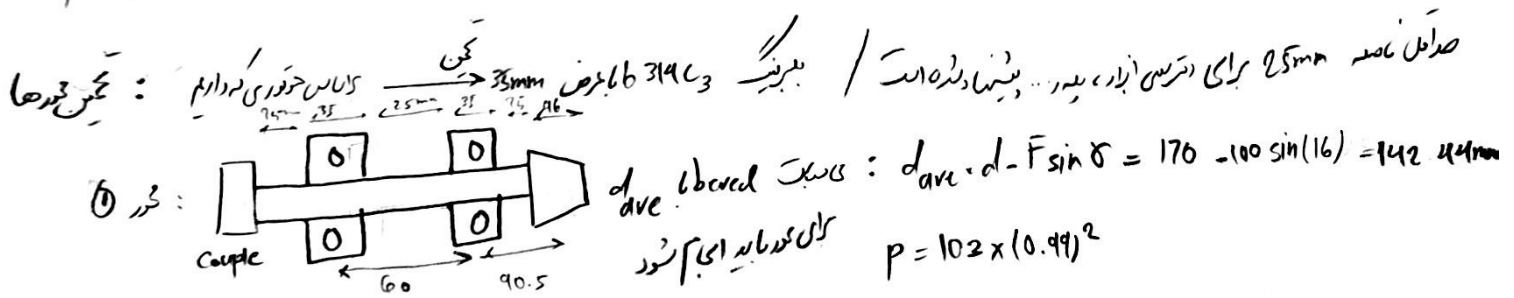
matlab

{ contact-helical-pinion: $m_f = 11.0144$ ^{mm}
 contact-helical-gear: $m_t = 10.854$ ^{mm}

$$\rightarrow m_n = m_t \cos(20^\circ) = 10.35 \text{ mm} \xrightarrow[\text{Series 2 (Secondary)}]{\text{निर्धारित}} \underline{m_n = 11 \text{ mm}}$$

Helical: pinion: $d_p = 199 \text{ mm}$, $N_p = 17$, $b_p = 99.5 \text{ mm}$, $m_n = 11 \text{ mm}$, $\phi = 20^\circ$, $\psi = 20^\circ$
 gear: $d_g = 760 \text{ mm}$, $N_g = 65$, $b_g = 99.5 \text{ mm}$, $m_n = 11 \text{ mm}$

۱) اے اللہ جانو کہ یہ حقیر و ذلیل بہت آجندہ کندار و داندانہ ہمارا نذرہ حاضر کرنا تھا لیکن تیرا اللہ ہی مومن اور امن پر عمل کرتا ہے جس نے اس کو جان بیکار و اندر ازنی نہیں



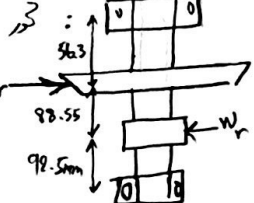
$$W_t = \frac{6 \times 10^7 \times P}{\pi d n} = 9176.7 \text{ N} \rightarrow W_r = W_t \tan \phi \cos \gamma = 3210.65 \text{ N}, T = \frac{P}{n \times \frac{2\pi}{60}} = 653.56 \text{ N.m}$$

steel carburized and case hardened $\rightarrow \rho = 7.85 \frac{\text{gr}}{\text{cm}^3} \rightarrow W = \rho \times \frac{\pi}{4} d^2 \times b \cos \theta \times g + 11$

$$\rightarrow W = 7.95 \times 10^{-6} \left(\frac{\text{kg}}{\text{mm}^3} \right) \times \frac{\pi}{4} \times (142.44)^2 \times 100 \times \cos(16) \times 9.81 = 118 \text{ N} \approx 12 \text{ kg}$$

$$\rightarrow W_f = 3210.65 + 118 = 3328.65 \text{ N} \rightarrow M_{\max} = 301242.8 \text{ Nmm}, T_{\max} = 653560 \text{ Nmm}$$

commercial steel with keys $\rightarrow P_t = 41 \text{ MPa}$, minor shocks $\rightarrow k_m = 2, k_t = 1.5 \rightarrow \underline{d_1 = 52.31 \text{ mm}}$

② β :  $d_{ave} = d - F \sin \gamma = 600 - 100 \times \sin(174) = 503.87 \text{ mm}$

$$P = 103 \times 0.99^2 \times 0.975, W_{t \text{ bevel}} = \frac{6 \times 10^7 P}{\pi d n} = 8852.6 \text{ N}$$

$$W_{r \text{ bevel}} = W_t \tan \phi \cos \gamma = 888 \text{ N}, T = \frac{P}{\omega} = 2230273.7 \text{ N}\cdot\text{mm}$$

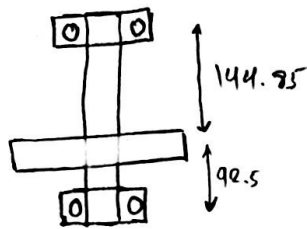
$$n = 1475 \times \frac{1}{3.5} = 421.43 \text{ rpm} \quad \text{helical: } P = 113 \times 0.99^2 \times 0.975 \times 0.99^2 \rightarrow W_t = \frac{6 \times 10^7 P}{\pi d n} = 21969 \text{ N}$$

$$\rightarrow W_r = W_t \times \frac{\tan \phi n}{\cos \phi} = 8509.23 \text{ N}, T = \frac{P}{\omega} = 2185891.2 \text{ N}\cdot\text{mm}$$

$$\rightarrow \begin{cases} T_{max} = 2230273.7 \text{ N}\cdot\text{mm} \\ M_{max} = 460870 \text{ N}\cdot\text{mm} \end{cases} \rightarrow \underline{d = 75.57 \text{ mm}}$$

③ β : $\bar{W}_{bevel} \rightarrow$

$$n = 1475 \times \frac{1}{3.5} \times \frac{1}{3.83} = 110.7 \text{ rpm}$$



$$P = 103 \times 0.99^2 \times 0.975 \times 0.99^2 \times 0.975$$

$$W_t = \frac{6 \times 10^7 P}{\pi d n} = 21487.32 \text{ N}$$

$$W_r = W_t \times \frac{\tan \phi n}{\cos \phi} = 8322.7 \text{ N}$$

$$T = \frac{P}{\omega} = 8165183.16 \text{ N}\cdot\text{mm} \rightarrow T_{max} = 8165183.16 \text{ N}\cdot\text{mm}$$

$$\rightarrow M_{max} = 469824.04 \text{ N}\cdot\text{mm} \rightarrow \underline{d = 115.17 \text{ mm}}$$

تکین های اندازه، جرم، بازه :

$$\gamma = 0.975 \times 0.975 \times (0.99)^2 \times 0.935 = 0.8284$$

0.975 ← helical, bevel بازه

0.99 ← بازه یاتس ها : 6 یاتس + 1 یاتس کفر

0.935 ← 100% در 225MX4

motor 225 MX4 → total length = 835 mm

total height = 555 mm

total width = 435 mm

m = 345 kg

clearance bevel gear bevel helical clearance bevel helical

$$\text{طول: } 25 \text{ mm} + 600 \text{ mm} + 760 \text{ mm} + 25 \text{ mm} = 280.5 \text{ mm} = 1129.5 \text{ mm}$$

helical gear clearance clearance

$$\text{عرض: } 760 \text{ mm} + 25 \text{ mm} + 25 \text{ mm} = 810 \text{ mm}$$

clearance helical, p space between projected width bevel clearance bearing

$$\text{عرض: } 25 \text{ mm} + 99.5 \text{ mm} + 25 \text{ mm} + 27.6 \text{ mm} + 25 \text{ mm} + 2 \times 35 \text{ mm} = 272.1 \text{ mm}$$

(f)

$\rho = 7.95 \times 10^{-6} \frac{\text{kg}}{\text{mm}^3}$ $\rightarrow$ gears : steel carburized and case hardened $\rightarrow$ $\rho = 7.95 \times 10^{-6} \frac{\text{kg}}{\text{mm}^3}$
 $\rho = 7.85 \times 10^{-6} \frac{\text{kg}}{\text{mm}^3}$ $\rightarrow$ shaft : commercial steel $\rightarrow$ $\rho = 7.85 \times 10^{-6} \frac{\text{kg}}{\text{mm}^3}$

shaft 1 : $L = 35 + 25 + 35 + 25 + 96 = 216 \text{ mm} \rightarrow m = L \times \frac{\pi}{4} \times d^2 \times \rho = 3.64 \text{ kg}$

shaft 2 : $L = 35 + 25 + 99.5 + 25 + 27.6 + 25 + 35 = 272.1 \text{ mm} \rightarrow m = 9.58 \text{ kg}$

shaft 3 : $L = 272.1 \text{ mm} \rightarrow m = 9.58 \text{ kg}$

bored pinion : ($d_{\text{ave}} = 142.44 \text{ mm}$) $\rightarrow m = \frac{\pi}{4} d_{\text{ave}}^2 \times b \times \cos \phi_p \times \rho = 12.02 \text{ kg}$

bored gear : ($d_{\text{ave}} = 503.87 \text{ mm}$) $\rightarrow m = \frac{\pi}{4} d_{\text{ave}}^2 \times b \times \cos \phi_g \times \rho = 43.15 \text{ kg}$

helical pinion : $m = 24.3 \text{ kg}$

helical gear : $m = 354.33 \text{ kg}$

$\rightarrow$ $m_{\text{total}} = 456.6 \text{ kg}$, $V_{\text{total}} = 0.2489 \text{ m}^3$

طراحی بزرگ جریز صفرون :

$\xrightarrow{+weight} 11.4894 \text{ kW}$

$w_z^w = w_a^g = 9.5485 \text{ kW}$, $n_w = 3$

$w_a^w = w_z^g = 15.8369 \text{ kW}$, $n_g = 41$

$w_r^w = w_r^g = 6.7213 \text{ kW}$, $d_g = 678.6367 \text{ mm}$

$d_g = 62.1522 \text{ mm}$

① $\xrightarrow{\text{بدین نوعی نمی سازد}}$ tapered roller bearings

② $\xrightarrow{\text{نمی آید چون نمی دهند مایل می}}$ spherical roller bearings

③ $\longrightarrow$ angular contact ball bearings

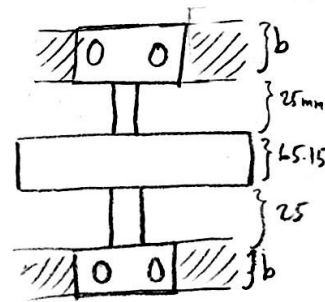
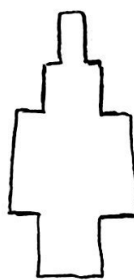
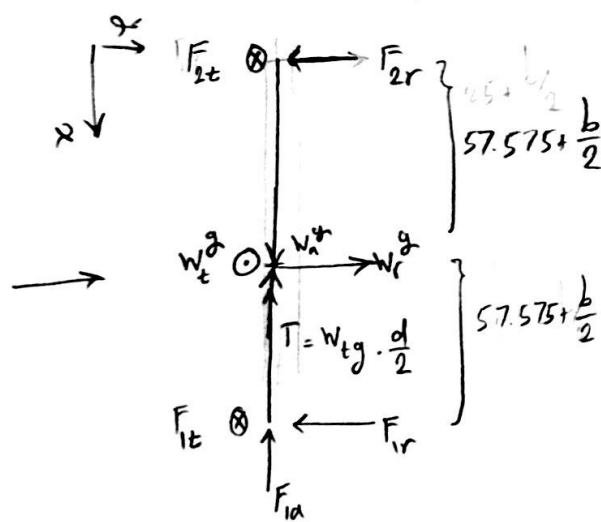
gear: centrifugal case bronze $\rightarrow$ gear mass = 197.85 kg

shaft: AISI 4140 steel $\rightarrow S_y = 1640 \text{ MPa}$, $S_t = 1770 \text{ MPa}$

hours: $10 \times 320 \times 12 \times 1.5 = 57600 \text{ hrs}$ باید عمر کند

Large roller bearing (high speed) $\rightarrow$ در نظر گرفتن ضریب ایمنی

بزرگ باریش می کند نیروی (وزن + w_a)



طراحی نهایی :

$F_{1r} = F_{2r} = \frac{w_r^g}{2} = 3.36 \text{ kW}$

$F_{1t} = F_{2t} = \frac{w_t^g}{2} = 7.92 \text{ kW}$

$F_{1a} = w_a^g = 11.5 \text{ kW}$

$M_{dy, \max} = 3.36(57.575 + \frac{b}{2}) \text{ Nm}$, $M_{xz, \max} = 7.92(57.575 + \frac{b}{2}) \text{ Nm}$, $T_{\max} = 15.84(\frac{678.6367}{2})$

$T_{\max} = 5374.8 \text{ N.m}$, $M_{\max} = \sqrt{M_{xz}^2 + M_{dy}^2}$, $p_t = \min(0.3S_y, 0.18S_{ut}) = (492, 318.6) = 318.6 \text{ MPa}$

$\longrightarrow$ که در EES باید د براس b

angular contact ball bearing → 7311BECBY

این کورن بیرینگ های SKF
در سایز 11
تعمیراتی کورن های 57600h

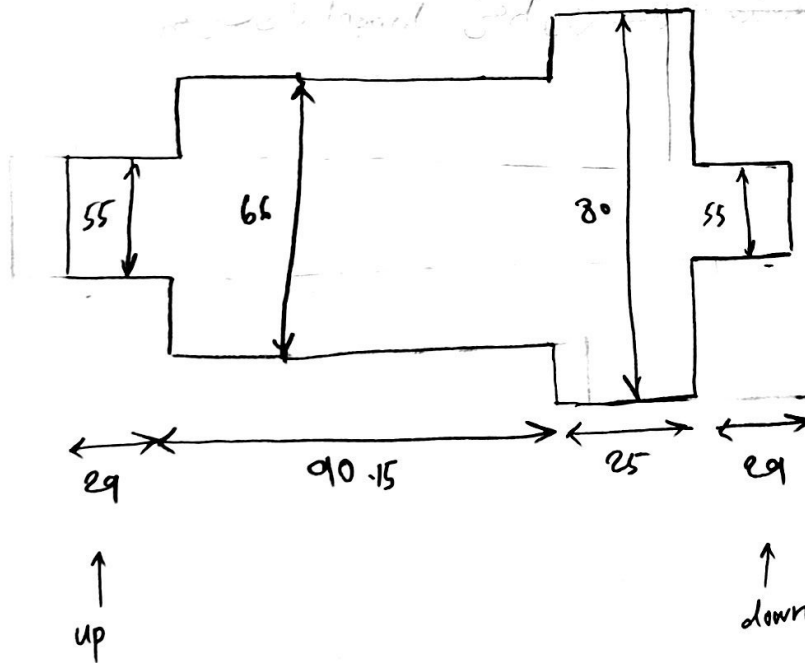
$L \rightarrow b = 29 \text{ mm}$, $d = 55 \text{ mm}$
نقطه $d_{\min} = 50.73 \text{ mm}$
(کمر 58800 سرعت 1)
کمر 58800 سرعت 1

سایز 11 SKF tapered, spherical, یا کورن های 57600h برای سایز 11 های 57600h
دی کورن های 57600h از سایز 11 های 57600h

$d = 55 \text{ mm}$, $B = 29 \text{ mm}$, $\text{mass} = 1.4 \text{ kg}$, $d_{\min} = 66 \text{ mm}$, $r_{\max} = 2 \text{ mm}$, $D = 120 \text{ mm}$

طبق جدول استاندارد
برای سایز 11 های 57600h
 $d_1 = 55 \text{ mm}$ $d_{\min} = 66 \text{ mm}$ $r_{\max} = 2 \text{ mm}$ $d_2 = 66 \text{ mm}$ $d_3 = 79.2 \text{ mm}$

برای سایز 11 های 57600h
از سایز 11 های 57600h
 $d_4 = 55 \text{ mm}$ $r = r_a = 2 \text{ mm}$ $79.2 \rightarrow 80$
نقطه 80
نقطه 80



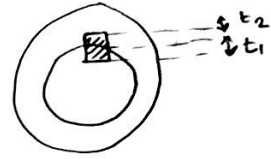
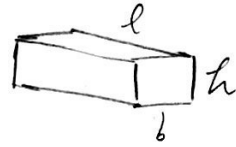
طراحی جایی خازر، خازر برای نیست فریاضفون

$S_y = 1640 \text{ MPa}$ / 200 MPa → جزیخ حفرین

بدین ندانش کمر دریت خازر
هم از جزیخ
استی ب ی کیم

براین اس اس در دیم خازر قدرت 66mm
است از جزیخ استیخار در دیم
کمای خازر استیخار استیخار کیم

$d_{sh} = 66 \text{ mm}$



$d_{sh} = 66 \text{ mm}$ → $b = 20 \text{ mm}$
 $h = 11 \text{ mm}$
 $t_1 = 7.5 \text{ mm}$
 $t_2 = 1.9 \text{ mm}$
 $l = 148 \text{ mm}$
 $l' = 49.3 \text{ mm}$

جزیخ خازر: $l = \frac{4T \cdot F_s}{d \cdot b \cdot S_y}$
جزیخ خازر: $l = \frac{4 \times 5374.8 \text{ N} \cdot \text{m} \times 1}{66 \times 10^{-3} \times 20 \times 10^{-3} \times 1640 \times 10^6} = 10 \text{ mm}$

جزیخ خازر: $l = \frac{4T \cdot F_s}{d \cdot h \cdot S_y} = \frac{4 \times 5374.8 \times 1}{66 \times 10^{-3} \times 11 \times 10^{-3} \times 200 \times 10^6} = 148 \text{ mm}$

200MPa سوجعت کیم

از آبی کیم جزیخ این طرز خازر
جزیخ تقسیم کیم $l' = l/3 = 49.3 \text{ mm}$

بدین اند کفر کفر کفر
کفر کفر کفر کفر کفر

⑩ کوئینک جی:

⑪ کوئینک موتور بدلت دردی:

کوئینک جدول بدلت دردی: $T_{\text{motor}} = 9250 \times \frac{55 \text{ kW}}{1475 \text{ rpm}} = 356 \text{ N.m} \rightarrow T_d = 356 \times 1.75 \times 1.5 = 667.5 \text{ N.m}$

دائری کند: 801 N.m
 سایه قرمز دایره ۱۰۰
 برش ۵۰۰ درشت
 کوئینک بدلت دردی
 راداشه بایر
 60mm bore
 60mm: نسبت شافت
 45mm: نسبت دردی

این کوئینک به جادادی خود تراکند
 یک بک آن را 60mm و دیگر را 40mm
 تراش می دهد (ای ۴۵، ۴۰ و ۴۱) چون کوئینک هم
 به لازم دارد
 چرا بجهت بران انداز، ۲ نفی که داریم
 دقت ده که داریم (برای جادادی)
 (۳) حساب بران فیط های شفتی هم هست
 (۴) عیب ارضاء تیره

Steel = $S_y = 1640 \text{ MPa}$ (نسبت شفتی به درشت است)

درای توانم فشارش به هم درشت کند

Cast iron: $S_y = 450 \text{ MPa}$ (درشت به هم شفتی)

طراحی خارجی کوئینک موتور دردی

key way width: $19 \text{ mm} = b$

دقیق تراد
 $t_1 = 11 \text{ mm}$
 $t_2 = 7 \text{ mm}$
 $t_2 = 4.4 \text{ mm}$

برای برش خارجی شافت: $l = \frac{4TFS}{dbS_y} = \frac{4 \times 801 \times 1.5}{60 \times 10^{-3} \times 19 \times 10^{-3} \times 1640 \times 10^6} = 9.3 \text{ mm} \xrightarrow{l < d} l = d = 60 \text{ mm}$

برای لیبی خارجی شافت: $l = \frac{4TFS}{dthS_y} = \frac{4 \times 801 \times 1.5}{60 \times 11 \times 450} = 16 \text{ mm} \xrightarrow{l < d} l = d = 60 \text{ mm}$

نسبت دردی جعبه دردی: $d = 40 \text{ mm}$ (۴۵ → ۴۰) (برای نیکو کوئینک بدلت) $d = 40, b = 12, h = 8, t_1 = 5, t_2 = 2.3$

برای لیبی خارجی شافت: $l = \frac{4TFS}{dthS_y} \approx 33.3 \text{ mm} \xrightarrow{l < d} l = 40 \text{ mm}$

برای برش خارجی شافت: $l = \frac{4TFS}{dbS_y} = 22.25 \text{ mm} \xrightarrow{l < d} l = 40 \text{ mm}$ (در مدل کوئینک جایی خود)

از بین تقسیم شده و مانده ابعاد را میگیریم → خارجوند ر کوبلنگ رکن سائر 135 → تقسیم کوبلنگ رهاهای دردی

→ ثابت دردی تبدیل 45 به 40 چون بعد کوبلنگ
 $(l=40\text{ mm}, b=12, h=8, t_1=5, t_2=3.3)$
 سائر 135، 70mm پس کوبلنگ خارجی است
 سائر تقسیم خارجیت

کوبلنگ امکان حساب بد برای این ابعاد ندار
 و از درستی خاص انتقال گشت در مبر با سائر در مبر است
 کوبلنگ امکان سائر 425

40 mm : تقسیم کوبلنگ تقسیم
 55 mm : تقسیم کوبلنگ
 $d=50\text{ mm}$ سائر بعد داریم

$T_{max} = 53741.8\text{ Nm}$ (فریب انجمن اردن جدید)

(برای این T_{max} سائر $d_{min}=47.76$ داریم که سائر سائر است)

دست مانت تقسیم برای این کوبلنگ را → تقسیم خارجی تقسیم
 تقسیم تقسیم

Cast-iron یا 450mpa دردی داریم

تقسیم خارجی تقسیم → $d=50\text{ mm}$ سائر $b=16, h=10, t_1=6, t_2=4.3$

خارجی جدید
 $l = \frac{4TFS}{dbs_y} = 16.4\text{ mm}$ برای $s_y = 1640$

$l = \frac{4TFS}{dhs_y} = 95\text{ mm}$ سائر 425، $l=114$ دارو سائر
 به تقسیم خارجیت
 $l = 95\text{ mm}$